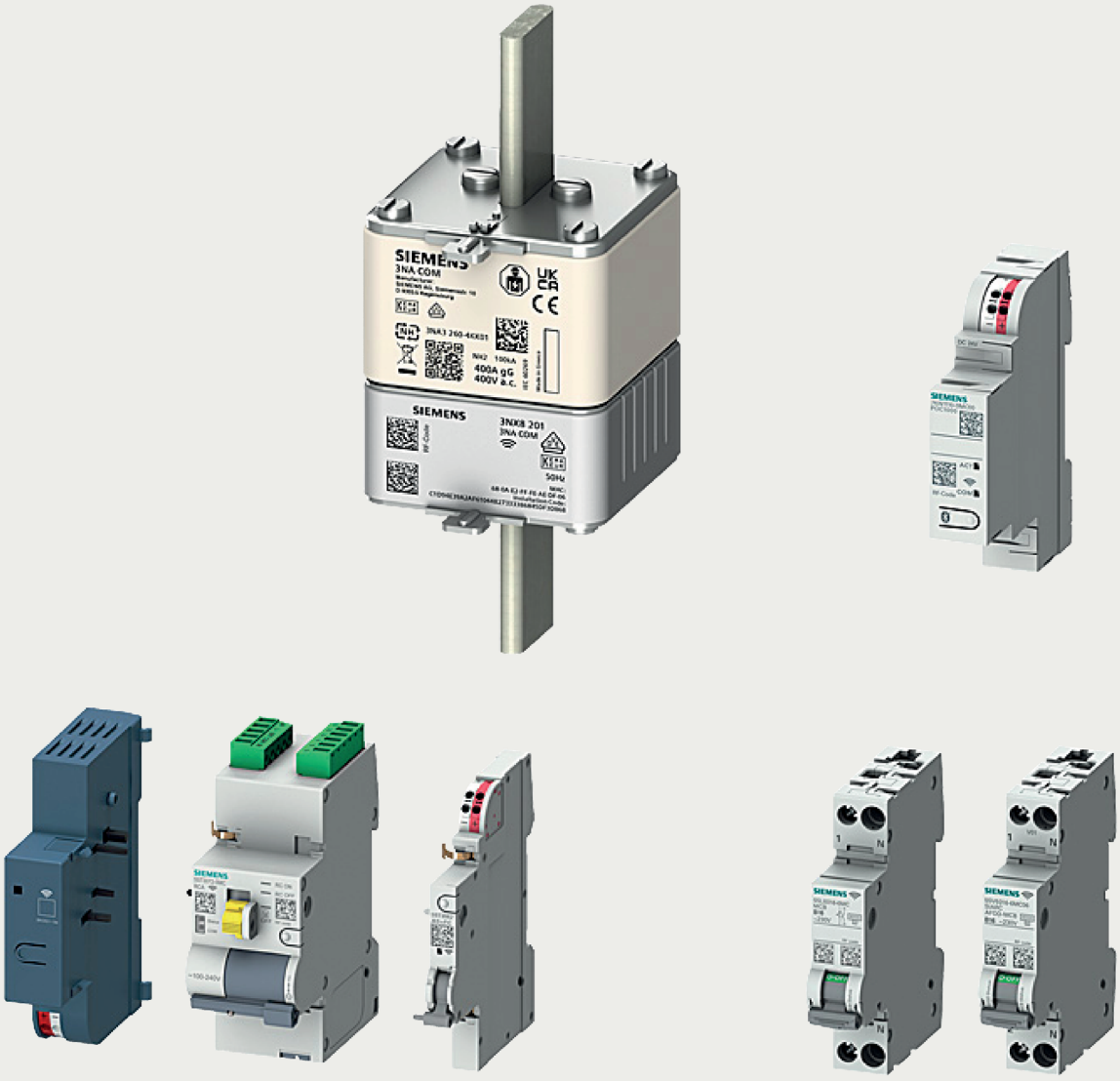
SIEMENS



SYSTEM MANUAL

SENTRON

Circuit protection devices with communication and metering function

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SENTRON

Circuit protection devices with communication and metering function

System Manual

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10/2023

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Legal information

Warning notice system

This manual contains notices you have to observe in order to ensure your personal safety, as well as to prevent damage to property. The notices referring to your personal safety are highlighted in the manual by a safety alert symbol, notices referring only to property damage have no safety alert symbol. These notices shown below are graded according to the degree of danger.

⚠ DANGER

indicates that death or severe personal injury **will** result if proper precautions are not taken.

⚠ WARNING

indicates that death or severe personal injury **may** result if proper precautions are not taken.

⚠ CAUTION

indicates that minor personal injury can result if proper precautions are not taken.

NOTICE

indicates that property damage can result if proper precautions are not taken.

If more than one degree of danger is present, the warning notice representing the highest degree of danger will be used. A notice warning of injury to persons with a safety alert symbol may also include a warning relating to property damage.

Qualified Personnel

The product/system described in this documentation may be operated only by **personnel qualified** for the specific task in accordance with the relevant documentation, in particular its warning notices and safety instructions. Qualified personnel are those who, based on their training and experience, are capable of identifying risks and avoiding potential hazards when working with these products/systems.

Proper use of Siemens products

Note the following:

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Siemens products may only be used for the applications described in the catalog and in the relevant technical documentation. If products and components from other manufacturers are used, these must be recommended or approved by Siemens. Proper transport, storage, installation, assembly, commissioning, operation and maintenance are required to ensure that the products operate safely and without any problems. The permissible ambient conditions must be complied with. The information in the relevant documentation must be observed.

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We have reviewed the contents of this publication to ensure consistency with the hardware and software described. Since variance cannot be precluded entirely, we cannot guarantee full consistency. However, the information in this publication is reviewed regularly and any necessary corrections are included in subsequent editions.

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Introduction

1

1.1 Reference documents

You can find more information in the following documents:

Title	Article number
Circuit protection devices with communication and measurement function Installation Instructions (https://support.industry.siemens.com/cs/ww/en/view/109791805)	L1V30827020A
SENTRON Powercenter 1000 Operating Instructions (https://support.industry.siemens.com/cs/ww/en/view/109793297)	L1V30610178A
5ST3 COM auxiliary switch and fault signal contact Operating Instructions (https://support.industry.siemens.com/cs/ww/en/view/109793296)	3338214103
5SL6 COM miniature circuit breaker Operating Instructions (https://support.industry.siemens.com/cs/ww/en/view/109793301)	3355484128
RCM&EM COM miniature circuit breaker Operating Instructions (https://support.industry.siemens.com/cs/de/de/view/109814723)	3355484120
5SV6 COM arc fault detection device Operating Instructions (https://support.industry.siemens.com/cs/ww/en/view/109793300)	3355484127
3NA COM electronic module Operating Instructions (https://support.industry.siemens.com/cs/ww/en/view/109793298)	2568024102
SENTRON Powercenter 3000 Equipment Manual (https://support.industry.siemens.com/cs/ww/en/view/109763838)	L1V30579222004
3RV2 COM wireless auxiliary and signaling switch for 3RV2 motor starter protectors Operating Instructions (https://support.industry.siemens.com/cs/de/de/view/109817772)	A5E51924610001A

1.2 Technical Support

You can find further support on the Internet at:

TechnicalSupport (https://www.siemens.com/support-request)

1.3 Advanced training courses

Find out about regional training courses on offer via the following link.

Training for Industry (https://www.siemens.com/sitrain-lowvoltage)

Here you can choose from:

- Web-based training courses (online, informative, free)
- Classroom training courses (course attendance, comprehensive, subject to fee)

If the correct training course is not shown, you can also get information from your local sales representative.

Safety instructions

2

2.1 Cybersecurity information

Siemens provides products and solutions with industrial cybersecurity functions that support the secure operation of plants, systems, machines and networks.

In order to protect plants, systems, machines and networks against cyber threats, it is necessary to implement – and continuously maintain – a holistic, state-of-the-art industrial cybersecurity concept. Siemens' products and solutions constitute one element of such a concept.

Customers are responsible for preventing unauthorized access to their plants, systems, machines and networks. Such systems, machines and components should only be connected to an enterprise network or the internet if and to the extent such a connection is necessary and only when appropriate security measures (e.g. firewalls and/or network segmentation) are in place.

For additional information on industrial cybersecurity measures that may be implemented, please visit (https://www.siemens.com/industrialsecurity).

Siemens' products and solutions undergo continuous development to make them more secure. Siemens strongly recommends that product updates are applied as soon as they are available and that the latest product versions are used. Use of product versions that are no longer supported, and failure to apply the latest updates may increase customer's exposure to cyber threats.

To stay informed about product updates, subscribe to the Siemens Industrial Cybersecurity RSS Feed at (https://www.siemens.com/cert):

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Keyword: Open Source Request (please specify Product name and version, if applicable)

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2.3 User policy for mobile devices

Tips for customers using mobile devices

A wide range of applications can be installed on mobile devices. A certain cyber security risk is associated with the use of these devices. The following information will allow the user to use the apps with the highest possible degree of cyber security.

What can go wrong?

The use of mobile devices increases the risk of data loss and data theft in different ways. Typical situations include:

- Loss or theft of devices, e.g. in the hotel room, on the airplane, in a rental car or taxi, in the conference room, in the lobby or during breaks.
- Leaving the mobile device in unsuitable hands. If children are allowed to play with the device, they can inadvertently perform operations or change configuration settings.
- Reading information off the screen or eavesdropping on conversations. When you are in public, you should be conscious of the fact that other people can be watching you or listening to you, thereby obtaining information which could damage your company, your customers, your business partners or colleagues.

How can I stay cyber safe?

- Use a virtualization software to separate private and business affairs in your mobile device.
- Activate security mechanisms, such as the PIN for your tablet, smartphone or mobile phone and the password for your laptop.
- Use passwords that cannot easily be guessed to protect your mobile device.
- When entering passwords, make sure that nobody is watching.
- Keep the operating system and the installed applications up to date.
- Mobile devices must be protected against malware and other threats. For this reason, no unnecessary apps and no apps from unknown origins should be installed. A malware scanner should also be installed if possible.
- Lock your mobile device as soon as you have finished using it.
- Every time you install an app, make sure to check which permissions the app requires. For example, a flashlight function does not require access to the internet or your contacts. Never use unprotected internet access, such as WLAN hotspots in airport lounges, hotels, conference centers, restaurants or cafés.
- It is imperative that you use a VPN solution every time you set up a connection to the World Wide Web.
- If possible, set up device tracking for your mobile device so that it can be found, or even locked, if it gets lost. This function varies from one mobile device to the other.
- Do you know your service provider's emergency number? You can ask your provider to lock your device or perform a remote wipe in the event of an emergency.

- Deactivate all unnecessary functions. If you do not need GPS tracking, for example, you can deactivate this function.
- Back up your data regularly. Save your backup to another safe place.
- Employees must be conscious of their responsibilities when it comes to cyber security. Cyber security training courses are advisable for this reason.

2.4 Five safety rules for working in or on electrical systems

A set of rules, which are summarized in DIN VDE 0105 as the "five safety rules", are defined for working in or on electrical systems as a preventative measure against electrical accidents:

1. Isolate
2. Secure against re-connection
3. Verify absence of operating voltage
4. Ground and short-circuit
5. Provide protection against adjacent live parts

These five safety rules must be applied in the above order prior to starting work on an electrical system. After completing the work, proceed in the reverse order.

It is assumed that every electrically skilled person is familiar with these rules.

Explanations

1. The isolating distances between live and de-energized parts of the system must vary according to the operating voltage that is applied.
   "Isolate" refers to the all-pole disconnection of live parts.
   All-pole disconnection can be achieved, e.g. by.:
   - Switching off the miniature circuit breaker
   - Switching off the motor circuit breaker
   - Unscrewing fusible links
   - Removing LV HRC fuses
2. The feeder must be locked against inadvertent reconnection to ensure that it remains isolated for the duration of the work. This can be achieved, for instance, by locking the motor and system circuit breakers in the OFF position or by unscrewing the fuses and using lockable elements to prevent them from being reinserted.
3. The de-energized state of the equipment should be verified using suitable test equipment, e.g. a two-pole voltmeter. Single-pole test pins are not suitable for this purpose. The absence of power must be established for all poles, phase to phase, and phase to N/PE.
4. Grounding and short-circuiting are only mandatory if the system has a nominal voltage greater than 1 kV. In this case, the system should always be grounded first and then connected to the live parts to be short-circuited.
5. These parts should be covered, or barriers erected around them, to avoid accidental contact during the work with adjacent parts that are still live.

Circuit protection devices with communication and metering function

3

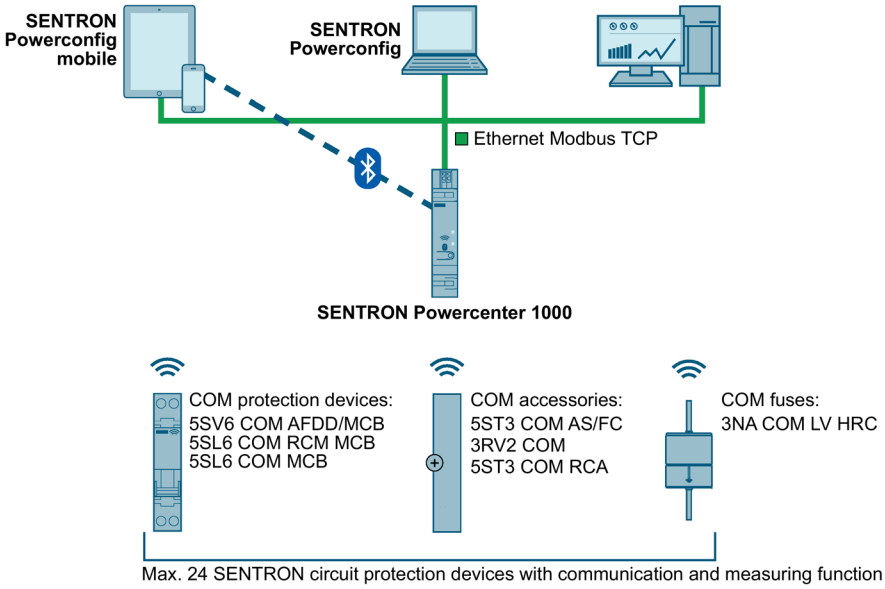
Circuit protection devices with communication and measurement functions are an integral part of SENTRON digitalization solutions.

System availability is increased by the early response to warning messages. The protection functions still offer secure, reliable protection in the final circuit. As a result, the communication and measurement function makes it easier to find the causes during malfunctions due to reporting the cause of the tripping operation, which in turn allows conclusions to be drawn about malfunctions of operating equipment. Another advantage is provided by the integrated operating hour and trip counters, which contribute to improved planning of maintenance work. In addition, circuit protection devices with measurement and communication function capture electrical parameters such as energy, active power, current, voltage, line frequency and temperature. This helps increase transparency and energy consumption in the final circuits can be broken down. By setting alarm threshold values and additionally measuring residual currents in several frequency ranges, it is possible to pinpoint the causes of errors.

Due to their compact construction, the circuit protection devices are ideal for the retrofit market or even new construction. The system can be simply put into operation via the SENTRON Powerconfig PC software or SENTRON Powerconfig mobile for mobile devices.

You can find more information in the Installation Manual - SENTRON circuit protection devices with communication and measurement function (https://support.industry.siemens.com/cs/de/en/view/109791805).

The SENTRON Powercenter 1000 data transceiver is the core of the system of communication-capable circuit protection devices. This gathers measured values of the paired circuit protection devices and transfers them to the higher-level systems. The measured values from up to 24 communication-capable SENTRON devices are wirelessly transferred to a SENTRON Powercenter 1000, which stores selected data for up to 30 days. Higher-level systems can access the data via the interfaces of the data transceiver. Either on site via Bluetooth or via Ethernet in the local network. In this case, a Modbus TCP protocol is used, which can easily be integrated by other systems.



The system of circuit protection devices with communication and measurement function increases system availability due to increased transparency through to the final circuit through wireless transmission and storage of measured values.

Because the devices communicate via radio, a radio approval is required for each country in which they are operated.

Existing country radio approvals
(https://support.industry.siemens.com/cs/de/en/view/109801197)

Further countries on request.

3.1 SENTRON Powercenter 1000



The SENTRON Powercenter 1000 data transceivers gather data from communication and measurement-capable 5SL6 COM miniature circuit breakers, 5SV6 COM arc fault detection devices, 5ST3 COM auxiliary switch and fault signal contacts, communication-capable 3NA COM fuses, and 3RV2 COM wireless auxiliary and signaling switches for 3RV2 motor starter protectors.

They communicate wirelessly with up to 24 terminal devices within a switchgear bay or a distribution board. The recorded data can be accessed via Bluetooth using a mobile terminal device on site or be forwarded to higher-level systems by means of Modbus TCP. Selected measured values are stored in the SENTRON Powercenter 1000 for up to 30 days and can be displayed via the Ethernet interface.

Energy flows can be visualized and optimized with the SENTRON Powermanager energy monitoring system. Via the SENTRON Powercenter 3000 IoT data platform, the recorded data can be viewed directly in a web interface or transferred to Cloud applications and evaluated.

The screwless plug-in terminals allow the 24 V DC (SELV) power supply to be looped through to other devices in the distribution board, and with its 1TE size, the space-saving SENTRON Powercenter 1000 enables easy mounting on the DIN rail.

3.2 5ST3 COM auxiliary switch and fault signal contact



5ST3 COM auxiliary switches and fault signal contacts are ideal for retrofitting devices that are not available as communication-capable versions. They are mounted on to the electromechanical master unit and extended by communication and metering functions for temperature, switch position and the number of shutdowns.

In order to work in the most space-saving manner, the 5ST3 COM auxiliary switch and fault signal contact is a measurement and communication-capable add-on module, which only uses 0.5 TE. This can be mounted on to the series 5SY, 5SP4 and 5SL miniature circuit breakers and the 5SV residual current operated circuit breaker and 5SV1 and 5SU1 RCBO switches.

The supply voltage of 24 V DC (SELV) can be bridged to other devices via the screwless plug-in terminals.

3.3 5SL6 COM miniature circuit breaker



5SL6..MC (EM)



5SL6..MF (EM+RCM)

5SL6 COM miniature circuit breakers do not just protect final circuits like conventional miniature circuit breakers during overload and short-circuits, they also gather information on the status and the faults in the electric circuit.

They wirelessly communicate the measured values such as current, voltage, temperature, line frequency, power, energy, operating hours, tripping operations and switching cycles to the higher-level SENTRON Powercenter 1000 data transceiver. Monitoring of the electric circuit enables predictive error detection by reporting pre-warnings if, for example, the set limit value of the load current is exceeded.

In addition to the power measurement function and the condition monitoring function, the 5SL6 COM version with RCM function (5SL6..-.MF) enables the measurement and monitoring of residual currents in accordance with IEC 62020-1 (residual current monitoring, RCM for short). This involves simultaneously monitoring residual currents in several frequency ranges, and enables the differentiation of error statuses in the system.

The device only requires the external and N-conductor connection (230 V AC) and no additional power supply (24 V DC) for the communication and measurement function. With a width of 1 TE, the 5SL6 COM miniature circuit breaker is as large as conventional miniature circuit breakers without communications capability. The 5SL6 COM is designed for nominal currents of 2 ... 32 A and available in the tripping characteristics B and C.

3.4 5SV6 COM arc fault detection device



With the communication-capable 5SV6 COM arc fault detection device, which has the integrated function of a miniature circuit breaker, the affected final circuit can be protected with regard to overload, short-circuit and fault arcs.

The measured values and statuses are wirelessly transmitted via the integrated communication and metering function. Thus, the final circuit can be monitored even further, by e.g. setting a limit value that sends a warning when a certain current or voltage value is exceeded. This predictive error detection ensures increased system availability. In addition to the electrical measured values (current, voltage, frequency, power, energy) of the circuit, values such as tripping type, temperature, switching cycles or operating cycles are also communicated.

Even here, the 1+N pole design of the switch offers the complete scope of protection, without additional connections for the communication function. As a result of the compact size of 1 TE, the 5SV6 COM arc fault detection devices offer a simple exchange of conventional arc fault detection devices or miniature circuit breakers. Devices with a compact structure (1+N) can be easily exchanged, for the single-pole devices the neutral conductor must also be connected. The 5SV6 COM is available for nominal currents of 6 A to 32 A and in tripping characteristics B and C.

3.5 3NA COM fuse



The 3NA COM fuses with communication and measuring function not only protect the circuit, but also enable early error detection by measuring the current and temperature.

Monitoring of the circuit makes it possible to combat the cause even before the fuse trips by being able to generate warning messages when the set over-current limit or even the temperature limit has been exceeded. This contributes towards the avoidance of power outages, or a reduction of power failures and thus, increases the availability.

The 3NA COM fuse is available in size NH2 and operating class gG or gFF. It can be easily retrofitted in the existing fuse-disconnector due to its standard-compliant dimensions and auxiliary power. The whole fuse module is composed of an electronic module and a fuse link with nominal currents between 80 ... 315 A. The fuse link can be easily replaced once the fuse has been tripped. It is not imperative to renew the electronic module, however, the functions should be checked after re-commissioning. A test must be performed with a suitable reference measuring instrument and the exchanged components must be documented in an appropriate manner (e.g. in the form of a test label and electronic recording).

The electronic module provides the communication and measuring function with an integrated instrument and supply transformer, where an additional connection for the supply voltage is not required. A min. current flow of 5 A is required in order to ensure the communication and measuring function (min. 10 A for a firmware update). Together with the fuse link, the electronic module forms the complete system of the 3NA COM fuse.

3.6 3RV2 COM wireless auxiliary and signaling switch for 3RV2 motor starter protectors



Wireless auxiliary switches and signaling switches are ideal for retrofitting motor starter protectors that are not available as communication-capable versions.

The devices (3RV2921-5M) can be mounted as an accessory on the electromechanical 3RV2 motor starter protector (size S00 - S3); the width is 18 mm.
The measured values and statuses of the basic unit are wirelessly transmitted via the integrated communication and measuring function. In this way, the temperature and the number of shutdowns are transmitted in addition to the switching statuses of the motor starter protector. Apart from the ON/OFF indication, a tripping operation is treated differently depending on whether it was caused by an overload or a short-circuit.

The supply voltage of 24 V DC (SELV) can be bridged to other devices via the screwless plug-in terminals. This device is supported in the Powercenter 1000 from firmware version 3.0.

3.7 5ST3 COM remote control auxiliary (RCA)

Note

Remote operating mechanism

(RCA)

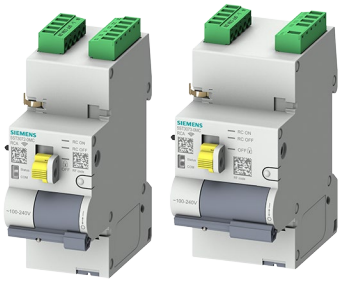


Figure 3-1 **5ST3 COM RCA Standard and 5ST3 COM RCA RCD test**

5ST3 COM remote control auxiliaries not only enable monitoring of the system/the attached master units, but they also allow remote switching. These devices are supported in the Powercenter 1000 from firmware version 3.0.

More functions are provided in addition to the standard measuring functions (device temperature, operating hours and the number of shutdowns), for example the exact breakdown of the switch position whether manual, automatic or remote switching or the switching command, an automatic reclosing function (ARD) after a tripping operation and its parameterization.

The RCD test version of the remote control auxiliary (5S3073-0MC) also allows one-off or cyclical FI (RCD) or insulation resistance (IR) tests to be carried out in accordance with IEC 63024, depending on the configured mount-on device. The time and date and the test results are stored and can be exported later.

Instead of the test function, the standard version (5S3072-0MC) provides wired connections for auxiliary and fault signal contacts.

Both versions are supplied with 100-240 V AC line voltage.

Installation and connection

4

Note

Only qualified personnel are permitted to install, commission or service the devices.

- Wear the prescribed protective clothing. Observe the general equipment regulations and safety regulations for working with high-voltage installations (e.g. DIN VDE, NFPA 70E as well as national or international regulations).
- The limits given in the technical specifications must not be exceeded even during commissioning or testing of the device.
- Before connecting the device, make sure that the line voltage matches the specifications on the rating plate.
- Before you start up the device, check that all the connections have been made correctly.
- Before power is applied to the device for the first time, it must have been located in the operating area for at least two hours in order to reach temperature balance and avoid humidity and condensation.
- Condensation on the device is not permissible during operation.

Note

During installation and connection, please note the five safety rules for work in or on electrical systems (Page 12).

Note

Prior to purchase or installation, it is necessary to check whether the equipment has the appropriate radio approval for the target country in which it will be operated.

The country radio approvals (https://support.industry.siemens.com/cs/de/de/view/109801197/en) are located in the SIOS portal (https://support.industry.siemens.com/cs/de/en/ps). Simply search for the product or the order number and select the Certificates option as the entry type.

4.1 Preparing installation

4.1.1 Checking delivery

Procedure

1. When you receive the delivery, check the packaging for visible damage in transit.
2. If you discover damage in transit, lodge a complaint with the carrier responsible. Have the carrier confirm the damage in transit immediately.
3. Unpack the device at its destination.
4. Keep the original packaging for reshipping the device. See information on damage.

NOTICE
Damage to the device during transportation and storage If a device is transported or stored without packaging, shocks, vibrations, pressure and humidity act on the unprotected device. Damaged packaging is an indication that the environmental conditions have already had a large impact on the device. The device may be damaged. Do not dispose of the original packaging. Pack the device for transportation and storage.

5. Check that the package contents are complete and undamaged.
6. If the package contents are incomplete or damaged, or not exactly what was ordered, inform the responsible delivery service immediately.

NOTICE
Damaged device • Ensure that the damaged device is not installed and commissioned. • Label the damaged device and keep it locked away. • Send the device for repair immediately.

NOTICE
Damage due to condensation If the device has been exposed to low temperatures or extreme temperature fluctuations during transportation, e.g. in cold weather, moisture may have formed as condensation on or inside the device. Moisture causes short-circuits in electrical circuits and damages the device. To avoid damage, proceed as follows: • Store the device in a dry place. • Equalize the temperature of the device with room temperature before starting it up. • Do not expose the device to the direct radiated heat of a heater. • In the event of condensation, only switch on the device when it has completely dried or after a delay of approx. 12 hours.

7. Also keep the supplied documentation in a safe place. It forms part of the device. When you commission the device for the first time, you will require the documentation.
8. Note the identification data of the device.

4.1.2 Identification of circuit protection devices

These circuit protection devices are usually used for final circuits and to secure individual phases and thus, a large number is used in a system. This is why quick recording and reliable archiving of the identification data is all the more important.

Identification of a circuit protection device is composed of two parts. The

1. Manufacturer identification
2. User identification

The electronic type plate is generated from both identification components.

4.1.2.1 Manufacturer identification

Manufacturer identification provides information on device-specific data such as:

- Device type
- Order number
- Serial number

These data are printed on the front or side of the device. An unlabeled Data Matrix Code is located on the device, this can be scanned using the Siemens Industry Online Support (SIOS) App (https://new.siemens.com/global/en/products/software/mobile-apps/industry-online-support.html). This allows technical or graphical data to be displayed directly.



① SIOS code for product information
② RF code for commissioning the communication function

The Data Matrix Code (DMC), which is marked with "RF Code", contains the following communication data in encrypted form:

- Device type
- MAC address
- Installation code

This communication information is necessary for commissioning so that the devices can connect to the SENTRON Powercenter 1000.

The RF code of the SENTRON Powercenter 1000 contains only the Bluetooth PIN code that is required for connecting to the mobile device.

The aforementioned information is also printed on the side of the device in a legible manner.

The manufacturer information can also be read out from the device after commissioning via the communication function and displayed, for example, via SENTRON Powerconfig mobile.

See also

Installation Manual - SENTRON circuit protection devices with communication and metering function (https://support.industry.siemens.com/cs/ww/en/view/109791805)

4.1.2.2 User identification

User information is used to identify or locate and distinguish the devices from each other. Usually this consists of:

- Plant identifier
- Location identifier
- Installation date

This information is noted by the customer, usually as an adhesive strip, on the device and identically in the system plan.

The communication-capable circuit protection devices offer the additional option of storing the aforementioned information electronically. This is also necessary to differentiate between the individual devices after commissioning.

During the installation of the SENTRON and SIRIUS circuit protection devices with a communication and measuring function, it should be ensured that the DMC codes and possibly the inscription on the devices are visible during commissioning and are accessible with a mobile device for commissioning. If this is not possible, the devices can be recorded prior to assembly using SENTRON Powerconfig mobile, whereby the necessary DMC codes are scanned prior to installation and added offline. Then the devices can be coupled with the SENTRON Powercenter 1000 when assembly and connection are complete. Further information can be found in the Installation Manual (https://support.industry.siemens.com/cs/ww/en/view/109791805).

Note

This is particularly necessary for 3NA COM fuses because after installation, these are no longer accessible without shutting down the main circuit.

In addition, the communication function offers a possibility of allowing each communication-capable device to flash with an LED for a certain time, which can facilitate the locating of a device during maintenance.

Note

If the Data Matrix Code is no longer legible because it has been covered or scratched during cleaning, this does not constitute a reason for a complaint.

4.1.3 Installation conditions

The installation devices can be installed on to any DIN rail in any installation position. The direction of incoming supply is arbitrary and can be set as a parameter for the miniature circuit breakers and arc fault detection devices.

In the SENTRON Powercenter 1000 data transceiver, it should be ensured that there is enough space for the Ethernet plug, pullout and the cable bending radius.

Furthermore, it is necessary to make sure that the data transceiver is mounted with a small distance to metallic surfaces in all directions (apart from the DIN rail) so that the efficiency of the integrated antenna is not impacted too much.

For the 3NA COM fuse, the usual installation position is vertical, as for the other LV HRC fuse links. The current load does not need to be reduced under normal ambient conditions. However, there are devices and applications, where the LV HRC fuse links are arranged horizontally. As per usual, the instructions of the device manufacturer, especially the maximum current carrying capacity, must be observed in this case.

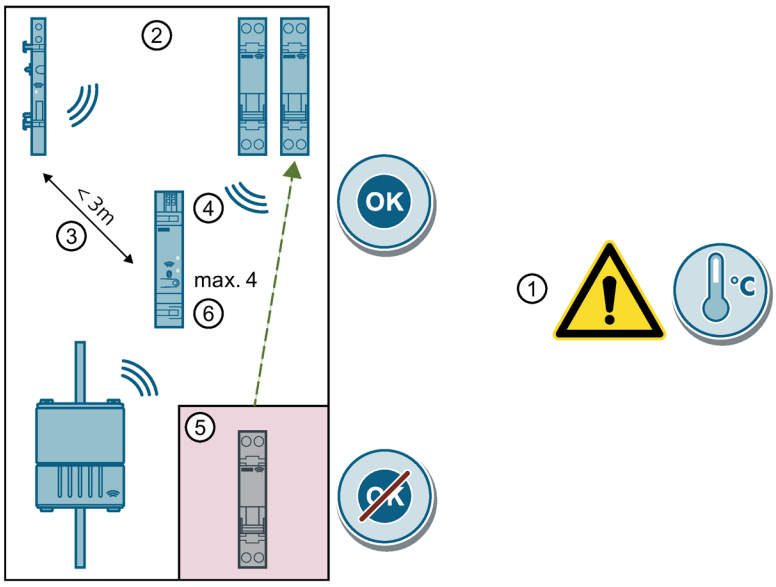
Note

Overhead installation

The upside down installation of the 3NA COM LV HRC fuse links, i.e. with the electronic module above the fuse unit is not permitted. It is possible to overheat the electronic module.

The ventilation slots must not be covered or closed. Follow the ESD guidelines (Page 137) and the mounting instructions of the operating instructions of the electronic module.

The spatial arrangement of the communication-capable circuit protection devices one below the other corresponds to the following recommendations:



(1) Prior to setting up the distribution board, pay attention to devices with a high temperature development. As usual, these are positioned away from smaller, more sensitive devices.

(2) The SENTRON Powercenter 1000 and the associated terminal devices should be installed together in a control cabinet or panel so that wireless communication is less disturbed by other devices or obstacles.

(3) The maximum recommended distance between the terminal devices and the SENTRON Powercenter 1000 should not exceed 3 m. The radio transmit power can be changed as a parameter, which can either decrease or increase the permitted distance between the devices. The standard setting of the transmit power of 0 dBm allows a distance of over 10 m without any other disturbances.

Important: The transmit power of each device in the system must be set the same so that the distance to another system can be reduced.

(4) If possible, the SENTRON Powercenter 1000 should be placed equally far away from all terminal devices. Therefore, a centrally located place is recommended.

(5) Metallic partitions or other radio devices that use the same frequency should not be installed between the individual terminal devices and the SENTRON Powercenter 1000 data transceiver so that radio transmission can be permanently ensured.

(6) If more than 24 communication-capable circuit protection devices are installed, further SENTRON Powercenter 1000 devices will be required. Up to 4 data transceivers can be used simultaneously. An equal distribution of the terminal devices to the different SENTRON Powercenter 1000s is recommended.

If several SENTRON Powercenter 1000 devices share the same radio channel, interferences may occur between the devices. To ensure that this does not happen, the expansion of the radio range can be reduced by lowering the transmit power of all devices in the system. The minimum transmit power amounts to -18 dBm. This allows the devices to be positioned closer together (< 10 m for metallic housings, approx. 50 m in the free field without obstacles). Further information can be found in the Chapter Commissioning several SENTRON Powercenter 1000 devices.

In safety-critical systems it is important to note that the wireless protocols can be disrupted from the outside, so suitable countermeasures are recommended such as adequate shielding.

4.1.4 Permitted ambient conditions

The SENTRON Powercenter 1000 and the communication-capable circuit protection devices are designed for contamination class 2. In the case of an application with an additional risk of contamination, heavy dust accumulation or aggressive atmosphere, the device must be installed in a suitable housing or installation distributor.

The 3NA COM fuse however, has a higher contamination class 3. Therefore, the environment must be examined more closely and the device must be protected using suitable measures (e.g. a separate housing).

The ambient temperature of the devices is specified at -25°C to +60°C or from -10°C to +55°C for the 3NA COM fuse.

The ambient temperature during storage or transport is around -40°C to +75°C or +85°C depending on the type of device. Another range of -10°C to +70°C applies to the 3NA COM fuse.

The maximum relative humidity amounts to 93% at 40°C without condensation. For the 3NA COM fuse, the relative humidity at 25°C without condensation during operation is maximum 95%.

If these requirements cannot be guaranteed, appropriate countermeasures must be taken.







Note

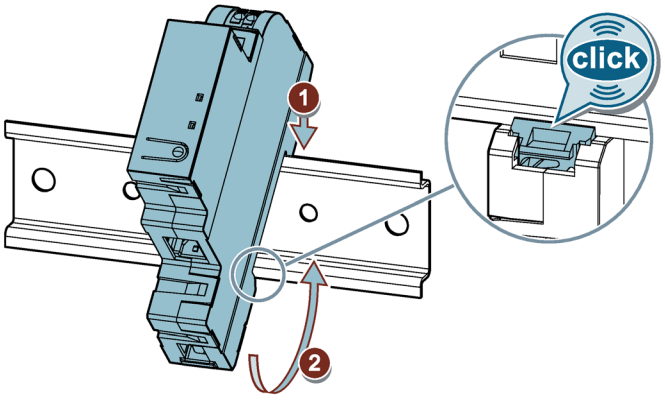
In case of increased ambient temperatures, a correction factor must be applied for the rated current. This applies to the 3NA COM fuse at a temperature of > 40°C and the 5SV6 COM arc fault detection device or the 5SL6 COM miniature circuit breaker at temperatures > 30°C according to the respective configuration manual of the product family. Information on this topic can be found in the SIOS portal (https://support.industry.siemens.com/cs/de/en/). Search for the corresponding product.

In operation, the 3RV2 COM wireless auxiliary and signaling switch for motor starter protectors has a minimum ambient temperature of -20°C and a maximum ambient temperature of +60°C.

4.2 Device installation

4.2.1 Install SENTRON Powercenter 1000

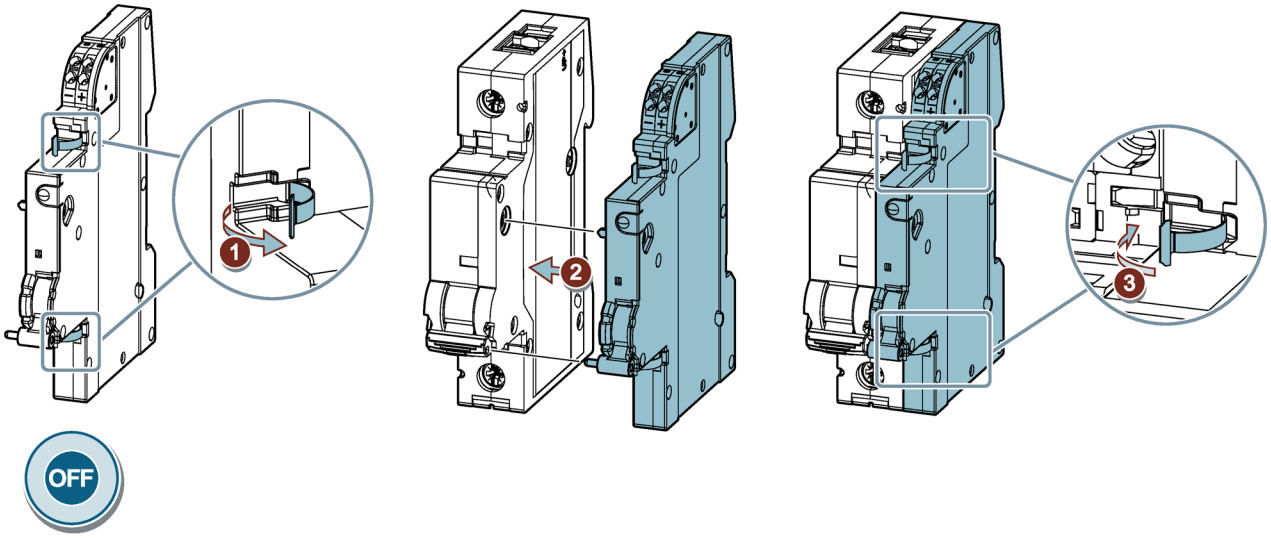
To install, simply snap the SENTRON Powercenter 1000 data transceiver onto the DIN rail and snap it into place.



4.2.2 Install 5ST3 COM auxiliary switch and fault signal contact

Prior to wiring, first mount the auxiliary switch/fault signal contact to the main unit or fasten together on to the DIN rail. This helps reduce the risk of injury.

The 5ST3 COM auxiliary switch/fault signal contact must be connected or disconnected from the supply voltage lines when completely assembled.



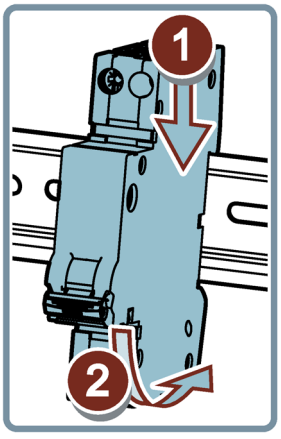
The 5ST3 COM auxiliary switch / fault signal contact can be mounted to the following main units:

Designation	Article numbers
Device protection switches or miniature circuit breaker	5SL, 5SY, 5SP
Residual current circuit breakers or RCBO	5SU1, 5SV
Other switching devices	5TL, 5ST30

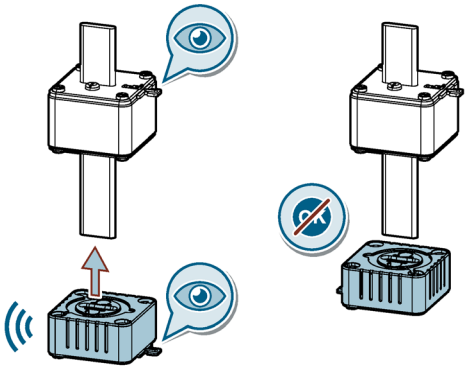
4.2.3 The 5SL6 COM miniature circuit breaker and 5SV6 COM arc fault detection device

Both devices are simply snapped on to the DIN rail.

Example 5SV6 COM arc fault detection device:



4.2.4 Install 3NA COM fuse



The 3NA fuse link can only be installed in an LV HRC fuse link holder together with the 3NX electronic module pushed on.

The LV HRC 3NA COM fuse link and the appropriate electronic module can be ordered as a unit, however they are delivered in separate packages and still have to be mounted together.

Installation takes place in all common LV HRC fuse bases or fuse switchgears of the appropriate size similar to the installation of a normal LV HRC fuse link according to IEC 60269-2 without the electronic module.

The 3NA COM electronic module does not require a separate power connection. It supplies itself from the primary current via the current transformer according to the Energy Harvesting principle, however, it requires a minimum current of 5 A for this purpose.

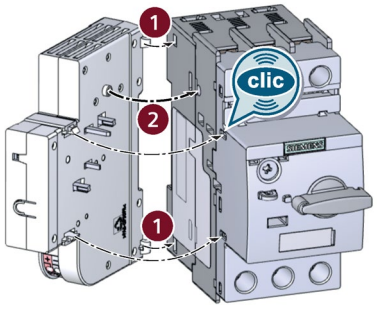
Note

The gFF operating class is only approved for the Netherlands. The relevant installation specifications apply.

In the event that the fuse link has switched off following a short-circuit or an overload, the fuse link must be replaced. Remove the LV HRV 3NA COM fuse link and the electronic module out of the device, pull the electronic module down off the contact blade and re-insert the electronic module on to the new 3NA COM fuse link in reverse order. The LV HRV 3NA COM fuse link can also be ordered separately.

The electronic module can be reused under certain circumstances, but this must be checked after recommissioning.

4.2.5 Install 3RV2 COM wireless auxiliary and signaling switch for 3RV2 motor starter protectors



① Attach the wireless auxiliary and signaling switch to the rear of the motor starter protector.

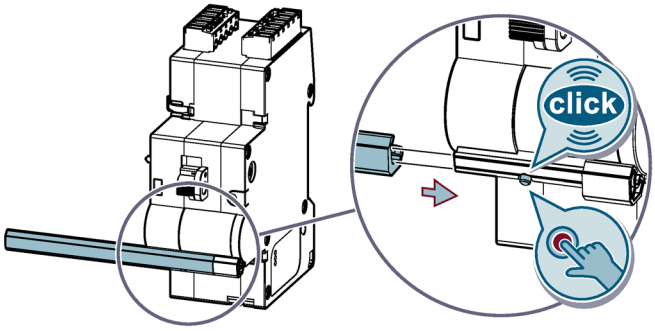
② Press the wireless auxiliary and signaling switch to the motor starter protector until you hear it engage (the motor starter protector must be switched off).

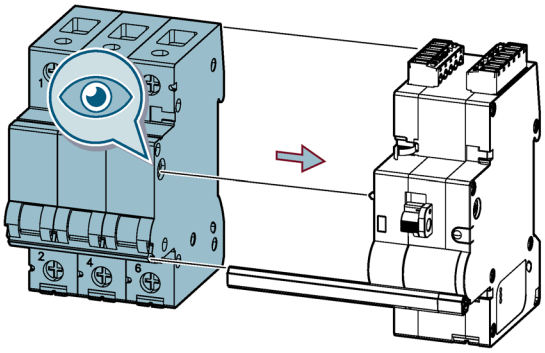
4.2.6 Install 5ST3 COM remote control auxiliary (RCA)

Mount the 5ST3 COM remote control auxiliary on the master unit. Depending on the master unit, a suitable handle adapter must be used.

	Suitable for	
5ST3820-1	5SY4/5/6/7/8 - 1/2 pole, 5SP4 - 1 pole	5SY60...CC
5ST3820-2	5SY4/5/6/7/8 - 3/4 pole, 5SP4 - 2/3/4 pole	
5ST3820-3	5SM2	
5ST3820-5	5SU1 (max. 3 width units)	

	Suitable for	
5ST3820-6	5SL6/4 - 1/2 pole, 5SY...CC, 5TL1 - 1/2 pole	5SV1/3/4/9, 5SL60, 5SV60
5ST3820-7	5SL6/4 - 3/4 pole, 5SY...CC, 5TL1 - 3/4 pole	





Mount the combination on the DIN rail.

Use the terminals included in the scope of supply to connect the remote control auxiliary. This step can also be carried out in the assembled status.

You will find more detailed information in the operating instructions (https://support.industry.siemens.com/cs/ww/en/view/109823192).

4.3 Connecting the devices

The communication-capable 5SL6 COM miniature circuit breaker and 5SV6 COM arc fault detection device have an integrated power supply unit for the electronics. Therefore, the phase and N-conductor (230 V AC) must be connected.

The SENTRON Powercenter 1000, the 5ST3 COM auxiliary/fault signal contact and the 3RV2 COM wireless auxiliary and signaling switch are supplied using an extra-low voltage of 24 V DC (SELV). The screwless connection terminals of the three devices enable a loop-through (daisy chain) of the supply voltage.

The 3NA COM fuses are supplied via Energy Harvesting from the main circuit and therefore do not require any additional wiring to the connections made as standard.

⚠ WARNING

Danger due to fire or electric shock.
May cause death, serious personal injury, or equipment damage.

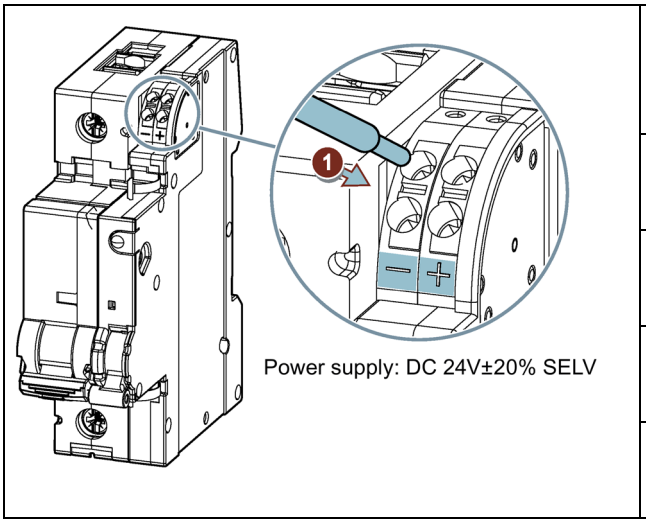
Only use leads that correspond to the local safety regulations.

4.3.1 Connection of SENTRON Powercenter 1000, 5ST3 COM auxiliary switch and fault signal contact, 3RV2 COM wireless auxiliary and signaling switch

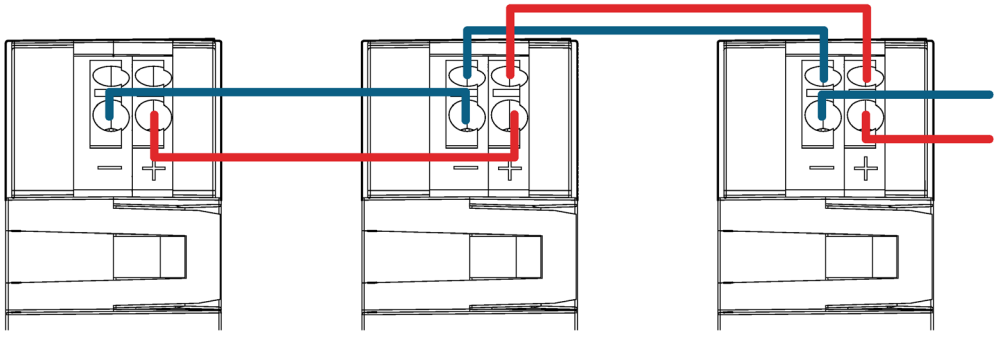
NOTICE
DIN rail mounting To avoid damage to property, mount the devices for connection firmly on to the DIN rail.

The screwless plug-in terminals are designed for the insertion of two conductors so that a loop-through is possible via the terminals to other devices in the control cabinet (daisy chaining).

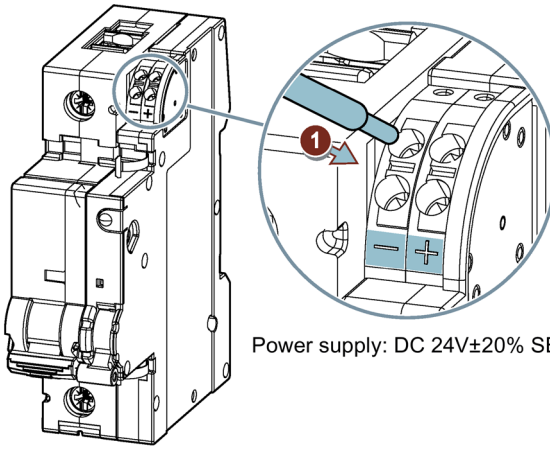
The cables may carry a maximum current of 5 A. The connecting cables can simply be plugged into the terminals without the need for any tools. The permitted conductor cross-sections are displayed below.



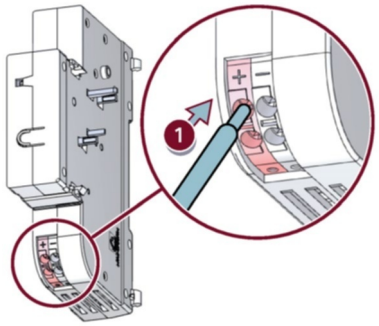
Cu		
	10 mm [0.39 in]	0.2 ... 1.5 mm² [AWG 24-16]
		0.2 ... 1.5 mm² [AWG 24-16]
		0.5 ... 0.5 mm² [AWG 20]
		0.5 ... 1.5 mm² [AWG 20-16]



To remove the cable, press on the slot of the terminal release with a 0.4 x 2.5 mm screwdriver, then the cables can be easily pulled out.

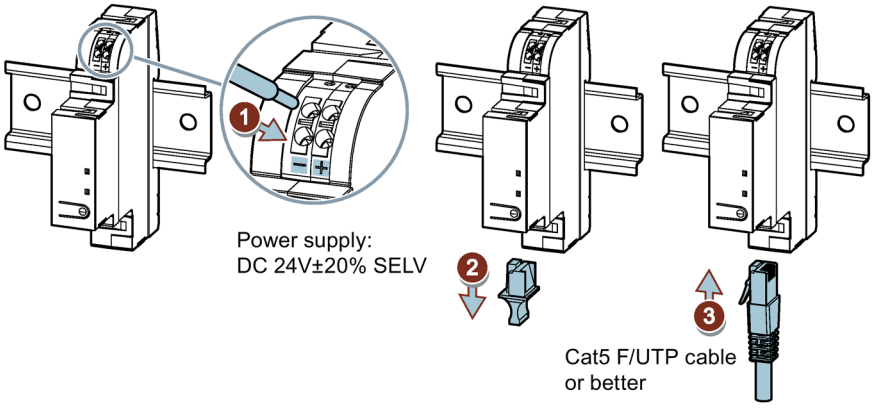


5ST3 COM



3RV2 COM

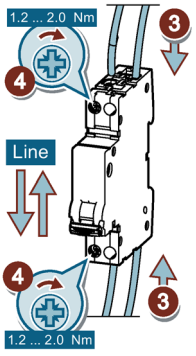
Powercenter 1000: For the data transceiver, the Ethernet cable (min. Cat5 F/UTP) must also be connected in addition to the supply voltage after the dummy plug has been removed. If only one Bluetooth connection is required, the dummy plug should remain in the device to protect the device from external influences.



4.3.2 Connecting the 5SL6 COM miniature circuit breaker and 5SV6 COM arc fault detection device

If the devices are mounted on the DIN rail, the terminals and the leads are tightened using a screwdriver.

This must be repeated on the incoming and outgoing side for the line conductor and the neutral conductor. The permitted conductor cross-sections are displayed below.



	⌀
	0.75 ... 16 mm²
	0.75 ... 10 mm²

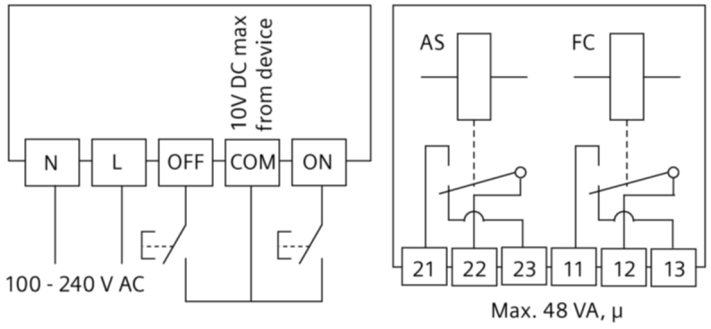
Note

To ensure the protection function of the 5SV6 COM arc fault detection device, it is essential for the neutral conductor to be wired on the incoming and outgoing sides. This also applies to the 5SL6 COM miniature circuit breaker with RCM function, so that the residual current is measured. For the 5SL6 COM (without RCM function), the neutral conductor can be dispensed with on the outgoing side.

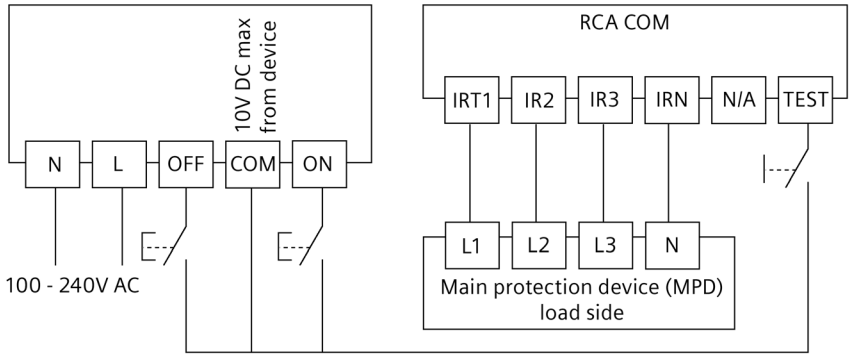
4.3.3 Connection of 5ST3 COM remote control auxiliary (RCA)

Both versions of the 5ST3 COM remote control auxiliary are connected using two plug-in terminal blocks.

5ST3072-0MC:



5ST3073-0MC:



4.4 Interfaces of the SENTRON Powercenter 1000

4.4.1 Ethernet interface

After mounting, the protection of the Ethernet plug must be removed in order to insert an Ethernet cable (Cat5 F/UTP or better). This connection enables IP communication of the SENTRON Powercenter 1000. If the data transceiver is connected to an Ethernet switch or a router, the data or parameters can be viewed or set in the whole local network via LAN or WLAN. Only the IP parameters, such as a static IP address, can be changed in the app via Bluetooth.

A VPN connection or another gateway can be used for advanced access beyond the local network. For further information, see Chapter Application Examples (Page 101).

The Modbus TCP protocol can be used via the Ethernet interface. For further information, see Chapter Modbus TCP Connection (Page 77).

4.4.2 Bluetooth interface

The Bluetooth interface enables local access to the data of the SENTRON Powercenter 1000 on site. This is the Bluetooth Low Energy 4.2 standard. However, data transmission performance can be significantly improved with Bluetooth Low Energy 5.1.

In order to establish the connection, the data transceiver must first be set to Bluetooth mode. This is either done after restarting the device or after a short button press (< 3 s). To do so, the mobile terminal device must be located near an active Bluetooth (approx. 5 m – 10 m). Only one active connection is supported. As of firmware version 1.1.0, the plant identifier, if entered, is displayed in addition to the device type. This makes it easier to distinguish between multiple devices.

In order to complete the connection and to be able to establish an encrypted connection between SENTRON Powercenter 1000 and the mobile terminal device, the 6-digit Bluetooth PIN of the data transceiver must be entered (encryption: AES CCM algorithm with 128 bits). This can either be scanned using the Data Matrix Code or manually typed in using the information printed on the side of the device. The PIN code can be changed for increased security after commissioning. The data transceiver needs to be reset in order to be able to reset the PIN to factory setting (button press ≥ 10 s). Similarly, the reception strength for Bluetooth can be decreased or increased so that remote access can be prevented or the range can be increased.

If the Bluetooth mode of the SENTRON Powercenter 1000 is not used for more than 180 seconds, it is switched off again. The LED will flash as long as the device is in the Bluetooth search mode. If the time without active connection has expired, the function is switched off or if a connection has been successfully established, the COM LED will stop flashing with 2 Hz.

If a mobile device is connected to the SENTRON Powercenter 1000 via Bluetooth, this connection can be disconnected by the mobile terminal device terminating the connection or by pressing the button again (< 3 s) on the SENTRON Powercenter 1000. The Bluetooth mode of the data transceiver is also switched if the PIN is entered incorrectly three times.

Note

Due to the performance, larger data packages such as Trends in SENTRON Powerconfig mobile are not displayed via the Bluetooth interface. Commissioning should only be carried out once in this manner.

The SENTRON Powercenter 1000 provides radio communication for the circuit protection devices and the Bluetooth communication for the mobile devices using the same radio module, which is why the performance of the Bluetooth connection is restricted. Therefore, the Modbus/TCP-connection via Ethernet is recommended for faster data transmission.

4.4.3 Radio interface to the devices

The circuit protection devices with communication and measuring function provide the SENTRON Powercenter 1000 with their data via radio. Data can be read from the devices and changed parameters can be sent to the devices.

Communication is possible via radio as soon as the devices are connected and supplied with power (minimum 5 A for the 3NA COM fuse).

A SENTRON Powercenter 1000 can communicate with up to 24 terminal devices. Each of these circuit protection devices must join the secure radio network of the Powercenter or, in other words, each terminal device must be paired with the SENTRON Powercenter 1000. The terminal devices do not communicate among one another, rather communication with the data transceiver is bidirectional/in a star configuration (no mesh-network).

In order to pair the devices, the RF code of the terminal devices must be scanned; this indicates the device type, MAC address and installation code of the device. Alternatively, this information can be entered manually. A description of the procedure can be found in the Installation Manual (https://support.industry.siemens.com/cs/ww/en/view/109791805).

Note

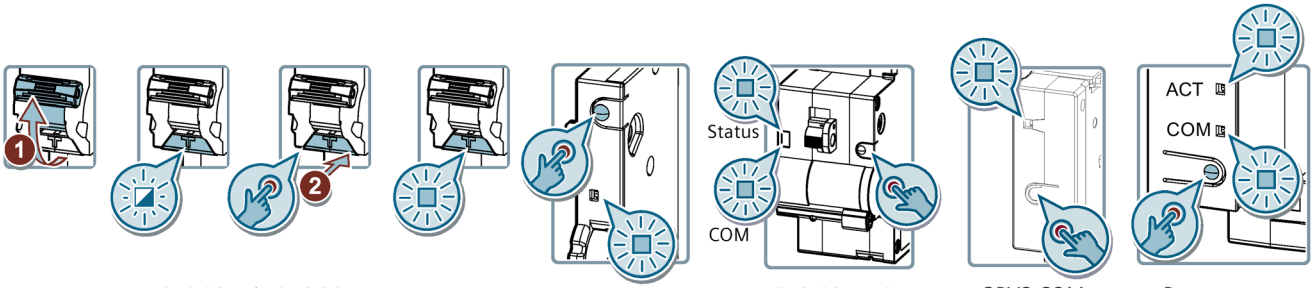
Once the terminal device has been paired with the data transceiver, the installation code is encrypted and replaced so that each communication connection is separately encrypted (encryption: AES CCM algorithm with 128 bits). Therefore it is necessary to reset the communication information in order to be able to carry out another pairing.

Radio transmission can be significantly influenced by other transmitting devices. It can result in losses of individual data packages. A recommendation on installation is stated in the Chapter Installation conditions (Page 28).

4.5 Operation of circuit protection devices

4.5.1 Standard operator controls

The circuit protection devices with measurement and communication function have different operating options in addition to the communication function.



5SL6 COM / 5SV6 COM | 5ST3 COM AS+FC | 5ST3 COM RCA | 3RV2 COM | Powercenter 1000

Only the 3NA COM fuse has neither LEDs for status display nor operating options on the device itself. The operation for this only works via the communications interface and a corresponding software, such as SENTRON Powerconfig mobile.

The following options are provided:

Description	Action	POC 1000	5ST3 / 3RV2 COM	5SL6 COM	5SV6 COM
Activating the Bluetooth mode (COM LED flashes with 2 Hz) or disconnecting an active Bluetooth connection	Button press < 3 s	✓	-	-	-
Confirming a tripping operation	Button press < 3 s	-	-	✓	✓
Confirming an RCM alarm (if alarm auto-reset is deactivated)	Button press < 3 s	-	-	✓[1)]	-
Executing a test button trigger/self-test	Button press < 3 s	-	-	-	✓
Executing an RCM function test	Button press < 3 s	-	-	✓[1)]	-
Resetting the radio or Bluetooth communication information (LED flashes after the first 3 s with 2 Hz for an additional 7 seconds. Then fast at 10 Hz for at least 2 s, which signals the end of resetting)	Button press ≥ 10 s	✓	✓	✓	✓
Switching the device on/off	Lever in Open or Closed position	-	-	✓	✓

1) Only for RCM versions

The activation of the Bluetooth mode or disconnecting an active Bluetooth connection is described in the Chapter Bluetooth interface (Page 41).

If a 5SL6 COM miniature circuit breaker or a 5SV6 COM arc fault detection device has tripped (overload, short-circuit or arcing fault), the device is switched off and the lever is in the off position; communication is not possible in this status. In order to switch on the device again, the lever needs to be pulled up to the On position. Communication is then possible and the LEDs indicate that the device has tripped via its protection function by flashing red. In order

to acknowledge this flashing, the tripping operation has to be confirmed with a short button press.

Furthermore, in the case of the 5SV6 COM arc fault detection device a test tripping operation is carried out after a short button press (< 3 s). This triggers the internal self-test of the electronics and a tripping operation examines the tripping mechanism at the same time. This is only possible if there is no other error on the device. Then the switch is tripped and subsequently shows the result of the self-test with either green or red. If the LED stays red even after renewed shutdowns, the device is faulty and needs to be replaced.

For the 5SL6 COM miniature circuit breaker with RCM function, the internal RCM function can be tested with a short button press. Each activated RCM alarm is tested, including LED activation, alarm and stored message. The parameters set in each case for hysteresis, time delay, etc. are taken into account here. The test function at the press of a button is only possible if neither a tripping operation nor an RCM alarm to be confirmed is displayed.

In addition, the RCM version has an alarm auto-reset function, in which by default an alarm is only displayed for as long as the measured value exceeds the limit value (i.e. the alarm is automatically reset). If this function is deactivated, an RCM alarm continues even after the measured value has fallen below the limit value, until it is confirmed with a button press or remote command.

In the case of the terminal devices, resetting the communication information means that the installation code is reset to the factory settings and in doing so, the devices lose their connection to the SENTRON Powercenter 1000. Alternatively, the devices can be unpaired from the data transceiver via the software without needing to reset the devices to factory setting (see Chapter Unpair (Page 55)). Resetting the SENTRON Powercenter 1000 resets the Bluetooth connections and the Bluetooth PIN to factory settings. The terminal devices remain paired with the data transceiver.

The communication information of the 3NA COM fuse can only be reset by ensuring communication of the fuse is possible for more than 24 hours but a connection to the SENTRON Powercenter 1000 cannot be established. To do so, the fuse must be operated with sufficient load current but the SENTRON Powercenter 1000 data transceiver must be switched off.

4.5.2 Other operator controls

In addition to the standard operator controls which are present in most of the communicative circuit protection devices (button, lever), there are other operator controls.

5ST3 COM remote control auxiliary:

Yellow slide to block the remote switching function and to reset various error/status messages.

To reset a message, move the slide to the OFF position (the operating lever must also be switched off) and then back to "RC ON".

You will find further information in the operating instructions (https://support.industry.siemens.com/cs/ww/en/view/109823192) and also in Chapter Switching operation (Page 75).

4.6 LED signaling of the SENTRON circuit protection devices

The circuit protection devices with measurement and communication function can display different statuses with the LEDs. The 3NA COM fuse is an exception here as well; it has no LEDs. The status of the fuse can only be read out using the communication function.

In the following table the SENTRON Powercenter 1000 distinguishes between the two LEDs ACT and COM. A ✓ would mean that both LEDs have this function. Similarly, the 5SV6 COM arc fault detection device also has two LEDs, a left one for the arc fault detection device part and a right one for the miniature circuit breaker part of the device. The ✓ shows that both have this function.

The 5SL6 COM miniature circuit breaker without RCM function has only one LED (right), whereas the version with RCM function has two LEDs for displaying the RCM status (left).

Description	LED	Response of the LED[1)]	POC 1000	5ST3 COM AS+FC / 3RV2 COM	5SL6 COM	5SV6 COM	5ST3 COM RCA
Device without power supply applied, function not provided		Off	✓	✓	✓	✓	✓
Device in working condition, function possible without restrictions		Permanently green	✓	✓	✓	✓	✓
No radio connection to the Powercenter 1000 or no connection to terminal devices		Flashing 0.75 Hz green	COM LED	✓	✓	LED on the right	COM LED
Manual shutdown or remote switching function off		Flashing 0.75 Hz green	---	---	---	---	Status LED
Internal process running, e.g. setting up a connection, firmware update		Flashing 2 Hz green	ACT LED	✓	✓	✓	COM LED
Active Bluetooth mode Search for mobile device		Flashing 2 Hz green	COM LED	---	---	---	---
Device is switched on (Charging)		Flashing 2 Hz green	---	---	---	---	Status LED
Reset Bluetooth or radio communication		Flashing 2 Hz green, changeover after 10 s to 10 Hz green if completed	COM LED	✓	✓	✓	COM LED
Device localization via software active for 10 s		Flashing 5 Hz green	ACT LED	✓	✓	LED on the right	Status LED
Locating: Switched off manually, remote control off		Flashing 0.75 Hz green	---	---	---	---	Status LED
Warning of an upper limit violation of a measured value (e.g. temperature, current or voltage). Warnings and limit values can be configured.		Flashing 2 Hz green/yellow	---	✓	✓	LED on the right	Status LED[3)]
Warning of an upper limit violation of service life parameters (e.g. operating hours, switching cycles, tripping operations) Warnings and threshold values can be configured.		Flashing 2 Hz yellow/red	ACT LED	✓	✓	LED on the right	Status LED[4)]
Manual switching off of the main unit		Flashing 2 Hz yellow/red	---	✓	---	---	Status LED 2 Hz green
Communication error		Flashing 2 Hz yellow	COM LED	✓	✓	LED on the right	COM LED
RCM pre-warning (only applies to RCM versions)		Flashing 2 Hz green/yellow	---	---	RCM LED	---	---
RCM alarm		Flashing 2 Hz yellow	---	---	RCM LED	---	---
Performing test		Flashing 5 Hz green/red	---	---	---	---	Status LED

Description	LED	Response of the LED[1)]	POC 1000	5ST3 COM AS+FC / 3RV2 COM	5SL6 COM	5SV6 COM	5ST3 COM RCA
Device error, e.g. self-test unsuccessful[2)]		Flashing 4 Hz red	ACT LED	✓	✓	Left 2 Hz red Right 4 Hz red	Status LED
Circuit protection device itself or attached circuit protection device has tripped		Flashing 2 Hz red	---	✓	✓	✓	Status LED[5)]

1) For example, a 2 Hz flashing frequency corresponds to a flashing of 2-times per second. In comparison, a 5 Hz flash (5 times per second) is faster
2) Displayed with 2 Hz yellow for firmware version 1.0.0
3) Or: IR test warning or ARD function failed after three attempts or a switch-on operation. Reset with the yellow slide.
4) Or: reset with the yellow slide failed. Reset must be repeated. After 5 failed attempts: Device error
5) Or: RCD/IR test failed. Reset with the yellow slide

Both for exceeding limit values and for warnings when the end of service life is reached, there are parameters which can be set to adjust the threshold value of the alarms or to switch off the alarm completely. If the alarm is switched off by the software, the flashing LED is also no longer displayed. The limit values for temperature, current, voltage are alarms which can be left without changing the alarm settings if the measured value falls below the set limit value. Once an alarm is active and it causes a yellow/red flashing, it can either be switched off or, if necessary, the threshold can be increased. See also Chapter Messages (Page 70).

A device error due to a failed self-test cannot be reset by the operator and must be replaced. A communication error that occurs for example when the signaling strength is not sufficient can be rectified by the operator e.g. by removing radio barriers such as partitions, switching off or moving other transmitting devices, or resetting the radio communication of the devices with a long button press (≥ 10 s).

Commissioning

5

Note

Only qualified personnel are permitted to install, commission or service the devices.

- Wear the prescribed protective clothing. Observe the general equipment regulations and safety regulations for working with high-voltage installations (e.g. DIN VDE, NFPA 70E as well as national or international regulations).
- The limits given in the technical specifications must not be exceeded even during commissioning or testing of the device.
- Before connecting the device, make sure that the line voltage matches the specifications on the rating plate.
- Before you start up the device, check that all the connections have been made correctly.
- Before power is applied to the device for the first time, it must have been located in the operating area for at least two hours in order to reach temperature balance and avoid humidity and condensation.
- Condensation on the device is not permissible during operation.

5.1 Commissioning with SENTRON Powerconfig mobile

The measurement and communication-capable SENTRON circuit protection devices are put into operation with the SENTRON Powerconfig commissioning app for mobile devices, also known as Powerconfig mobile.

Note

An exact description on how to use the app can be found in the Installation Manual (https://support.industry.siemens.com/cs/ww/en/view/109791805).

First the SENTRON Powercenter 1000 data transceiver must be added. This can happen in the following ways:

- WLAN searches if the SENTRON Powercenter 1000 and the mobile device are located in the same network.
- Bluetooth searches of all available devices. To do so, the Bluetooth mode must be activated with a short button press and the Bluetooth PIN code must be entered after scanning the Data Matrix Code. The PIN code can also be entered manually with the code printed on the side of the device.
- Manual entry of the IP address of the SENTRON Powercenter 1000. This may be necessary if at first all devices are to be added offline or in case of a VPN connection the IP address must be entered manually for security reasons. In order for the data transceiver to set up a communication with the smartphone or tablet, it must be connected to the 24 V DC-power supply. After the supply voltage has been applied and the IP address has been determined, it can be subsequently changed in the app.

Then the individual communication-capable terminal devices must be added. To do so, the DMC must be scanned with the "RF code" marking of the individual devices, this contains the device type, MAC address and the installation code. Alternatively the scanning step can be skipped and the aforementioned information can be entered manually. These are printed on the side of the device.

A clear name or plant identifier must be provided when adding the communication-capable terminal devices so that the devices can be distinguished from one another.

Note

Scanning can also be performed without powering the devices. This is necessary if the RF codes are not accessible when fully installed (e.g. possible with the 3NA COM fuse) or the devices are not allowed to be switched off to scan the RF code. The RF code of the 5SL6 COM miniature circuit breakers and 5SV6 COM arc fault detection devices can only be scanned if the operating lever is set to the Off position.

Two Data Matrix Codes need to be scanned for the 3NA COM fuse. First the one of the electronic module and then the one of the fuse link

The 3NA COM fuse link and the electronic module can still be scanned when installed, e.g. in a 3NP fuse switch disconnector, if necessary, the flashlight function of the mobile device must be used to illuminate the code.

Then the communication-capable devices must be paired with the SENTRON Powercenter 1000. This is only possible if the devices are powered, contrary to the previous steps, which can also be carried out offline. Pairing can be carried out for each individual

device or simultaneously for several devices. Here, the terminal devices are connected one after the other in groups.



DANGER

Hazardous voltage

Will cause death, serious personal injury, or equipment damage.

In order to be able to establish communication, the devices must be switched on and supplied with power. In doing so, attention must be paid to sufficient safety measures (among other things, touch protection).

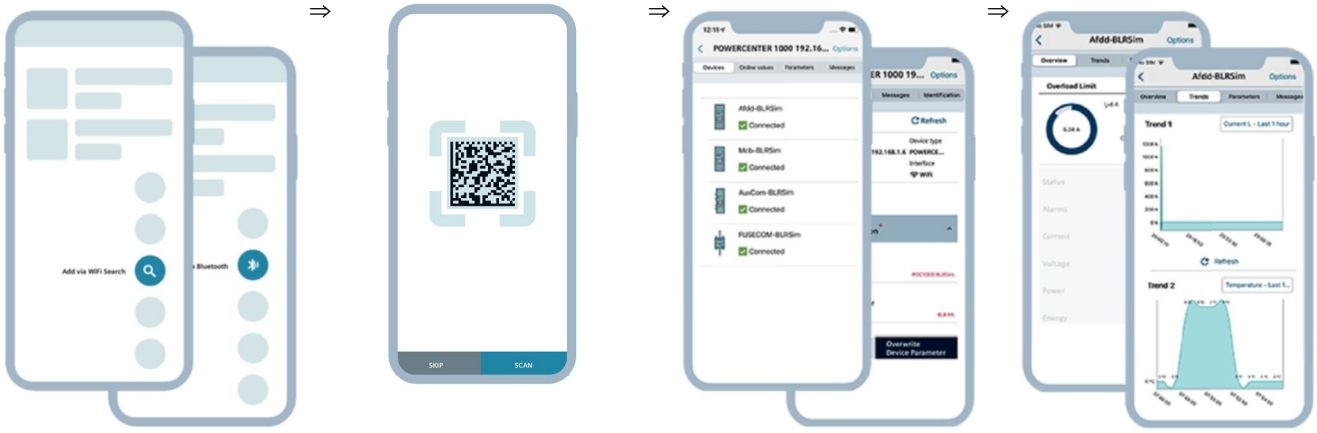
The measured values can be examined if the devices are paired. The trends are only displayed with an existing WLAN (Ethernet) connection. Similarly, the parameters of the individual devices can now be adapted, for example, the plant identifier which is required for the identification of the devices. For example, the various different alarm messages (see Chapter Messages (Page 70)) can be changed, switched on or off or the cyclic time synchronization of the SENTRON Powercenter 1000 (see Chapter Time synchronization (Page 76)) can be activated. In addition, there are also views of the device identification and the stored messages such as exceeding an alarm limit or the tripping of a device.

Note

The communication parameters for the IP connection (DHCP and static IP address) can only be changed in SENTRON Powerconfig mobile via Bluetooth. Alternatively, the SENTRON Powerconfig PC version can be used via Ethernet.

During commissioning, the devices display different statuses on the LEDs; these were explained in the Chapter LED signaling of the SENTRON circuit protection devices (Page 45). Upon completion of the commissioning process, all LEDs should be static green, i.e. the commutative circuit protection devices are fully functional.

1. Adding the Powercenter 1000 (Bluetooth or Ethernet) ⇒ **2. Adding and pairing the communicative circuit protection devices** ⇒ **3. Display and parameterization of the connected devices** ⇒ **4. Visualization of the recorded values**



If a SENTRON Powercenter 1000 is added to another project with another mobile device, the terminal devices do not need to be scanned and paired again if they have already been paired.

5.2 Commissioning with SENTRON Powerconfig for PC

Alternatively, the circuit protection devices with communication and measuring function can also be (re)configured with the SENTRON Powerconfig PC software. First the SENTRON Powercenter 1000 must be added from the same network. Then the individual terminal devices that have already been paired must be added in the online view "Overview": Unpaired terminal devices can be added from the library and paired by entering the device address, the MAC address and the installation code in the "Communication" view.

An alternative is to export an existing project from SENTRON Powerconfig mobile once commissioning has been completed. In doing so, a .splx file is stored on the smartphone, which can be imported as a PC version. In addition, the project data and parameters can be easily viewed and further processed on the PC. The extended range of functions of the SENTRON Powerconfig PC version also enables a firmware update of the entire system via the SENTRON Powercenter 1000.

Similarly, a .splx file could be exported from the PC version, archived on the PC and imported in SENTRON Powerconfig mobile.

5.3 Setting of parameters

All devices have parameters that can be set. Some parameters are found on different device types, others only on specific ones. The availability of the parameters depending on the device is given in Chapter Parameter categories of the SENTRON circuit protection devices (Page 83).

The most important parameters that can be set are:

- The plant identifier, as a unique name of each device
- The system time on the SENTRON Powercenter 1000
- The Ethernet settings on the SENTRON Powercenter 1000 (static IP address can be set only via Bluetooth or Powerconfig PC)
- The order number of the fuse link for the 3NA COM fuse, without which the rated current of the device is not known
- The direction of incoming supply for the 5SL6 COM and 5SV6 COM. This influences the counters for exported energy and imported energy and the sign of the power factor. If the set and physical direction of incoming supply match, the sign is positive
- The radio transmit power, which must match for all devices in the system if changes are necessary
- The automatically selected radio channel of the SENTRON Powercenter 1000
- All alarm settings that must be selected depending on the application
- Parameters for remote control (wired or wireless), for the reclosing function (ARD) and for the RCD/IR test with the 5ST3 COM remote control auxiliary.

Note

Not all alarms are activated by default.

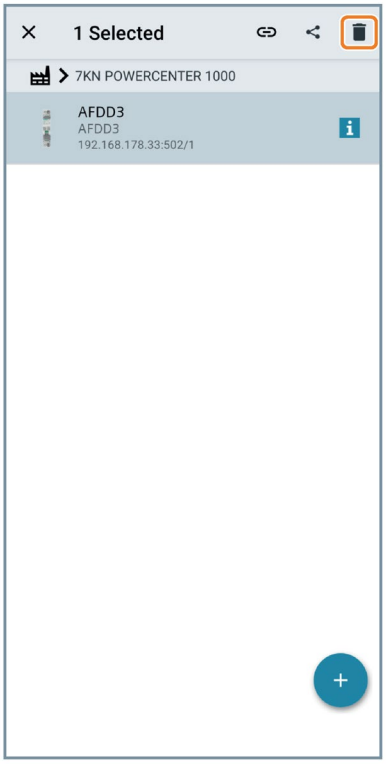
Once the parameters have been changed, it is important to save these to the device. If it is necessary to check which parameters are current in the device, the parameters must be downloaded so that the values in the project are overwritten.

5.4 Removing devices

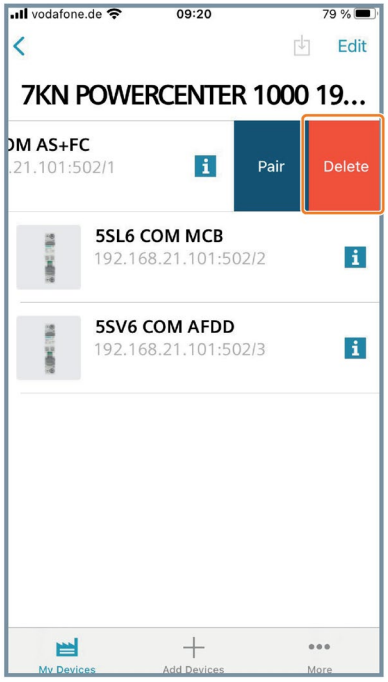
5.4.1 Delete

Added devices can be deleted from the list view of the project in the software. In the app, this can only take place in the project view via the gear icon to the right of a SENTRON Powercenter 1000. This means that this can no longer be displayed in the project. If the SENTRON Powercenter 1000 is selected, the devices continue to be paired and are displayed in the list of connected devices. The terminal devices are disconnected from the SENTRON Powercenter 1000 or from this mobile device only by resetting communication by pressing the button on the devices or by using the Powerconfig decoupling function.

If a device was deleted, it does not need to be rescanned, selecting it in the SENTRON Powercenter 1000 view of paired devices is enough. This device is then displayed in the project view once again.



Android



iOS

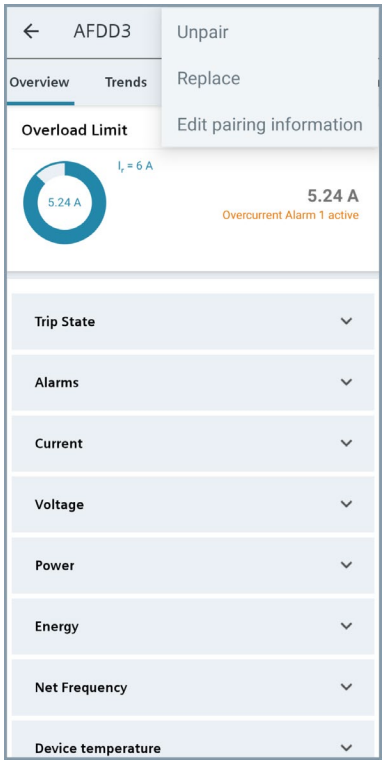
5.4.2 Decouple

The Unpair function can be used in the detailed view of the individual devices. In doing so, the terminal devices are unpaired from SENTRON Powercenter 1000, similar to a long button press on the terminal devices. To ensure that both the terminal device and Powercenter 1000 are unpaired on both sides, both devices must first be switched on and communicating with each other. This applies both for the command via the app and for the button press.

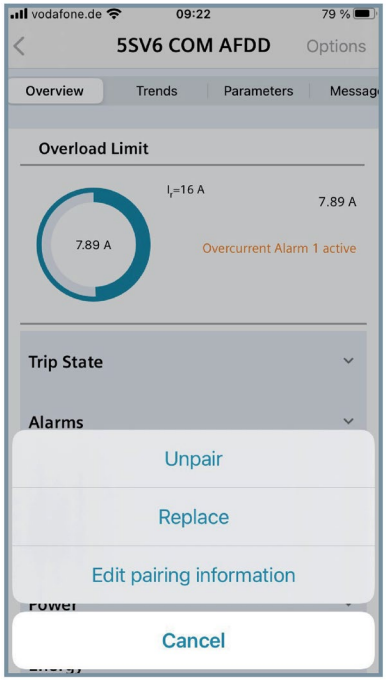
Example 1: If the terminal device is not switched on but is unpaired via the app, Powercenter 1000 forgets the terminal device. Following restart, however, the terminal device attempts to connect Powercenter 1000 and then returns a communication error. The long button press must then be performed on the terminal device.
Example 2: If the button on the terminal device is pressed for ≥ 10 s while Powercenter 1000 is switched off, the latter will continue to search for the terminal device following restart and display it in the list. The terminal device must then be removed in the app.

If a terminal device was unpaired, it is no longer displayed in the list of connected devices of the Powercenter 1000. In the app's project list, which can be accessed via the gear icon, the device is still displayed and can either be re-paired or simply deleted.

If unpairing is not possible because the terminal device cannot communicate with Powercenter 1000 at the moment, a message is displayed with an option of deleting the terminal, which removes the terminal in Powercenter 1000 and in SENTRON Powerconfig mobile.



Android



iOS

5.4.3 Replace

In addition to unpairing, the Replace function can also be selected in the detailed view of the individual devices. To do so, a new device of the same type must be scanned, which then replaces the current devices. In addition, the old device is removed from the radio network, the new device is added, and the parameters included in the app project at this time are transferred to the new device. If the device to be replaced is no longer communication-capable and the last parameters are not known to the project in the app, then only the default parameters are used.

The Replace function only works if the same type of device is scanned and SENTRON Powercenter 1000 is online. If the new device has already been added to the project, it must be deleted if the Replace function is to be used.

The app shows if the process takes longer because many steps are carried out one after another. In addition, if the radio channel is heavily loaded, this may take longer. The app cannot be used while using the Replace function. The procedure can be interrupted and continued later on.

For the 3NA COM fuse there are different possibilities available to replace the device. On the one hand, only the fuse link can be replaced by inserting a fuse link with the same rated current together with the previously paired electronic module. A change in the software is then no longer necessary.

Note

If another rated current is used, the system must be adjusted to the new rated current and the new rated current must be set in the software.

If it is only the electronic module or the electronic module and the LV HRC fuse link being replaced, the Replace function in the software can be used by scanning the new Data Matrix Code.

5.4.4 Change communication information

The third function, which can be selected in the device detailed view is changing the communication or pairing information. Here, the MAC address or the installation code can be adapted as long as the devices have not been paired with the SENTRON Powercenter 1000. This may be necessary if the information was entered manually and an error occurred, whereby pairing is not possible.

Similarly, the IP address can also be changed if a SENTRON Powercenter 1000 was added manually. This can be useful, for example, if the Powercenter 1000 was added offline.

5.5 Special features of the 3NA COM fuse

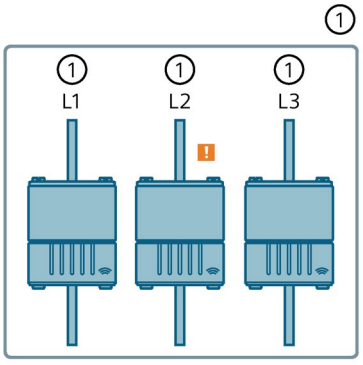
If several fuses are added, these are displayed together in the app. This is referred to as "Device Group".

In order to summarize fuses in a device group, the plant identifier of the devices must match. The respective installation position / phase should be specified for each fuse. As soon as the plant identifier no longer matches, the devices are no longer displayed as a device group.

Note

A name can be given when adding the fuses. If this is not done, the default designation of the devices remains. The unnamed devices are also summarized.

The representation in the app includes the number of grouped fuses (2 to max. 3 pieces), the status (connected or not connected) and a possible indication of at least one present alarm.



① Line

If a red circle is displayed in the overview, the rated current of the fuse link is not known to the project and/or the electronic module. This problem can be resolved by saving the order number of the fuse link in the parameters.

Note

If the rated current is not known to the electronic module, no alarm can be generated if the upper limit for the current is violated.

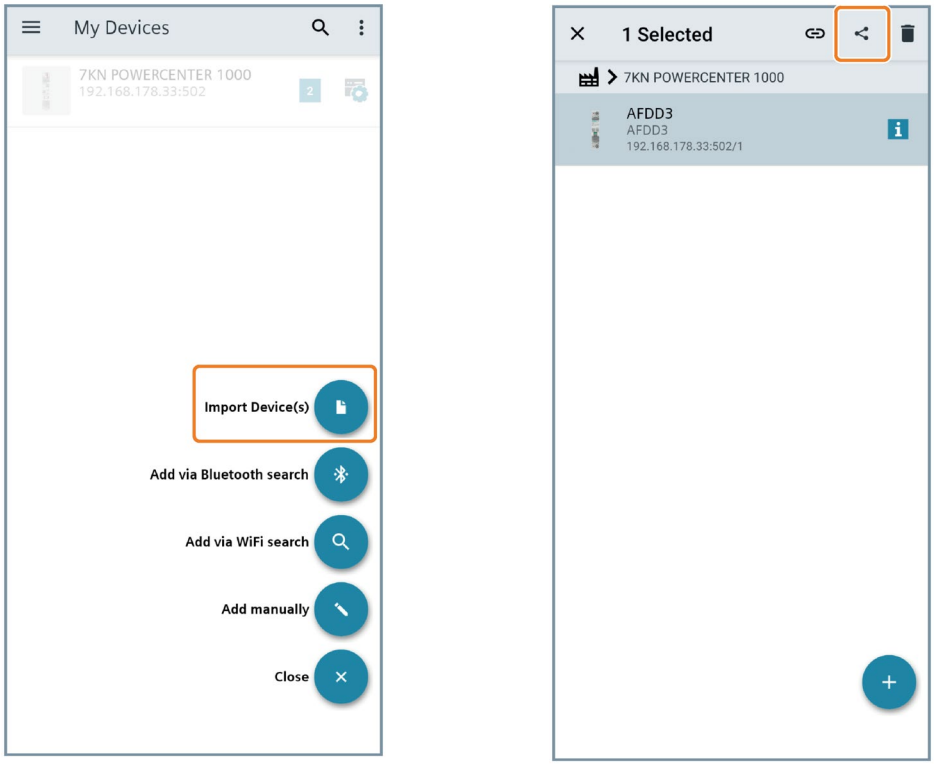
5.6 Import and Export

Several devices can be added in SENTRON Powerconfig mobile and displayed in a project. If it is necessary to summarize devices in different projects, it is recommended to export the .splx file to the mobile device for each project after commissioning. Several projects can be stored with different names in this manner and can be imported again at a later time.

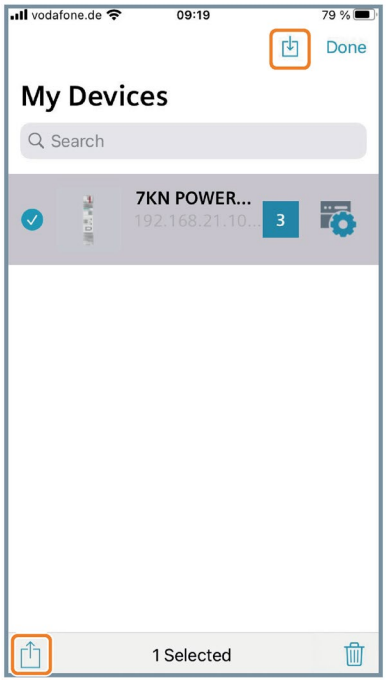
During import, it must be ensured that the imported devices do not overwrite the existing ones. If only the imported project is to be displayed, it is recommended to export the existing project and then delete the devices. Alternatively, the different devices can also be distinguished with a specific name.

Note

The PC version of SENTRON Powerconfig must be used to get an overview of different projects or to move devices to different projects. The mobile app only supports a listing of all added devices.



Android



iOS

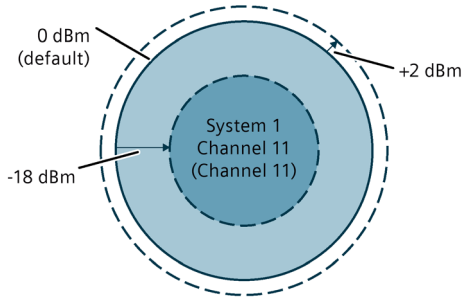
5.7 Commissioning several Powercenter 1000 devices

5.7.1 Automatic radio channel selection

Note

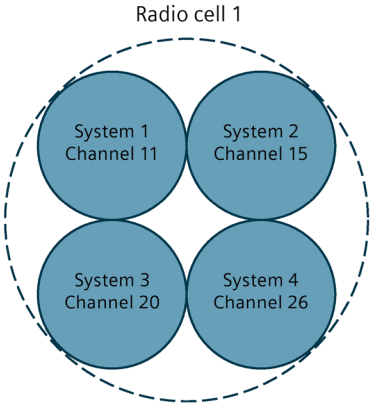
Only qualified personnel are permitted to install, commission or service the devices.

Up to four SENTRON Powercenter 1000 devices can be operated in parallel in one radio cell. Several data transceivers can also be used by increasing the distance of the devices among each other (or reduced signal strength) so that the devices do not disturb one another. It makes sense to distribute all communication-capable circuit protection devices as equally as possible to the various data transceivers.

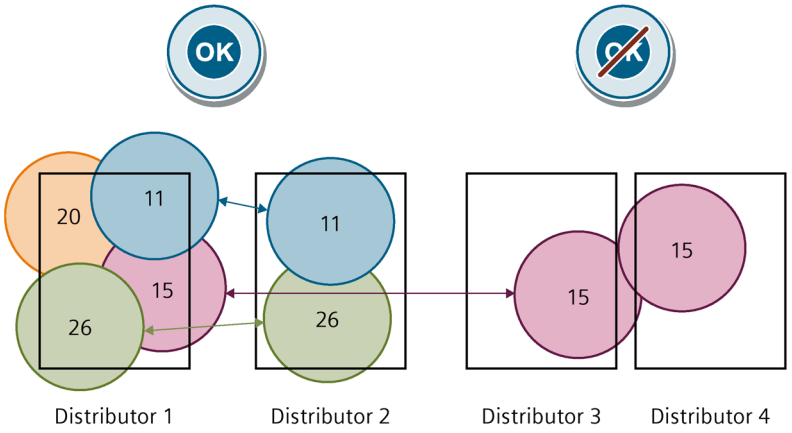


The signal strength must be set in the SENTRON Powercenter 1000 and in the associated terminal devices. This can be set between -18 ... +2 dBm.

In the used frequency band between 2400 MHz and 2483.5 MHz there are 16 individual channels (11-26) with separate bandwidths. Each Powercenter 1000 independently selects the channel with the lowest capacity utilization. The selected radio channel is visible in the parameters. The following four channels can be automatically selected: 11, 15, 20 or 26.



If four Powercenter 1000 devices are operated in different radio channels, a larger distance does not need to be maintained between the devices. They can be set up in one or neighboring distribution boards. However, if more than four data transceivers are operated, it can result in a double assignment of radio channels. In this case the distance between the two systems including the terminal devices must be big enough so that they do not cause interference. See Installation conditions. The radio signals of the systems, which are operated in different radio channels, can spatially overlap.



Note

A specific order must be maintained during commissioning so that not all Powercenter 1000 devices radio on one channel at the same time, thus putting too much strain on the channel and causing data to be lost. This is because the more radio-capable stations there are, the greater the risk that individual data packages from the different stations will be lost.

During the whole commissioning process, all Powercenter 1000 devices can be switched with the associated circuit protection devices either all at once or sequentially. Channel selection takes place as soon as the first communication-capable circuit protection device is paired with the SENTRON Powercenter 1000. All of the devices can be added to the app when switched off, only then is the sequential switching on of the terminal devices required and commissioning can be shortened by pairing 24 terminal devices at the same time.

The following steps should be followed in sequence:

1. Switching on all SENTRON Powercenter 1000 devices.
2. Switching on the circuit protection devices, which are paired with the first Powercenter 1000. The fewer other, unpaired circuit protection devices are active, the faster the initial pairing process will be.
3. Pairing the circuit protection devices with the SENTRON Powerconfig mobile app.
4. Once all devices have been successfully paired with the first Powercenter 1000, they need to be operated further for the next step so that the previously assigned channel does not appear to be free/unused for the next data transceiver.
   Recommendation: Restart the devices that have just been paired so as to ensure reliable connection to the wireless network.
5. The next communicative circuit protection devices should only be switched on now in order to pair them with the second SENTRON Powercenter 1000.

Repeat steps 2 to 4 until all devices have been paired with the different SENTRON Powercenter 1000 devices.

Note

Due to the conditional disconnection possibility of 3NA COM fuses and 5ST3 COM auxiliary/fault signal contacts and the 3RV2 COM wireless auxiliary and signaling switch, it is recommended to pair them with the first Powercenter 1000. If necessary, an additional switch must be placed for the power supply of the auxiliary switch.

5.7.2 Manual radio channel selection

As of Version 2.0 on the SENTRON Powercenter 1000, it is possible to switch off automatic radio channel selection via the parameters. As a result, it is necessary to make a manual selection between channels 11-26. This setting must be made before the first terminal device is paired. If a device is paired with the SENTRON Powercenter 1000, the radio channel is no longer changed on the device, even if a different setting is stored. If a different radio channel is required, all terminal devices must first be unpaired. This function is supported for all terminal devices as of Version 2.0. Older device versions can only be paired on the 4 standard channels. To use the full scope of this function, a firmware update (see Chapter Firmware update (Page 103)) to Version 2.0 must be performed.

Important:

- Radio channel selection must take place before the first terminal device is paired
- Manual radio channel selection is possible only for devices as of Version 2.0

Manual radio channel selection makes it possible to operate a larger number of SENTRON Powercenter 1000 devices simultaneously. The exact number depends on the real ambient conditions (e.g. interference with WLAN). Before selecting the channel, it is recommended that the capacity utilization of the radio channels be determined by means of spectrum analysis so that the smooth transmission of data can be ensured. It is recommended that a radio spectrum analysis be carried out so that the radio channels having the lowest capacity utilization are selected. The commissioning of several systems must be carried out in a certain sequence in accordance with Chapter (Automatic radio channel selection (Page 59)).

5.8 Time-Outs

5.8.1 Reclosing

If the SENTRON circuit protection devices with communication and measuring function were already paired with the Powercenter 1000, the connection is automatically re-established after the devices have been switched off.

After reclosing, an individual circuit protection device connects with the Powercenter 1000 after approx. two seconds. If several circuit protection devices are simultaneously reclosed, it will take longer for them all to be connected again (up to 20 s).

If a Powercenter 1000 is reclosed, the connection establishment can take longer. Up to 1 minute at full load with 24 terminal devices.

5.8.2 Pairing

The simultaneous pairing process of several circuit protection devices with one SENTRON Powercenter 1000 takes place sequentially. Due to the increased data volume, the pairing of 24 terminal devices can take longer compared to a single device.

During the pairing of one or several devices, a pairing timeout can be displayed in the app after 60 seconds. However, the devices will continue (unlimited) to try to pair even after this timeout provided the process is not terminated by the software.

Depending on the capacity utilization of the radio channel and depending on the number and types of terminal devices, the pairing duration can vary. A single device can be paired in a few seconds (< 10 s). In many devices, especially 5SV6 COM arc fault detection devices, which transmit the most data in comparison, this can also take several minutes. A waiting time of up to two minutes is to be expected, depending on the capacity utilization of the radio channel.

5.8.3 Unpairing

The unpair command is followed by a timeout of 60 s, as for pairing. If this is exceeded, the SENTRON Powercenter 1000 will attempt to unpair the device indefinitely.

If the operation is however not possible, e.g. because the terminal device has been destroyed or is no longer available, the unpair operation can be terminated and as a result the device unpaired from the Powercenter 1000 without the knowledge of the terminal device. The terminal device would display a communication error, as no response would be received from the corresponding Powercenter 1000.

5.9 Use of third-party software

Both the SENTRON Powerconfig, SENTRON Powerconfig mobile and the SENTRON circuit protection devices with communication and measuring function use third-party software licenses.

The OSS licenses of Powerconfig are displayed via the Info and Help menu items. You can find the OSS licenses of the SENTRON circuit protection devices here (https://support.industry.siemens.com/cs/de/en/view/109797242).

Functions

6

The SENTRON circuit protection devices with communication and metering function offer different functions depending on the respectively recorded measured values and setting options.

6.1 Recorded measured values and storage

6.1.1 Measured value acquisition

The following measured values are recorded for the individual device types and e.g. displayed in SENTRON Powerconfig mobile in the device overviews. A more detailed description of all measured values and parameters can be found in Chapter Information categories of the circuit protection devices (Page 83).

Measured value	POC 1000	5ST3 COM AS+FC	3RV2 COM	5SL6 COM	5SV6 COM	3NA COM	5ST3 COM RCA
Temperature (incl. mean value)	✓	✓	✓	✓	✓	✓	✓
Current (incl. mean value and maximum value)	---	---	---	✓	✓	✓	---
Voltage	---	---	---	✓	✓	---	---
Line frequency	---	---	---	✓	✓	---	---
Apparent, reactive, active power and power factor	---	---	---	✓	✓	---	---
Reactive, active energy	---	---	---	✓	✓	---	---
Residual currents in several frequency ranges	---	---	---	✓[1)]	---	---	---
Switching status (for 5ST3 COM/3RV2 COM of the attached device)	---	✓	✓	✓	✓	---	✓
Operating hours	✓	✓	✓	✓	✓	✓	✓
Operating hours with load current	---	---	---	✓	✓	✓	✓
Mechanical switching cycles	---	✓	✓	✓	✓	---	✓
Counter for short-circuit tripping operations	---	---	✓	✓	✓	---	---
Counter for tripping operations general	---	✓	✓	✓	✓	---	✓
System time	✓	---	---	---	---	---	---
Number of test executions	---	---	---	---	---	---	✓
Status for ARD, test	---	---	---	---	---	---	✓
Detail status for slide and lever	---	---	---	---	---	---	✓

1) Applies only to versions with RCM function

Measured values are displayed in the software as invalid, if they are not received for a specific period of time.

All pairable terminal devices also have a value specifying the radio received signal strength (RSSI). This indicates the quality of the communication via radio with the SENTRON Powercenter 1000. If the value is greater than -90 dBm (e.g. -60 dBm), a stable connection can be assumed. At values below -100 dBm, dropouts and interruptions can occur.

6.1.2 Accuracy

The accuracy and measuring range of the individual measured values with a reference temperature of 23°C are described as follows:

Measured value	5ST3 COM and 3RV2 COM	5SL6 COM and 5SV6 COM	3NA COM
Temperature	±2°C from -2°C ... 100°C	±2.5°C from -25°C ... 100°C	±2.5°C from +20°C ... 120°C
Current	--	0.5% of 0.02 ... 2 x I_n	2% of 2.5 A ... < 8 A and 1% of 8 A ... 440 A[1)]
Voltage	--	0.5% at U_n and 1% of 0.9 ... 1.1 x U_n	--
Line frequency	--	0.5% of 45 Hz ... 60 Hz	--
Power values	--	1% at 0.9 ... 1.1 x I_n and U_n	--
Power factor	--	±0.1 of -1 ... 1	--
Energy	--	1% at 0.9 ... 1.1 x I_n and U_n	--

1) Reference temperature here 25°C relative to mean value

The accuracy classes are in accordance with the IEC 61557-12, IEC 62053-22 and IEC 62053-23 standards.

For the 5SL6 COM RCM versions, the RMS values at the rated frequency are also subject to the following absolute accuracy of differential currents:

±30% from 3 mA ... 5 mA

±15% from > 5 mA ... 1000 mA

6.1.3 Measured value transmission frequency

The measured values are transmitted to the SENTRON Powercenter 1000 in different transmission frequencies.

This leads to the fact that not every measured value is updated equally as fast as in the software.

The following measured values are transmitted every x seconds:

- Temperature: 2 s
- Temperature mean value: 60 s
- Current: 2 s
- Current mean value: 2 s
- Current maximum value: 60 s
- Residual current (all measurement channels): 2 s
- Voltage: 60 s
- Line frequency: 60 s
- Active, reactive and apparent power: 2 s
- Power factor: 60 s
- Active and reactive energy: 60 s
- Operating hours counter with/without load current: 60 s
- Operating cycles counter, test execution counter: 10 s and in the event of a change
- Counter for tripping operations: 60 s and in the event of a change
- Number of short-circuit tripping operations: 60 s
- Switching status: 10 s and in the event of a change
- Alarm status: 60 s and in the event of a change
- Radio RSSI: 60 s

6.1.4 Measured value memory

The different measured values are saved in the SENTRON Powercenter 1000 and can be displayed in the trends of the respective devices in SENTRON Powerconfig mobile (not via Bluetooth). A distinction can be made between the following trends depending on whether the aforementioned measured value is supported by the respective device.

Stored measured value	Storage duration	Interval	Representation type
Temperature mean value	1 hour	1 minute	Line diagram
Temperature mean value	7 days	15 minutes	Line diagram
Current mean value	1 hour	10 seconds	Line diagram
Current mean value	7 days	15 minutes	Line diagram
Imported active energy	7 days	15 minutes	Line diagram
Imported active energy	30 days	1 day	Bar diagram
Voltage min/max	10 days	1 day	Bar diagram
Line frequency min/max	10 days	1 day	Bar diagram
Active power min/max	10 days	1 day	Bar diagram
Apparent power min/max	10 days	1 day	Bar diagram
Temperature min/max	10 days	1 day	Bar diagram

As of Version 2.0, the stored measured values can be adjusted individually for each trend. The storage duration and the intervals remain unchanged.

6.1.5 Residual current measurement (RCM)

The 5SL6 COM miniature circuit breaker with RCM function corresponds to the IEC 62020-1 standard. This is a type F device and determines sinusoidal residual currents (AC) up to 100 kHz and residual currents from pulsating DC current. The measuring range is between 3 mA and 1000 mA for AC residual currents or between 3 mA and 300 mA for pulsating DC residual currents.

The measured value of the residual current is measured in several frequencies simultaneously and distributed over the following measurement channels:

- Line frequency (50 Hz)
- Harmonic of line frequency
- Low-pass - AC
- Low-pass - AC and pulsating DC (also referred to as "RMS", i.e. AC and DC combined)
- Band-pass
- High-pass

All ranges of the measurement channels can be set except for the base frequency, which is determined by the device itself. You will find more detailed information about setting options in Chapter RCM measured values and parameters (Page 96).

By subdividing the residual current into different frequency ranges, it is possible to identify different error causes, such as defective insulation and humidity in the low frequency range or electromagnetic interference at high frequencies.

The two low-pass channels also have a configurable alarm and pre-alarm, including threshold value, hysteresis, ON and OFF-delay. Here, the pre-alarm is set as a percentage of the main alarm.

A general ON-delay of the RCM alarms after a restart of the device can also be set, for example to wait until a motor has ramped up before the measurement or alarm is once again active.

In addition, the automatic resetting of alarms can be switched off if the measured value is below the threshold value, so that an RCM pre-alarm / alarm can be confirmed only by a button press or via a remote command. This is useful so as not to miss an upper limit violation of the RCM measured value.

Here, too, all setting options can be found in RCM measured values and parameters (Page 96).

6.1.6 Special characteristics

The sign of the power factor and the counting of the exported or imported energy are dependent on the set and physical direction of incoming supply.

If the physical incoming supply is from above, the intrinsic consumption of the device is also counted in the energy counter.

The threshold value from which the energy counter starts to accumulate is 1.5 Wh or 1.5 varh.

The energy counters can only be reset together.

The rms value of the current is measured even if the current curve does not exhibit true sinusoidal behavior. In this case, the standard relationship ($S^2=P^2+Q^2$) between apparent, reactive and active power cannot be used.

6.2 Messages

6.2.1 Measured values and upper limit violation

The following messages are possible due to upper limit violations:

Possible message sources:							
Description	**Default**	**POC 1000**	**5ST3 COM**	**5SL6 COM**	**5SV6 COM**	**3NA COM**	**3RV2 COM**
End of maximum set operating periods	Inactive	✓	✓	✓	✓	✓	✓
End of maximum set operating duration with definable load	Inactive	---	---	✓	✓	✓	---
End of maximum set switching cycles	Inactive	---	✓	✓	✓	---	✓
End of maximum set short-circuit trips	Active	---	---	✓	✓	---	✓
End of maximum set tripping operations	Inactive	---	✓	✓	✓	---	✓
Reaching set temperature[1)]	Active	✓	✓	✓	✓	✓	✓
Overcurrent alarm 1 in % of rated current[2)]	Active	---	---	✓	✓	✓	---
Overcurrent alarm 2 in % of rated current[2)]	Inactive	---	---	✓	✓	✓	---
Undercurrent alarm 1 in % of rated current[2)]	Inactive	---	---	✓	✓	✓	---
Undercurrent alarm 2 in % of rated current[2)]	Inactive	---	---	✓	✓	✓	---
Overvoltage alarm 1 in volts	Inactive	---	---	---	✓	---	---
Overvoltage alarm 2 in volts	Inactive	---	---	---	✓	---	---
Undervoltage alarm 1 in volts	Inactive	---	---	---	✓	---	---
Undervoltage alarm 2 in volts	Inactive	---	---	---	✓	---	---
Voltage underflow AFDD trip[3)]	Active	---	---	---	✓	---	---
RCM pre-alarm low-pass channel AC	Inactive	---	---	✓[4)]	---	---	---
RCM alarm low-pass channel AC	Inactive	---	---	✓[4)]	---	---	---
RCM pre-alarm low-pass channel RMS	Active	---	---	✓[4)]	---	---	---
RCM alarm low-pass channel RMS	Active	---	---	✓[4)]	---	---	---
Number of RCD tests achieved	Inactive	---	✓[5)]	---	---	---	---

1) Base line is an average temperature that is determined over an adjustable time.
2) Base line is an average current that is determined over an adjustable time.
3) Base line is the average voltage which is fixed according to the standard, constant at 195 V and a hysteresis of 10%.
4) Applies only to RCM versions
5) Applies only to 5ST3 COM remote control auxiliary with RCD test function

Note

Not all alarms are activated by default. All alarms can each be switched on or off.

The respective alarms can be activated or even deactivated. A message is not displayed if they are deactivated and the device LEDs do not flash according to the pattern described above

(see Chapter LED signaling (Page 45)). Furthermore, there are adjustable threshold values that generate messages when they are exceeded. In addition, a hysteresis value can be set for the alarms for current, voltage and temperature. An alarm is activated if a threshold value is exceeded. If the measured value drops below the threshold value including the set hysteresis, the alarm status is exited again. As of firmware version 1.1.0, exiting an alarm when the limit value is no longer exceeded is also stored as a message.

For measured values relating to service life (e.g. operating cycles counter, operating hours, number of tripping operations), an alarm status reached can only be left if the threshold value is increased or the alarm is deactivated once the device has been tested. If it is no longer possible to ensure that a device is in perfect operating condition (e.g. visible burns after too many short-circuit trips), it is recommended that the device be replaced.

Chapter Measurement settings (Page 90) described which parameters are available with the respective units, value ranges and default settings.

The aforementioned warnings are displayed in the device overviews for the respective measured values. Furthermore, these are displayed in the Messages view with a time stamp.

In addition to the messages that can be controlled by the user, listed in the table above, there are others events captured during operation, e.g.

- Resetting the energy counter
- Changing the time on the SENTRON Powercenter 1000
- Time synchronization of the terminal device with the data transceiver after a restart
- DHCP network errors on the SENTRON Powercenter 1000
- Different statuses of a firmware update
- Tripping operations with type of trip, e.g. short-circuit, serial or parallel arcing fault
- Confirmation of a tripping operation [1)]
- Manual switching operation of the auxiliary switch [1)]
- Restart of the devices [1)]
- Activation/deactivation of an alarm [1)]
- Activation/deactivation of the ARD function in the 5ST3 COM remote control auxiliary

[1)] Not available in firmware version 1.0.0

Up to 50 of these messages are displayed in the app under the Messages view, sorted chronologically. Other message details can be accessed to some extent. In order to avoid an unnecessary flood of message and the loss of important messages, the limit values for the alarms must be set in an expedient manner.

6.2.2 Tripping operations in the event of a fault

The circuit protection devices with measuring and communication function have different tripping functions/protection functions. These protection functions are independent of the communication function, i.e. the electromechanical devices continue to provide protection even if there is a communication failure.

The following trip causes are reported:

Trip cause	5ST3 COM	5SL6 COM, 3RV2 COM	5SV6 COM	3NA COM
Attached device has tripped (reason unknown)	✓	---	---	---
Manual/mechanical shutdown	✓	✓	✓	---
Overload (thermal tripping)	---	✓	✓	---[1)]
Short-circuit	---	✓	✓	---[1)]
Arcing faults	---	---	✓	---
Overvoltage	---	---	✓	---
Test tripping	---	---	✓	---

[1)] The protection function is provided, but the communication function does not send a trip message

Note

Short-circuit detection in 5SL6 COM and 5SV6 COM

In the case of rapidly rising short-circuits or a reconnection to an existing short-circuit, the short-circuit is not detected by the measuring function In these situations, the integrated miniature circuit breaker mechanically trips the device so quickly that there is no entry in the messages, there is no alarm and no trip counter is activated.

The 5ST3 COM auxiliary components communicate the status of the installed device even in the switched-off status. They distinguish between a manual shutdown and a tripping operation due to error. The trip cause is not detected. The manual shutdown is displayed in the device overview and in the operating cycles counter. In the event of a tripping operation, this is also displayed in the trip counter and as a message.

The 3RV2 COM wireless auxiliary and signaling switch for motor starter protectors distinguishes between a manual shutdown and a trip caused by short-circuit or overload. The trip cause is detected and can be output. The external power supply means that communication is possible even with the operating lever in the OFF position. A short-circuit trip increments both counters (trip counter and short-circuit trip counter) whereas only the trip counter is incremented for an overload and the test function.

Communication is not possible if the 5SL6 COM miniature circuit breaker and 5SV6 COM arc fault detection device are switched off. If the device is switched off manually, the operating cycles counter is incremented. If the device trips, both the operating cycles counter and the trip counter are incremented. In addition, a message containing more detailed information including the reason for the trip operation and the time is generated. If a trip is detected, it must be confirmed by pressing the button briefly when the devices are reclosed. From version V1.1, the switching status is displayed.

It is possible that a trip message/the change of switching status is no longer transmitted or received by the SENTRON Powercenter 1000 in time before the 5SL6 COM miniature circuit breaker and 5SV6 COM arc fault detection device are switched off.

In order to monitor a trip message or the switching status of the 3NA COM fuse, a commercially available external fuse monitoring system must be installed via an additional device. If the fuse element has tripped, communication can no longer take place and the fuse itself indicates this via a front indicator. The electronic module does not distinguish whether

there is a disconnection of faults, a power failure or the deactivation of a consumer. In any case, the electronic module reports the “Connection disconnected” status.

6.2.3 Test execution and memory

The execution of test functions is recorded in a further memory area. Here, an entry is created for each test execution and includes the result and the time stamp, among other information. It is possible to export the test results. Device self-tests (e.g. for the AFDD) are not documented.

The following device tests are stored for each device:

5SL6 COM miniature circuit breaker with RCM function

RCM test:

This simulates an internal residual current that triggers all activated alarms and pre-alarms, and the associated LED flashing, messages in the message memory and setting of the alarm status. The set parameters such as the auto-reset function and the ON and OFF-delay are taken into consideration here.

5ST3 COM remote control auxiliary with RCD test function

RCD/IR test:

A test procedure normally includes an RCD test and an insulation resistance test (= IR test). Each can be disabled and parameterized individually. The two tests are only available if an RCD, RCBO or RC unit is set as the attached device.
For the RCD/FI test, the correct RCD (30 mA, 100 mA, 300 mA selective or non-selective) must be set. The comparison resistance value for the IR test is defined at the same time. For the IR test, the number of poles must also be set if the IR test is to be carried out.

Note

The IR test is deactivated by default. The IR test takes place with < 2 mA at nominal voltage in accordance with IEC 63024.

The following test results are logged:

- Time of day
- RCD test result
- Set RCD type
- RCD test voltage
- Tripping time and tripping current
- IR test result
- Set resistance value
- Set number of poles
- Insulation resistance values.

Note the following:

Note

- A stand-alone IR test, not following an RCD test, can only be started in the switched-off status. As a result, if an MCB is attached, the switch must first be turned off before a (cyclical) test can be performed.
- An IR test is performed after every tripping operation of the mount-on device unless the IR test is disabled.
- After a successful test execution, the device is switched on again. If an IR test fails, the system must firstly be checked, the error message reset and the switch turned back on again manually. Remote switching/ARD function are not possible in this situation. An IR test error occurs if the measured resistance Rd is below the requirement for the value Rd0 in the IEC 63024 standard (8000 Ω at 30 mA RCD; 2500 Ω at 100 mA RCD; 800 Ω at 300 mA RCD).
- An IR test warning occurs if the measured resistance is below the setting Rd but not yet below the requirement Rd0 in the standard. This improves system safety. After a warning, the device can be switched on remotely, or automatic reclosing (ARD) may be possible depending on the configuration.

An RCD/IR test can be set and executed individually or automatically (cyclically).

Note

The system time of the Powercenter 1000 is used for cyclical test execution. It is strongly recommended to configure a time server to ensure that the time is set correctly even after an outage. You will find more information in Chapter Time synchronization (Page 76).

6.3 Switch-off operation

With devices such as the 5ST3 COM remote control auxiliary, an active switching command can be sent. However, if the device is switched off manually with the lever, it must also be switched on again manually or a separate switch-on command must be sent so that a technician on site is not in any danger.

⚠ WARNING

Death or serious injuries are possible

There is dangerous voltage after an external switching command.

Make sure that a technician on site, for example, is not in any danger from an external switching command.

With the 5ST3 COM remote control auxiliary, a switching command can be sent either via the wired interface at the plug-in terminals or via the software and the Powercenter 1000. Both methods are not possible at the same time. The corresponding setting must be made in the configuration software. To prevent commands such as a switching command or a test execution being skipped, they must not be re-sent within 10 s. It takes at least 1 s to execute a switching command.

Note

The remote switching function can be blocked with the yellow slide on the remote control auxiliary (positions "RC OFF" or "OFF").

This block prevents remote switching and also test execution. This is necessary especially if secure, restricted access cannot be guaranteed in the network.

A Modbus TCP command is able to switch the device even unintentionally. Make sure that the system is protected against unauthorized access.

In addition to the remote switching function, the remote control auxiliary supports the automatic reclosing function (ARD) after a tripping operation of the mount-on device. This function can be configured: on/off, three delay times for repeated reclosing attempts and a waiting time after which three new attempts will be made. It is also possible to specify whether the device is automatically reclosed after an IR test warning.

If the device cannot be reclosed after three or six attempts, the ARD function is blocked and must be reset with the yellow slide. ("RC ON" → OFF → "RC ON").

6.4 Time synchronization

SENTRON Powercenter 1000 devices and all connected communication and measuring circuit protection devices can take part in time synchronization via SNTP (Simple Network Time Protocol) like many other SENTRON devices. This can only take place with an existing Internet connection.

This allows events and measured variables to be assigned a point in time.

UTC time is used in the SENTRON Powercenter 1000. The time stamps are displayed in the software according to the time zone set on the device.

Note

The system time of the SENTRON Powercenter 1000 must be synchronized during the initial commissioning procedure. Either once with the mobile device via Powerconfig or cyclically via a time server in the parameter view. The latter is recommended because the time is automatically synchronized again once the SENTRON Powercenter 1000 has been switched off.

The one-off time synchronization of the SENTRON Powercenter 1000 device is possible via Powerconfig. However, the time setting gets lost when the supply voltage is switched off. Therefore, time synchronization via an SNTP server is recommended.

Using the selected synchronization mode, the SENTRON Powercenter 1000 cyclically receives the current time of the selected time server/SNTP server and continues this internally. If the SNTP client is activated on the SENTRON Powercenter 1000, an IP address of a web server must be entered. If the broadcast function is activated, the server IP address must be set to 0 in order to receive all broadcasts in the network.

A typical SNTP server is the SENTRON Powercenter 3000, if it has been configured accordingly. Alternatively, the PC with SENTRON Powermanager can also be used for this purpose.

Many systems offer the standardized SNTP server function, e.g. router. If necessary, the appropriate ports (123) must be opened for this in the firewalls.

There are also country-specific time servers, which can be reached via the Internet. For example, in Germany it is a time server of the National Metrology Institute of Germany (PTB) with the following IP address: 192.53.103.108

Note

The contents of the trends are deleted in version 1.0 after a time step > 30 min on the SENTRON Powercenter 1000. As of version 1.1, unavailable data records are filled with NaN. Messages remain intact with the appropriate time stamp of the entry.

See also

Equipment Manual - SENTRON Powercenter 3000
(https://support.industry.siemens.com/cs/ww/en/view/109763838)

How do you configure your PC as NTP server?
(https://support.industry.siemens.com/cs/ae/en/view/22144502)

6.5 Modbus TCP connection

The Modbus TCP protocol is based on unencrypted communication. Security measures such as access restrictions must be implemented in the higher-level system/network.

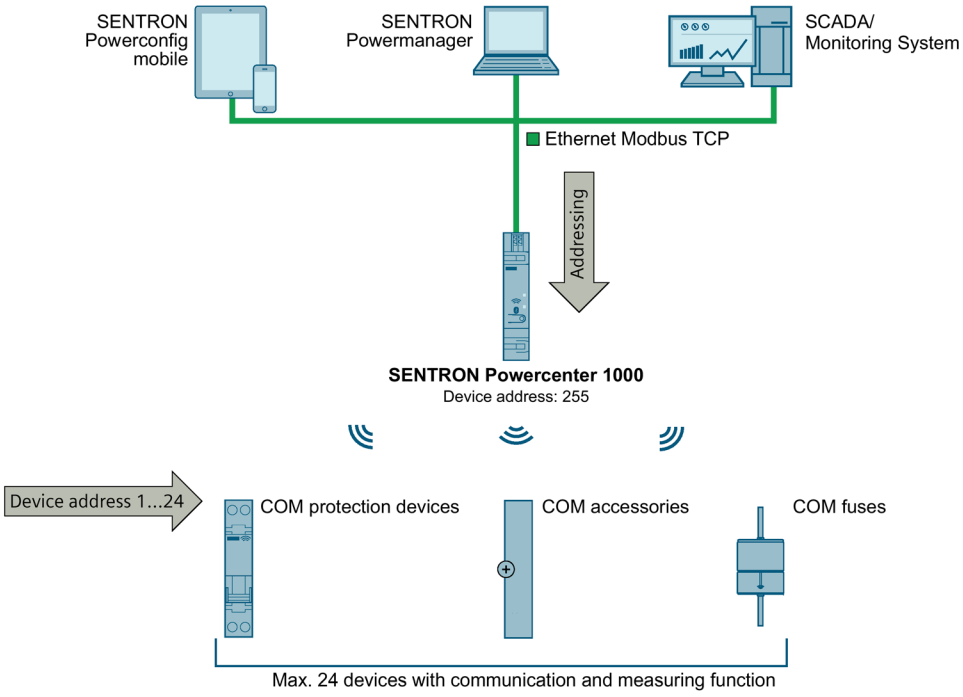
A SENTRON Powercenter 1000 supports up to 3 Modbus TCP connections simultaneously via the Ethernet interface and 1 additional Bluetooth connection in parallel. This means that different software applications would be able to communicate with the data transceiver. This however, is advised against.

Note

It is recommended to always only use one Modbus TCP connection operatively so that it does not result in an overlap of commands.

The information of the individual measurement and communication-capable SENTRON circuit protection devices via Modbus TCP is addressed in 2 levels:

1. Device addresses of the individual devices (255 for SENTRON Powercenter 1000, 1-24 for subordinate devices)
2. Register address of the required data point



Registration address	Measured variable	Format	...
0xC00	Temperature	FP32	...
0xC04	Current L1	FP32	...

The device address is automatically issued (see Chapter Device addressing via Modbus TCP (Page 78)) in SENTRON Powerconfig continuously. If required, it can be adjusted e.g. for an existing user program.

A measured variable or status is addressed via a register address. In addition to the register address, other description information such as format, is provided.

6.5.1 Device addressing via Modbus TCP

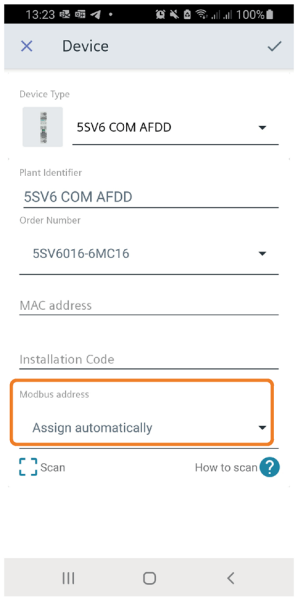
SENTRON Powercenter 1000 devices are addressed via their IP address and the device address. Even the paired, communication-capable terminal devices are individually addressed via their device address. The data can also be transmitted to higher-level systems.

Note

The information of the SENTRON Powercenter 1000 is addressed with 255 (0xFF) in the Unit ID.

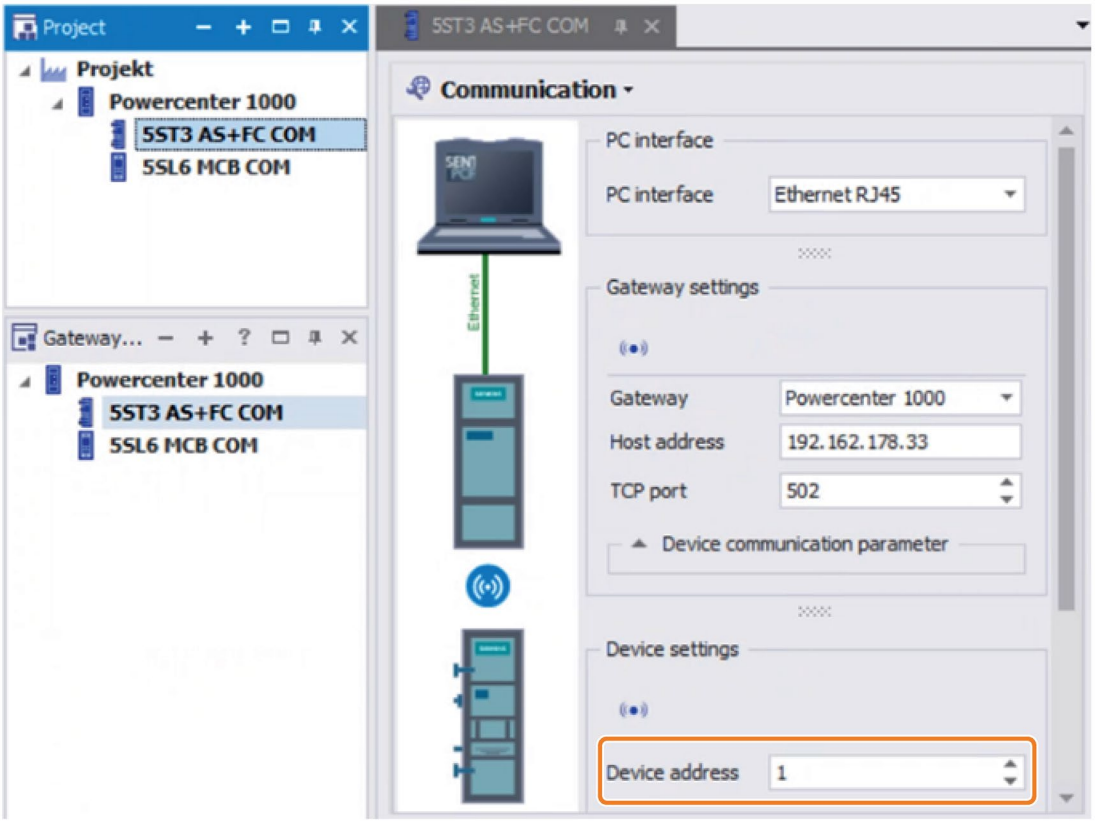
The paired circuit protection devices are assigned unit IDs 1 ... 24 (or 0x01 ... 0x18).

The device addresses are issued in SENTRON Powerconfig mobile. The device address can be chosen manually after a Data Matrix Code of the terminal device has been scanned. These addresses are automatically assigned from 1-24 by default.



Once the devices have been added, the IP and device addresses of the individual devices are displayed in the list, where the devices can be paired, removed or added again.

In the Powerconfig PC version, the device addresses can be set in the communication view.



If the Unit ID information is not available, a device can be identified by querying the identification parameters (order number, serial number and plant identifier) if all 24 possible unit IDs are checked.

The device addresses of the communication-capable circuit protection devices incl. the SENTRON Powercenter 1000 are transmitted to the "Unit_ID" (Unit Identifier) field of the Modbus protocol.

Modbus Protocol Header (7 Bytes)				Protocol Data Unit (PDU)	
Transactions Identifier	Protocol Identifier	Lenghtfield	Unit Identifier	Function Code	Data
(2 Bytes)	(2 Bytes)	(2 Bytes)	(1 Byte)	(1 Byte)	(varies)

Modbus TCP/IP Application Data (ADU)
(This information is embedded into the data portion of the TCP frame)

For the information of the SENTRON Powercenter 1000 itself, such as its operating hours or the system time, this means that 255 (0xFF) must be entered in the Unit ID. The first device is addressed with 0x01. If e.g. a 3NA COM fuse has been added at the first position, a valid voltage measurement value cannot be read out here, because this is only supported by 5SL6 COM and 5SV6 COM.

See also

Messaging on TCP / IP Implementation Guide V1.0b
(https://www.modbus.org/docs/Modbus_Messaging_Implementation_Guide_V1_0b.pdf)

Modbus Application Protocol Specification V1.1b
(https://modbus.org/docs/Modbus_Application_Protocol_V1_1b.pdf)

6.5.2 Protocol information

Register addressing

According to the Modbus specification the registers are numbered starting from 1 but addressed starting with 0. Subsequently, the start address in the protocol must be decremented by -1 when reading out a register.

Function codes

Read processes (Read = R) are carried out alternatively with the function codes 0x03 or 0x04 according to the Modbus specification.

Write processes (Write = W) take place with function codes 0x06 or 0x10 according to the Modbus specification.

Data formats

The following data formats are possible:

Abbreviation	Description
U8	unsigned 8 Bit
U16	unsigned 16 Bit
U32	unsigned 32 Bit
S16	signed 16 Bit
UCHAR	unsigned Character with x Bytes
FP32	floating point 32 Bit (according to IEEE-754)
FP64	floating point 64 Bit (according to IEEE-754)
TS	time stamp
ST	system time - System time

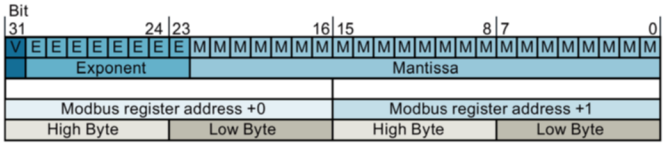
Storage of the time stamp according to the Unix format (UNIX_TS) in seconds since January 1, 1970

Byte arrangement with big-endian data transmission

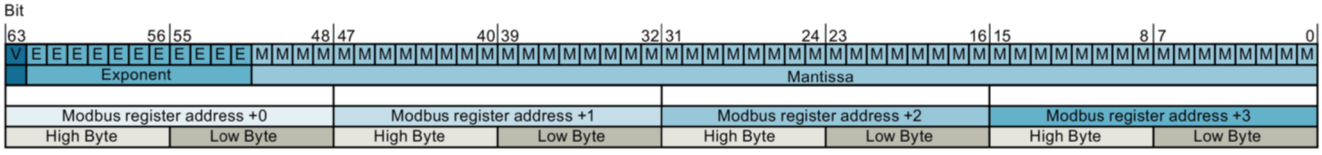
Register		U8	U16	U32	FP32
Register address	High byte	0x00	High data byte	1st data byte	1st data byte (sign bit)
	Low byte	Data byte	Low data byte	2nd data byte	2nd data byte
Register address + 1	High byte	-	-	3rd data byte	3rd data byte
	Low byte	-	-	4th data byte	4th data byte
Number of registers		1	1	2	2

Individual units of information are identified by register addresses. A register is 16 bits in size. If a unit of information is larger than 16 bits, it will require the corresponding number of registers.

Example of the representation by means of a 32-Bit floating point number FP32 according to IEEE 754



Example of the representation of a 67-Bit floating point number FP67 according to IEEE 754


Bit
63
56
55
48
47
40
39
32
31
24
23
16
15
8
7
0
V E E E E E E E E E E M M M M M M M M M M M M M M M M M M M

Intervals

It is recommended to query each device no more than 1x per second. The terminal devices must be addressed individually and should be processed sequentially. The measured values are updated at least every 2 s (see Measured value transmission frequency (Page 67)).

If possible, several registers should always be polled in blocks instead of sending one protocol per register.

6.5.3 Delayed Response and parallel accesses

For many devices there is enough time to acknowledge the Modbus access within the required response time.

The system however, usually composed of one SENTRON Powercenter 1000 and the connected circuit protection devices, is a spatially distributed system. This may lead to a delayed response (Delayed Acknowledge DA) during write accesses via Modbus. If a write access to a corresponding register is executed by the Modbus master, the following reactions may occur. This behavior can occur with all registers from the following chapters with a write access (0x06 or 0x10).

Request	Response	Example	Further measures
Write (Register)	OK	Write response	None
Write (Register)	Delayed Acknowledge	Exception Code 05	Function is formally correct, is brought to execution.
Write (Register)	Slave Device (= device address) Busy	Exception Code 06	Occurs if a Write has already been sent for this terminal device. Renewed write access is only possible if the previous Write is completed.

On the one hand, the statuses (Idle → Exception Code 05, Busy → Exception Code 06) are monitored separately for each device or device address; on the other hand, a device can be accessed in parallel by the three simultaneous connections, so Exception Code 06 must generally be expected if the same terminal device receives several write requests from different connections.

A terminal device should also only be parameterized from a physical connection. The write requests should be executed sequentially on the same terminal device.

Note

To support multiple applications at the same time, it is necessary to clarify the Write permissions at application level so that the applications do not influence each other, i.e. changes are not overwritten by each other.

Therefore it is recommended to use only one application and to set parameters only via Powerconfig (mobile or PC) and to clearly define the responsibility.

6.5.4 Parameter categories of the circuit protection devices

The metering and communication-capable SENTRON circuit protection devices have different categories of information.

Many of these are common to all or many devices. There is also information that is specific for individual device types. In the following chapters this is marked with an "x" for all devices, if the register is available. Furthermore, the scope of information can expand in other versions.

The following categories are accessible to the applications via Modbus TCP.

Category	Description
Identification	Device characteristics according to manufacturer (e.g. order number, rated current, tripping characteristic) Usage environment (plant identifier, location identifier, installation date)
Communication parameters	Parameters for communication path settings: Ethernet, Bluetooth, radio
Communication information	MAC addresses and installation codes of the radio stations
Measured and status variables	Electrical measured variables (e.g. current, voltage, line frequency, power, energy), Temperature Switching status
Measured value settings	Times for the averaging Limit value monitoring Alarm triggering thresholds Expected energy flow direction
Diagnostics	Measured variables of the diagnostics functions (e.g. operating hours, switching cycles, tripping operations) Parameters for alarm and status information
RCM	Measured values and set values for RCM function

This information can be displayed or changed with SENTRON Powerconfig and SENTRON Powerconfig mobile.

The following description is used to be able to access this with other applications and not just with SENTRON applications such as Powermanager, Powermind, Powerconfig, Powerconfig mobile and SENTRON Powercenter 3000.

The information is described in the data point with register number, data format and access possibility.

The accesses mean the following function codes:

- R: 0x03, 0x04
- RW: 0x03, 0x04 or 0x10
- WO: 0x10
- CMD: 0x06

6.5.4.1 Information on identification

Register	Length	Designation	Format	Value range	Access	POC 1000	5ST3 COM/ 3RV2 COM	5SL6 COM	5SV6 COM	3NA COM
2	1	Manufacturer identification	U16	42 = Siemens	R	x	x	x	x	x
3	10	Order number	UCHAR[20]	ASCII characters	R	x	x	x	x	x
13	8	Serial number	UCHAR[16]	ASCII characters	R	x	x	x	x	x
21	1	Hardware version	U16		R	x	x	x	x	x
22	2	Software version	UCHAR[4]		R	x	x	x	x	x
29	16	Plant identifier	UCHAR[32]	ASCII characters	RW	x	x	x	x	x
45	11	Installation location	UCHAR[22]	ASCII characters	RW	x	x	x	x	x
56	8	Installation date	UCHAR[16]	ASCII characters	RW	x	x	x	x	x
94	1	Target market	U16	1= IEC, 2 = UL, 3 = IEC+UL	R	x	x	x	x	x
95	1	Rated current I_n in A	U16		R	---	---	x	x	---
96	1	Tripping characteristic	U16	0 = not defined 1 = Characteristic A 2 = Characteristic B 3 = Characteristic C	R	---	---	x	x	---
97	1	Setting the flashing mode for device localization	U16	0 = Stop flashing 1 = Flashing for 10 seconds	CMD	x	x	x	x	---
98	1	Installation location of the fuse in device frames	U16	0 = Not available 1 = L1, 2 = L2, 3 = L3, 4 = N	RW	---	---	---	---	x
99	10	Order number fuse fused part (for storage in the electronic module)	UCHAR[20]	ASCII characters	RW	---	---	---	---	x
109	1	Hardware revision level of the electronics module	U16	0 = Not available 1 - 10 = AS1 - AS10	R	X	x	x	x	x
110	1	Rated current fused part I_n in A	U16	0 ... 1000	RW	---	---	---	---	x

Register	Length	Designation	Format	Value range	Access	POC 1000	5ST3 COM/ 3RV2 COM	5SL6 COM	5SV6 COM	3NA COM
111[1)]	1	Type of mount-on device	U16	0 = Unknown 1 = RCD - residual current device 2 = MCB - miniature circuit breaker 3 = SCD - switch and control device 4 = AFD - arc fault detection device 5 = RCBO - residual current operated circuit breaker (FI-LS)	RW	---	x	---	---	---
112[1)]	1	Device version	U16	0 = Unknown 1 = 5ST3 COM AS+FC 2 = 3NA COM Fuse 3 = 5SL6 COM MCB 4 = 5SV6 COM AFDD	R	---	x	x	x	x
113[1)]	16	Plant identifier, mount-on device	UCHAR[32]	ASCII characters	RW	---	x	---	---	---

1) Not available in firmware version 1.0.0

6.5.4.2 Communication parameters IP

Register	Length	Designation	Format	Value range	Access	POC 1000	5ST3 COM/ 3RV2 COM	5SL6 COM	5SV6 COM	3NA COM
512	3	Ethernet MAC address	UCHAR[6]		R	x	---	---	---	---
515	2	DHCP on/off	U32	0 = Off, 1 = On	RW	x	---	---	---	---
517	2	SNTP server IP address	U32	0 ... 0xFFFFFFFF[1)]	RW	x	---	---	---	---
519	2	SNTP client mode	U32	0 = off (default) 1 = active 2 = broadcast	RW	x	---	---	---	---
521	2	Firewall on/off	U32	0 = Off, 1 = On	RW	x	---	---	---	---
523	2	IP port number	U32	0 ... 0xFFF (Default 502)	RW	x	---	---	---	---
525	2	Static IP address	U32	0 ... 0xFFFFFFFF[1)] (Default 0.0.0.0)	RW	x	---	---	---	---
527	2	Subnet mask	U32	0 ... 0xFFFFFFFF[1)] (Default 255.255.255.0)	RW	x	---	---	---	---
529	2	Gateway	U32	0 ... 0xFFFFFFFF[1)] (Default 0.0.0.0)	RW	x	---	---	---	---
669	2	Current IP address	U32	0 ... 0xFFFFFFFF[1)]	R	x	---	---	---	---

Register	Length	Designation	Format	Value range	Access	POC 1000	5ST3 COM/ 3RV2 COM	5SL6 COM	5SV6 COM	3NA COM
671	2	Current subnet mask	U32	0 ... 0xFFFFFFFF[1)]	R	x	---	---	---	---
673	2	Current gateway address	U32	0 ... 0xFFFFFFFF[1)]	R	x	---	---	---	---
675	2	Activation of a changed IP/Ethernet configuration	U16	0x0	CMD	x	---	---	---	---

1) The U32 value is interpreted as an IP address. E.g. 0xC0A815A1 represents the address C0.A8.15.A1 or in decimals 192.168.21.161

6.5.4.3 Bluetooth communication parameters

Register	Length	Designation	Format	Value range	Default	Access	POC 1000	5ST3 COM/ 3RV2 COM	5SL6 COM	5SV6 COM	3NA COM
768	1	Bluetooth switch on/off or reset	U16	1 = BLE switch on 2 = BLE switch off 3 = Passkey reset		CMD	x	---	---	---	---
769	1	Bluetooth Status	U16	0 = inactive 1 = connectable 2 = connected		R	x	---	---	---	---
770	1	Bluetooth transmit power in dBm	S16	-18 4	2	RW	x	---	---	---	---
771	3	Bluetooth device address	U8[6]			R	x	---	---	---	---

6.5.4.4 Communication parameter radio

Register	Length	Designation	Format	Value range	Default	Access	POC 1000	5ST3 COM/ 3RV2 COM	5SL6 COM	5SV6 COM	3NA COM
1024	2	Date/time (UTC)	U32	UNIX_TS since 01.01.1970	01.01.2000 at 00:00:00	RW	x	---	---	---	---
1026, 1027, ... 1049	1	Pairing status of terminal devices 1 ... 24	U16	0 = Information not available (MAC/Installation code missing) 1 = Pairing running 2 = Pairing completed 3 = Pairing unsuccessful 5 = Pairing timeout 9 = Remove running 11 = Remove timeout		R	x	---	---	---	---
1050	1	Transmit power in dBm	S16	-18 ... 2	0	RW	x	x	x	x	x
1051, 1052, ... 1074	1	Device status of terminal devices 1 ... 24	U16	0 = IDLE (no terminal device paired) 1 = Offline 2 = Connect 3 = Connected		R	x	---	---	---	---

6.5.4.5 Communication information radio

Register	Length	Designation	Format	Value range	Access	POC 1000	5ST3 COM/3RV2 COM	5SL6 COM	5SV6 COM	3NA COM
1536	9	Install. Code device 1	UCHAR[18]	0xFFF...	WO	x	---	---	---	---
1545	4	MAC address device 1	UCHAR[8]	0xFFF...	RW	x	---	---	---	---
1549	9	Install. code device 2	UCHAR[18]	0xFFF...	WO	x	---	---	---	---
1558	4	MAC address device 2	UCHAR[8]	0xFFF...	RW	x	---	---	---	---
...										
1835	9	Install. code device 24	UCHAR[18]	0xFFF...	WO	x	---	---	---	---
1844	4	MAC address device 24	UCHAR[8]	0xFFF...	RW	x	---	---	---	---
1848	12	Disconnects the terminal device from Powercenter 1000	U8[24]	0 = no action 1 = start pairing 3 = Remove device, terminate pairing *(example: Start pairing for terminals 1 and 3: Set byte 1 & 3 to value 1, the rest are 0)*	WO	x	---	---	---	---
1860[1)]	1	Manual radio channel selection on / off (can only be set prior to pairing)	U16	0 = Off (default) 1 = On	RW	x	---	---	---	---
1861[1)]	1	Manual radio channel (can only be set prior to pairing)	U16	11 ... 26	RW	x	---	---	---	---

[1)] Available as of Version 2.0

6.5.4.6 Measured and state variables

Register	Length	Designation	Format	Value range	Unit	Access	POC 1000	5ST3 COM/ 3RV2 COM	5SL6 COM	5SV6 COM	3NA COM
3072	2	Current temperature	FP32		°C	R	x	x	x	x	x
3074	2	Average temperature	FP32		°C	R	x	x	x	x	x
3076	2	Actual current L	FP32		A	R	---	---	x	x	x
3078	2	Average current L	FP32		A	R	---	---	x	x	x
3080	2	Maximum current L	FP32		A	R	---	---	x	x	x
3082	2	L-N Voltage	FP32		V	R	---	---	x	x	---
3084	2	Line frequency	FP32		Hz	R	---	---	x	x	---
3086	2	Active power {L}	FP32		W	R	---	---	x	x	---
3088	2	Apparent power {L}	FP32		VA	R	---	---	x	x	---
3090	2	Total reactive power Qtot {L}	FP32		V	R	---	---	x	x	---
3092	2	Power factor {L}	FP32			R	---	---	x	x	---
3094	4	Imported active energy	FP64		Wh	R	---	---	x	x	---
3098	4	Exported active energy	FP64		Wh	R	---	---	x	x	---
3102	4	Imported reactive energy	FP64		varh	R	---	---	x	x	---
3106	4	Exported reactive energy	FP64		varh	R	---	---	x	x	---
3110	1	Switching status of the (attached) circuit protection device	U16	0 = Status unknown 1 = Off, without tripping operation 2 = On 3 =Tripped 4 = Tripped, but lever on/blocked		R	---	x[1)]	x	x	---

[1)] Not applicable to 5ST3 COM remote control auxiliary. For the separate data point see Chapter "Supplements for remote control auxiliary"

NAN is shown as "0xFFFFFFFF"

6.5.4.7 Measurement settings

Register	Length	Designation	Format	Value range	Default	Unit	Access	POC 1000	5ST3 COM/ 3RV2 COM	5SL6 COM	5SV6 COM	3NA COM
3586	2	Averaging period temperature	U32	3600 ... 86400	21600	sec	RW	x	x	x	x	x
3588	1	Activating the temperature alarm	U16	0 = Off 1 = On	1	---	RW	x	x	x	x	x
3589	2	Limit values for the temperature alarm	FP32	20 ... 110 20 ... 120 3NA 20 ... 85 POC	105 115 3NA 80 POC	°C	RW	x	x	x	x	x
3591	2	Hysteresis of the temperature alarm	FP32	0 ... 10	10	%	RW	x	x	x	x	x
3597	1	Averaging period current	U16	3 ... 3600	10	sec	RW	---	---	x	x	x
3598	1	Switching Alarm1 on and off for upper current violation	U16	0 = Off 1 = On	1	---	RW	---	---	x	x	x
3599	2	Limit value of upper current violation for Alarm1 in % of In	FP32	0 ... 150	80	%	RW	---	---	x	x	x
3601	2	Hysteresis for upper current violation Alarm1	FP32	0 ... 10	5	%	RW	---	---	x	x	x
3607	1	Switching Alarm2 on and off for upper current violation	U16	0 = Off 1 = On	0	---	RW	---	---	x	x	x
3608	2	Limit value of upper current violation for Alarm2 in % of In	FP32	0 ... 150	90	%	RW	---	---	x	x	x
3610	2	Hysteresis for upper current violation Alarm2	FP32	0 ... 10	5	%	RW	---	---	x	x	x
3616	1	Switching Alarm1 on and off for lower current violation	U16	0 = Off 1 = On	0	---	RW	---	---	x	x	x

Register	Length	Designation	Format	Value range	Default	Unit	Access	POC 1000	5ST3 COM/ 3RV2 COM	5SL6 COM	5SV6 COM	3NA COM
3617	2	Limit value of lower current violation for Alarm1 in % of In	FP32	0 ... 105	10	%	RW	---	---	x	x	x
3619	2	Hysteresis for lower current violation Alarm1	FP32	0 ... 10	5	%	RW	---	---	x	x	x
3625	1	Switching Alarm2 on and off for lower current violation	U16	0 = Off 1 = On	0	---	RW	---	---	x	x	x
3626	2	Limit value of lower current violation for Alarm2 in % of In	FP32	0 ... 105	5	%	RW	---	---	x	x	X
3628	2	Hysteresis for lower current violation Alarm2	FP32	0 ... 10	5	%	RW	---	---	x	x	x
3634	1	Switching Alarm1 on and off for upper voltage violation	U16	0 = Off 1 = On	0	---	RW	---	---	---	x	---
3635	2	Limit value for upper voltage violation for Alarm1	FP32	220 ... 270	250	V	RW	---	---	---	x	---
3637	2	Hysteresis for upper voltage violation Alarm1	FP32	0 ... 10	10	%	RW	---	---	---	x	---
3643	1	Switching Alarm2 on and off for upper voltage violation	U16	0 = Off 1 = On	0	---	RW	---	---	---	x	---
3644	2	Limit value for upper voltage violation for Alarm2	FP32	220 ... 270	270	V	RW	---	---	---	x	---
3646	2	Hysteresis for upper voltage violation Alarm2	FP32	0 ... 10	10	%	RW	---	---	---	x	---

Register	Length	Designation	Format	Value range	Default	Unit	Access	POC 1000	5ST3 COM/ 3RV2 COM	5SL6 COM	5SV6 COM	3NA COM
3652	1	Switching Alarm1 on and off for lower voltage violation	U16	0 = Off 1 = On	0	---	RW	---	---	---	x	---
3653	2	Limit value for lower voltage violation for Alarm1	FP32	150 ... 230	195	V	RW	---	---	---	x	---
3655	2	Hysteresis for lower voltage violation Alarm1	FP32	0 ... 10	10	%	RW	---	---	---	x	---
3661	1	Switching Alarm2 on and off for lower voltage violation	U16	0 = Off 1 = On	0	---	RW	---	---	---	x	---
3662	2	Limit value for lower voltage violation for Alarm2	FP32	150 ... 230	185	V	RW	---	---	---	x	---
3664	2	Hysteresis for lower voltage violation Alarm2	FP32	0 ... 10	10	%	RW	---	---	---	x	---
3670	1	Resetting the energy counter	U16	0x0815	---	---	CMD	---	---	x	x	---
3674	1	Energy flow direction	U16	0 = down → up, 1 = up → down	0	---	RW	---	---	x	x	---
3675	1	Reset the extreme values	U16	0x0815	---	---	CMD	---	---	x	x	---
3676	1	Switching the alarm on and off for low AFDD trip threshold violation	U16	0 = Off 1 = On	1	---	RW	---	---	---	x	---

6.5.4.8 Diagnostic information

Register	Length	Designation	Format	Value range	Default	Unit	Access	POC 1000	5ST3 COM/ 3RV2 COM	5SL6 COM	5SV6 COM	3NA COM
2560	2	Alarm status	U32	See table Alarm status details below.			R	x	x	x	x	x
2562	4	Operating hours with load current	FP64			sec	R	---	---	x	x	x
2566	1	Switching the operating hours violation with load current alarm on and off	U16	0 = Off 1 = On	0		RW	---	---	x	x	x
2567	1	Minimal load current based on I_n, from the point at which the operating hours counter is active	U16	5...90	70	%	RW	---	---	x	x	x
2568	4	Limit value of the operating hours alarm with load current	FP64	60 ... 360,000,000	8640000	sec	RW	---	---	x	x	x
2578	4	Total operating hours (with and without load current)	FP64			sec	R	x	x	x	x	x
2582	1	Switching the operating hours violation alarm on and off	U16	0 = Off 1 = On	0		RW	x	x	x	x	x
2583	4	Operating hours limit value	FP64	60 ... 864000000	158000000	sec	RW	x	x	x	x	x
2593	2	Number of mechanical operating cycles	FP32				R	---	x	x	x	---
2595	1	Switching the mechanical operating cycles alarm on and off	U16	0 = Off 1 = On	0		RW	---	x	x	x	---
2596	2	Mechanical operating cycles limit value	FP32	0 ... 10000	1000		RW	---	x	x	x	---
2602	2	Number of tripping operations	FP32				R	---	x	x	x	---

Register	Length	Designation	Format	Value range	Default	Unit	Access	POC 1000	5ST3 COM/ 3RV2 COM	5SL6 COM	5SV6 COM	3NA COM
2604	1	Switching the trip violations alarm on and off	U16	0 = Off 1 = On	0		RW	---	x	x	x	---
2605	2	Trip counter limit value	FP32	1 ... 10000	30		RW	---	x	x	x	---
2621	1	Bluetooth received signal strength (RSSI)	S16	-127 ... 20	-127	dBm	R	x	---	---	---	---
2622	1	Radio received signal strength (RSSI)	S16	-127 ... 127		dBm	R	---	x	x	x	x
2624	2	Number of short-circuit tripping operations	FP32				R	---	x[1)]	x	x	---
2626	1	Closing and opening the trip violations short-circuit	U16	0 = Off 1 = On	1		RW	---	x[1)]	x	x	---
2627	2	Short-circuit tripping operation limit value	FP32	1 ... 50	6		RW	---	x[1)]	x	x	---
2629	4	Current system time and synchronization status {uint32_t system time, uint32_t Synch. Status}	U8[8]				R	x	---	---	---	---
2633	1	Used radio station (11-26)	U16	0 ... 26	0		R	x	---	---	---	---
2652	1	RCM test status	U16	0 = unknown 1 = passed 2 = failed 3 = not executed 4 = canceled			R	---	---	x[2)]	---	---

1) Applies only to 3RV2
2) Applies only to RCM versions

Alarm status details (address 0x0A0): U32 - bit sequence

Bit value	Designation	Value range
0	Alarm operating hours with load current	0 = inactive, 1 = active
1	Alarm operating hours	0 = inactive, 1 = active
2	Alarm operating cycles	0 = inactive, 1 = active
3	Alarm tripping operations	0 = inactive, 1 = active
4	Alarm violation of upper temperature limits	0 = inactive, 1 = active
5	Alarm 1 overcurrent	0 = inactive, 1 = active
6	Alarm 2 overcurrent	0 = inactive, 1 = active
7	Alarm 1 undercurrent	0 = inactive, 1 = active
8	Alarm 2 undercurrent	0 = inactive, 1 = active
9	Alarm 1 overvoltage	0 = inactive, 1 = active
10	Alarm 2 overvoltage	0 = inactive, 1 = active
11	Alarm 1 undervoltage	0 = inactive, 1 = active
12	Alarm 2 undervoltage	0 = inactive, 1 = active
13	Breaker tripped	0 = inactive, 1 = active
14	Tripping arcing faults	0 = inactive, 1 = active
15	Overvoltage tripping operation	0 = inactive, 1 = active
16	Short-circuit tripping operation	0 = inactive, 1 = active
17	Alarm: Low AFDD trip threshold violation	0 = inactive, 1 = active
18	Self-test failed	0 = inactive, 1 = active
24	RCM pre-alarm low-pass channel AC/RCD test counter alarm with remote control auxiliary	0 = inactive, 1 = active
25	RCM alarm low-pass channel AC	0 = inactive, 1 = active
26	RCM pre-alarm low-pass channel RMS	0 = inactive, 1 = active
27	RCM alarm low-pass channel RMS	0 = inactive, 1 = active

6.5.4.9 RCM measured values and parameters

The following data points apply only to RCM versions of the 5SL6 COM MCB. The data points are supported from firmware version V2.0:

Register	Length	Designation	Format	Value range	Default	Unit	Access
3328	2	Measured value residual current channel: Harmonic	FP32			A	R
3330	2	Measured value residual current channel: Low-pass AC	FP32			A	R
3332	2	Measured value residual current channel: High-pass AC	FP32			A	R
3334	2	Measured value residual current channel: Band-pass AC	FP32			A	R
3336	2	Measured value residual current channel: Low-pass RMS (AC+ pulsating DC)	FP32			A	R
3338	2	Measured value residual current channel: Fundamental frequency (determined by device, 50 Hz or 60 Hz)	FP32			A	R
5120	1	Setting range measurement channel harmonic of the fundamental frequency	U16	1 = 3rd harmonic 2 = 5th harmonic 3 = 7th harmonic 4 = 9th harmonic 5 = 11th harmonic 6 = 13th harmonic 7 = 15th harmonic 8 = 21st harmonic	1 (3rd harmon.)		RW
5121	1	Setting range measurement channel low-pass (applies to AC & RMS measured value)	U16	1 = to 60 Hz 2 = to 180 Hz 3 = to 500 Hz 4 = to 1 kHz 5 = to 2 kHz 6 = to 5 kHz 7 = to 10 kHz 8 = to 20 kHz 9 = to 50 kHz 10 = to 100 kHz	5 (to 2 kHz)		RW
5122	1	Setting range measurement channel high-pass	U16	1 = 1 kHz - 100 kHz 2 = 2 kHz - 100 kHz 3 = 5 kHz - 100 kHz 4 = 10 kHz - 100 kHz 5 = 20 kHz - 100 kHz 6 = 50 kHz - 100 kHz	5 (10 kHz - 100 kHz)		RW
5123	1	Setting range measurement channel band-pass	U16	1 = 60 Hz - 180 Hz 2 = 180 Hz - 500 Hz 3 = 500 Hz - 1 kHz 4 = 1 kHz - 2 kHz 5 = 1 kHz - 10 kHz 6 = 2 kHz - 5 kHz 7 = 2 kHz - 20 kHz 8 = 2 kHz - 50 kHz 9 = 5 kHz - 10 kHz 10 = 10 kHz - 20 kHz 11 = 20 kHz - 50 kHz 12 = 50 kHz - 100 kHz	7 (2 kHz - 20 kHz)		RW

5124	2	General RCM alarm ON-delay after restart	FP32	0 ... 600 s	0	s	RW
5126	2	Hysteresis residual current RCM pre-alarm	FP32	0 ... 25	15	%	RW
5128	2	Hysteresis residual current RCM alarm	FP32	0 ... 25	15	%	RW
5130	2	ON-delay time RCM alarm	FP32	0 ... 10	0.01	s	RW
5132	2	OFF-delay time RCM alarm	FP32	0 ... 3600	1	s	RW
5134	1	Activation and deactivation of automatic RCM alarm reset (auto-reset)	U16	0 = Off 1 = On (automatic reset activated)	1		RW
5135	1	Activation and deactivation of RCM pre-alarm - low-pass AC	U16	0 = Off 1 = On	0		RW
5136	1	Activation and deactivation of RCM alarm - low-pass AC	U16	0 = Off 1 = On	0		RW
5137	2	Limit value of RCM pre-alarm - low-pass AC	FP32	50 ... 100% of associated alarm	50	%	RW
5139	2	Limit value of RCM alarm - low-pass AC	FP32	0.01 ... 0.3	0.015	A	RW
5141	1	Activation and deactivation of RCM pre-alarm - low-pass RMS	U16	0 = Off 1 = On	1		RW
5142	1	Activation and deactivation of RCM alarm - low-pass RMS	U16	0 = Off 1 = On	1		RW
5143	2	Limit value of RCM pre-alarm - low-pass RMS	FP32	50 ... 100% of associated alarm	50	%	RW
5145	2	Limit value of RCM alarm - low-pass RMS	FP32	0.01 ... 0.3	0.015	A	RW
5225	1	Confirmation of RCM alarm	U16	0 x 0815			CMD
5226	1	Start RCM test	U16	0 x 0815			CMD

6.5.4.10 Supplements for remote control auxiliary

The following data points are applicable to both versions (standard and with RCD test) of the remote control auxiliary, in addition to the above tables with the column "5ST3 COM". These data points are supported from firmware version 3.0.

Register	Length	Designation	Format	Value range	Default	Unit	Access	RCD test	Standard
2634	1	ARD status	U16	0 = Unknown 1 = ARD ok 2 = ARD blocked after IR test error 3 = ARD blocked after 3 failed attempts 4 = ARD blocked after 6 failed attempts 5 = ARD canceled			R	x	x
2635	1	RCD test status	U16	0 = Unknown 1 = RCD test successful 2 = Failed: Mount-on device not tripped 3 = Failed: Tripping time exceeded 4 = Failed: Configuration 5 = Error: Lever status 6 = Error: Slide status 7 = Error: Trip-arm status 8 = Error: Hardware 9 = Test not performed 10 = Test canceled			R	x	
2636	1	IR test status	U16	0 = Unknown 1 = IR test successful after trip/RCD test 2 = Individual IR test successful 3 = IR test resistance error 4 = IR test resistance warning 5 = Error: Configuration 6 = Error: Lever status 7 = Error: Slide status 8 = Error: L-N reversed 9 = Error: Zero crossing timeout 10 = Test not performed 11 = Test canceled			R	x	
2637	2	RCD test counter	FP32		0 = Off		R	x	
2639	1	RCD test counter on and off	U16	0 = Off, 1 = On			RW	x	
2640	2	RCD test counter limit value	FP32	1 ... 10000	100		RW	x	
2642	1	RCD test counter reset	U16	0x0815 void			CMD	x	

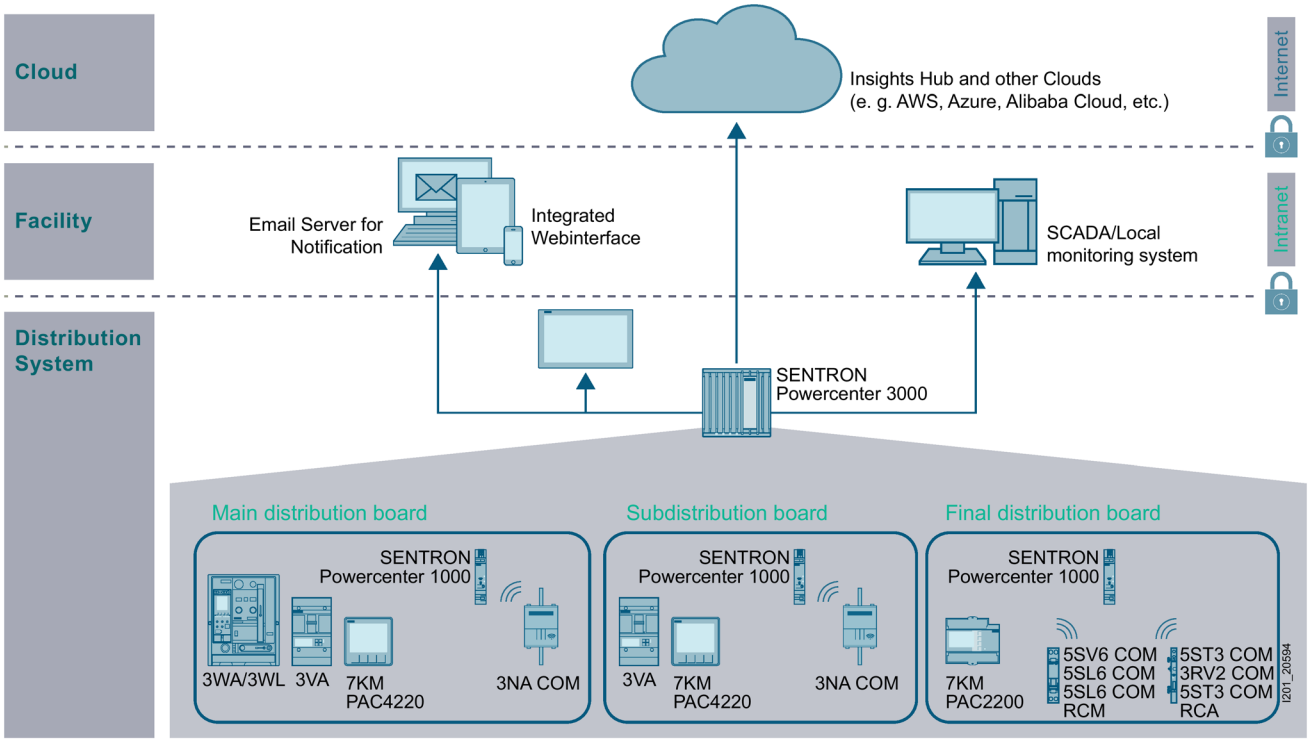
2643	1	Execution of configured tests	U16	0x0815 void			CMD	x	
2645	1	Execution of a device restart.	U16	0x0815 void			CMD	x	x
3111	1	Status of yellow slide	U16	0 = Unknown 1 = OFF, switched off 2 = RC OFF, no remote switching possible 3 = RC ON, remote switching possible			R	x	x
3112	1	Status of handle	U16	0 = Unknown 1 = Off: manual 2 = On manual 3 = Off: No manual off reset 4 = On: Remote switching 5 = Off: Remote switching 6 = On: ARD 7 = Off: Tripping			R	x	x
3677	1	First ARD delay time	U16	10 ... 1800	10	s	RW	x	x
3678	1	Second ARD delay time	U16	10 ... 1800	60	s	RW	x	x
3679	1	Third ARD delay time	U16	10 ... 1800	600	s	RW	x	x
3680	1	ARD waiting time for second pass	U16	0 ... 43200	0	s	RW	x	x
3681	1	Control of ARD function	U16	0 = ARD off 1 = ARD on	1		RW	x	x
3682	1	IR test setting	U16	0 = IR test off 1 = 1-pole IR test 2 = 3-pole IR test	0		RW	x	
3683	1	RCD/IR test setting	U16	0 = RCD test of (Rd=8kΩ) 1 = 30mA RCD, Rd=13kΩ 2 = 30mA RCD, Rd=16kΩ (default) 3= 100mA RCD, Rd=5kΩ 4= 100mA RCD, Rd=8kΩ 5= 100mA RCD selective, Rd=5kΩ 6= 100mA RCD selective, Rd=8kΩ 7= 300mA RCD, Rd=5kΩ 8= 300mA RCD, Rd=8kΩ 9= 300mA RCD selective, Rd=5kΩ 10= 300mA RCD selective, Rd=8kΩ	2		RW	x	
3685	2	Automatic test start time	U32	Unix time (seconds since January 1, 1970, 00:00)	946681200 (= January 1, 2000, 00:00)		RW	x	

3687	1	Automatic test repetition interval	U16	0= never 1 = once at start time 2 = every hour 3 = every 2h 4 = every 4h 5 = every 8h 6 = every 12h 7 = every 24h 8 = every 7 days 9 = every 14 days 10 = every 30 days 11 = every 60 days 12 = every 90 days 13 = every 180 days	0		RW	x	
3688	1	ARD after IR test warning	U16	0 = ARD off after IR warning 1 = ARD on after IR warning	1		RW	x	
3689	1	Remote control switching command	U16	0 = Off 1 = On			CMD	x	x
3690	1	Interlock reset after Manual Off	U16	0x0815 void			CMD	x	x
3691	1	Remote control cause (wired or wireless)	U16	0 = Wired remote control 1 = Wireless remote control	0		RW	x	x
3696	2	Last execution time of automat. test	U32	Unix time (seconds since January 1, 1970, 00:00)	946681200 (= January 1, 2000, 00:00)		R	x	

Application examples

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The SENTRON circuit protection devices with communication and metering function protect the final circuit and gather its data to increase system availability. The devices are used in a main or sub-distribution board. In doing so, the increased transparency allows predictive fault detection through early warnings or targeted, simplified maintenance through permanent status recording. The gathered data still have to be processed in higher-level applications. The SENTRON portfolio offers different digitalization solutions for this purpose.



This can be directly connected to different applications via the Modbus TCP connection of the SENTRON Powercenter 1000, for example:

- the energy monitoring system SENTRON Powermanager
- the IoT (Internet of Things) data concentrator SENTRON Powercenter 3000 with web-based display
- the SENTRON Powermind cloud-based application in the Insights Hub via SENTRON Powercenter 3000

The SENTRON Powercenter 3000 can be used to connect the measurement and communication-capable SENTRON circuit protection devices with SCADA, energy monitoring and maintenance systems. The SENTRON Powercenter 3000 has an integrated web server to visualize the status and measured variable display of all connected devices without requiring other software to be installed.

Furthermore, it also offers the possibility of archiving the data and thus, a more profound data analysis. Similarly, the SENTRON Powercenter 3000 makes it possible to send warnings via email. It also enables connection to other cloud systems via MQTT for fully comprehensive and long-term data availability, even outside the local network. You will find more information on the functions of the SENTRON Powercenter 3000 in the Equipment Manual.

Other types of devices can be connected and visualized via SENTRON Powercenter 3000. For example SENTRON PAC measuring instruments, other Modbus TCP devices or air and compact circuit breakers (3WA and 3VA). Up to 32 devices of the low-voltage power distribution board can be connected with a SENTRON Powercenter 3000 device. In doing so, the SENTRON Powercenter 1000 data transceiver (including its up to 24 devices) count as 1 device.

The SENTRON circuit protection devices with metering and communication function are included in the local Ethernet network via Ethernet switches or router. This is where the other Modbus TCP-capable applications are to be connected in order to visualize the data. The data can be accessed from any location via the SENTRON Powercenter 3000 and the cloud applications.

Alternatively to cloud solutions, a secure VPN connection via the router provides access to the data in the local network from any location.

Communication via Modbus TCP also enables the setting up of own applications. For example via a LOGO! 8.3 or suitable systems. The information from the Chapter Modbus TCP Connection (Page 77) must be used for this purpose.

See also

Equipment Manual - SENTRON Powercenter 3000 (https://support.industry.siemens.com/cs/ww/en/view/109763838)

SENTRON Powermanager (https://support.industry.siemens.com/cs/ww/en/view/109771760)

SENTRON Powermind (https://support.industry.siemens.com/cs/ww/en/view/109793862)

Insights Hub (https://www.plm.automation.siemens.com/global/en/products/mindsphere/)

Service and maintenance

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8.1 Repair instructions

The standard warranty obligations apply to the SENTRON circuit protection devices. You are exempt from a repair.



⚠ WARNING

Hazardous voltage.
Will cause death or serious injury.

This device/unit can result in hazardous voltages.

Touching live parts will result in death or serious bodily injury.

Installation, commissioning and maintenance only by qualified personnel.

8.2 Firmware update

The firmware updates of all SENTRON circuit protection devices with communication and measurement function are provided in the SIOS portal.

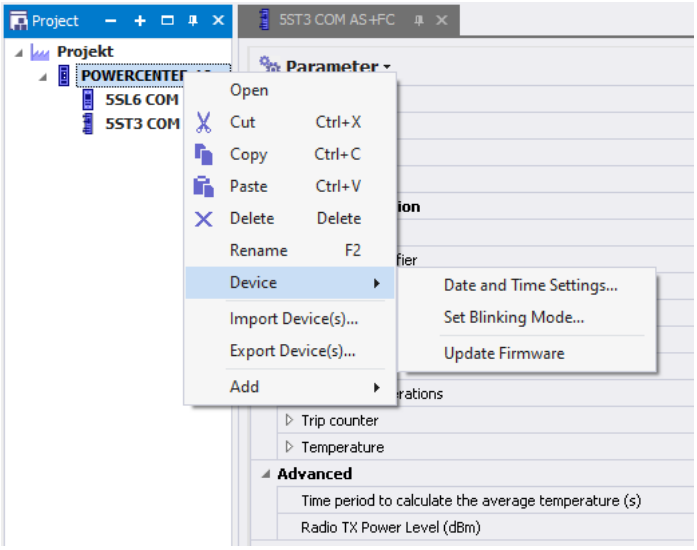
The current firmware update can be found here (https://support.industry.siemens.com/cs/ww/en/view/109797242).

The firmware update comprises the entire device network so as to avoid a mixture of different firmware versions within a system.

Note

A safe, signed update can only be carried out with the SENTRON Powerconfig PC version.

Each communication-capable circuit protection device has the possibility of a firmware update. This can be carried out via the device menu of the SENTRON Powercenter 1000.



The updates are transferred to the SENTRON Powercenter 1000 via the Ethernet cable. During this process, any active Bluetooth connection is terminated. The connected terminal devices receive their update automatically via wireless transmission from the SENTRON Powercenter 1000, even if they are not set up in the Powerconfig project. As a result of wireless communication, it can result in longer wait times. The progress of the system update can be checked in Powerconfig or restarted in the event of an error. Set parameters are maintained in the devices after a firmware update.

The duration of a firmware update depends on the capacity utilization of the radio channels. Usually an update of a SENTRON Powercenter 1000 takes approx. 2 minutes. It takes about 5 minutes per terminal device, although it may take 10-15 minutes for the 3NA COM fuse. To save time here, the firmware update is started for several devices simultaneously.

Note

All devices must be permanently powered during the firmware update for it to be successful. In the case of the 3NA COM fuse, a continuous current flow of at least 10 A through the fuse must also be guaranteed.

It is recommended that an update on the SENTRON Powercenter 1000 be carried out first, before pairing new terminal devices with it. This serves to ensure that a new terminal device or its firmware version is supported by the SENTRON Powercenter 1000.

8.3 Disposal of waste electronic equipment

Disposal of waste electronic equipment



Waste electronic equipment must not be disposed of as unsorted municipal waste, e.g. household waste. When disposing of waste electronic equipment, the current local national/international regulations must be observed.

FAQs

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9.1 Error on commissioning

Fault description	Solutions
The LEDs are not illuminated	Check the power supply units.
No SENTRON Powercenter 1000 found in the list of available Bluetooth devices	• Check whether the mobile device activated Bluetooth and the GPS data • Check if the SENTRON Powercenter 1000 has been set to the Bluetooth mode with a short button press (LED flashes 2 Hz green for max. 180 s, if a connection cannot be established) • If another device is connected to the SENTRON Powercenter 1000, or tries to connect, the Bluetooth connection can be disconnected via the button. After this try again • Switch the Bluetooth mode on again and check whether the manual entry of the PIN code is correct
SENTRON Powercenter 1000 can no longer be connected to a mobile device that has already been paired	Perform a new Bluetooth search. If the device is displayed but cannot be paired, reset everything and perform the Bluetooth commissioning again: • Unpair SENTRON Powercenter 1000 with a long button press (> 10 s) and reset the Bluetooth keys • Remove SENTRON Powercenter 1000 in the operating system under the connected devices • Delete SENTRON Powercenter 1000 from the app project • Activate Bluetooth on the mobile device and SENTRON Powercenter 1000 via a button press < 3 s (COM LED flashes green with 2 Hz) • Perform Bluetooth search and add device incl. scan process and PIN entry
No SENTRON Powercenter 1000 found in the WLAN search	• Check whether the mobile device is also connected to the same network • Update the list once again • Check the router settings • Unplug and plug the Ethernet cable on the SENTRON Powercenter 1000 and perform search again

Fault description	Solutions
The pairing is running in a timeout	• Checking of the power supply of the device • Pairing is running in the background, even if the timeout is displayed after 60 seconds • If wait times are too long, unpair devices, reset the devices via a long button press ≥10 s and perform pairing again
SENTRON Powercenter 1000 no longer provides any data after commissioning	• Check whether the mobile device is also connected to the same network as the SENTRON Powercenter 1000 • Check whether the SENTRON Powercenter 1000 has received another IP address, by carrying out the WLAN search again • Check the power supply. Are the LEDs illuminated? • Check whether the LEDs of the devices are permanently illuminated green • Unplug and plug the Ethernet cable on the SENTRON Powercenter 1000 • Restart the devices (disconnect power supply or switch off devices using the operating lever)
The devices no longer deliver any data	• Check the functioning communication to the SENTRON Powercenter 1000. Is the data transceiver providing values? • Check the power supply and the LEDs • Restart the devices (disconnect power supply or switch off devices) • Disconnect the terminal device from the SENTRON Powercenter 1000 via a long button press on the terminal device or carry out the Unpair function with the app and pair the terminal device again
WLAN search is not finding any devices with the existing VPN connection	For safety reasons, the IP address of the SENTRON Powercenter 1000 can only be entered manually via a VPN connection
Time line of the trends and time stamp of the messages do not match	Check the system time of the SENTRON Powercenter 1000 and re-synchronize the time with the mobile terminal device or check the SNTP server
Devices flash yellow permanently (device error)	• Switch power supply or device on and off • Remove possible radio interference to rectify a communication error • Resetting via a long button press ≥ 10 s to exclude communication error • Replace devices if device error persists

Fault description	Solutions
Device flashes red continuously (tripping operation active)	Confirm the tripping operation with a short button press
Device flashes green/yellow (warning limit value) or yellow/red (warning end of service life)	• Check the active alarm messages • Check the alarm parameters that are switched on and their threshold values (increase threshold value if necessary) • Check the status of the final circuit (an overload tripping operation may occur soon)

9.2 Error with Modbus TCP connection

Fault description	Solutions
Devices not reachable	• Check power supply (communication only possible if the device is supplied with power. If necessary, the device has tripped) • Check the correct device address • Check the correct register
Data point provides incorrect value	• Check the correct device address and register of the desired device (each terminal device does not support each data point) • Check whether the start address of the register has been decremented by -1 for readout
Parameters cannot be modified	• Check the correct function code • Check the value range

9.3 Error with firmware update

Fault description	Solutions
Update not possible	• Check the power supply • Check the error-free communication • Check the correct firmware version (this must be higher than the current one) • Check the correct device type • Restart firmware update
Update runs into a timeout	• Restart update

Technical specifications

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10.1 Powercenter 1000

Designation	Value
Order number	7KN1110-0MC00
Product name	SENTRON Powercenter 1000
Execution of the housing	Standard rail instrument
Suitability for application	5ST3 COM, 5SL6 COM, 5SV6 COM, 3NA COM, 3RV2 COM From firmware V2.0: 5SL6 COM with RCM function; From firmware V3.0: 3RV2 COM, RCA 5ST3 COM
Number of supported devices	24
Supply voltage	DC 24 V SELV
Minimum supply voltage at DC rated value	19.2 V
Maximum supply voltage at DC rated value	28.8 V
Number of interfaces in accordance with wireless	2 (radio, Bluetooth)
Number of interfaces in accordance with Ethernet	1
MODBUS protocol is supported	Yes
IP degree of protection	IP20
Standard for wireless communication Bluetooth	V5.1
Insulation degree of pollution in accordance with EN 60664-1	2
Shock acceleration during operation EN 60068-2-27	150 m / s², 11 ms
Environmental effects vibrations in accordance with IEC 60068-2-6	1-6 Hz, 15 mm, 6-500 Hz, 20 m/s²
Maximum relative humidity with condensation in accordance with IEC 60068-2-38	93%
Minimum ambient temperature during operation	-25°C
Maximum ambient temperature during operation	60°C
Minimum ambient temperature during storage and transport	-40°C
Maximum ambient temperature during storage and transport	85°C
Depth	70 mm
Height	90 mm
Width	18 mm
Net weight	52 g
Radio regulations EU	2014/53/EU
Radio transmission wireless protocol	2400 - 2483.5 MHz
Radio transmission power	< 4.5 dBm (BLE), < 1.5 dBm (Wireless)
max. power consumption	100 mA
Terminal type	screwless
Maximum connectable conductor cross-section	1.5 mm²

Designation	Value
Minimum connectable conductor cross-section	0.2 mm^2
Type of connectable conductor cross-sections with AWG cables	24 ... 16

10.2 5ST3 COM auxiliary switch and fault signal contact

Designation	Value
Order number	5ST30620MC
Product name	5ST3 COM auxiliary switch / fault signal contact
Operating voltage rated value	24 V DC (SELV)
Design of the product factory fitted	No
Available measurement data	Manual ON/OFF, tripping operation, temperature, switching cycles, operating hours, warnings
Product expansion can be mounted	universal (LS, FI, FI/LS, AFDD ON/OFF 5TL1 remote operating mechanism)
Number of modular width units	0.5
Product component auxiliary switch	Yes
Product component fault signal contact	Yes
IP degree of protection	IP20
Reference ID in accordance with IEC 81346-2:2009	SF
Maximum operating voltage for DC	28.8 V
Minimum operating voltage for DC	19.2 V
Power loss [V_A] during standby	0.5 W
Maximum current consumption	20 mA
Terminal type	screwless
Maximum connectable conductor cross-section	1.5 mm^2
Minimum connectable conductor cross-section	0.2 mm^2
Type of connectable conductor cross-sections with AWG cables	24 ... 16
Implementation of the LED display	Yes
Typical electrical endurance (operating cycles)	10000
Maximum permissible ambient temperature during storage	85°C
Minimum permissible ambient temperature during storage	-40°C
Minimum ambient temperature during operation	-25°C
Maximum ambient temperature during operation	60°C
Height	90 mm
Width	9 mm
Depth	76.3 mm
Net weight	28.5 g
Radio regulations EU	2014/53/EU

Designation	Value
Radio transmission frequency	2400 - 2483.5 MHz
Radio transmission power	2.5 dBm

10.3 5SL6 COM miniature circuit breaker

Designation	Value
Order number	5SL60xx-yMC (xx = 02, 04, 06, 08, 10, 13, 16, 20, 25, 32; y = 6 or 7) / 5SL60xx-yMF for RCM version
Product name	5SL6 COM compact miniature circuit breaker
Tripping characteristic class	B and C
Operational current for AC rated value	2 A, 4 A, 6 A, 8 A, 10 A, 13 A, 16 A, 20 A, 25 A, 32 A
Supply voltage for AC rated value	230 V
Breaking capacity current acc. to EN 60898 rated value	6 kA
Available measurement data	Voltage, current, active power, apparent power, reactive power, active energy, line frequency, power factor, temperature, operating cycles, operating hours, tripping operation, warnings, residual current in several frequency ranges (for RCM version)
Energy limitation class	3
Minimum connectable conductor cross-section, solid	0.75 mm^2
Maximum connectable conductor cross-section, solid	16 mm^2
Minimum connectable conductor cross-section, stranded	0.75 mm^2
Maximum connectable conductor cross-section, stranded	16 mm^2
Minimum connectable conductor cross-section, finely stranded with end sleeve	0.75 mm^2
Maximum connectable conductor cross-section, finely stranded with end sleeve	10 mm^2
Minimum tightening torque	1.2 Nm
Maximum tightening torque	2 Nm
Product expansion can be installed	Yes
Number of poles	1P+N
Product function with switching neutral conductor	Yes
IP degree of protection	IP20, with connected conductors
Pollution degree	2
Overvoltage category	III
Number of modular width units	1
Minimum ambient temperature during storage	-40°C
Maximum ambient temperature during storage	75°C
Maximum ambient temperature during operation	60°C
Minimum ambient temperature during operation	-25°C
Ambient operating temperature	-25 ... 45°C: max. 85%rF; 45 ... 60°C: max. 43%rF
Number of test cycles for environmental testing in accordance with IEC 60068-2-30	6
Insulation voltage for AC rated value	250 V

Designation	Value
Impulse withstand voltage, rated value	4000 V
Minimum operating voltage	195 V
Maximum operating voltage with single-phase operation with AC	253 V
Insulation voltage with single-phase operation with AC rated value	250 V
Mounting type	DIN rail
Reference ID in accordance with IEC 81346-2:2009	F
Product standard leading	EN 60898-1
Supply voltage frequency rated value	50 Hz
Power loss [W] (max.)	Between 1 ... 5.7 W depending on rated current and characteristics
Power loss [W] at rated current in warm operating condition per pole	Between 0.9 ... 4.6 W depending on rated current and characteristics
Power loss [W] on N pole at rated current during operation	Between 0.02 ... 0.9 W depending on rated current and characteristics
Vibration resistance according to IEC 60068-2-6	Yes
Product features touch protection	Yes
Position of network connection cable	Any
Mounting position	Any
Mounting depth	70 mm
Height	90 mm
Width	18 mm
Depth	76 mm
Net weight	120 g
Electrical endurance (operating cycles) at AC load on average	7500
Typical mechanical endurance (operating cycles)	10000
Product feature silicone-free	Yes
Product feature halogen-free	Yes
Product feature sealable	Yes
Suitability for use amperemeter	Yes
Suitability for use reactive power meter	Yes
Suitability for use frequency meter	Yes
Suitability for use voltmeter	Yes
Suitability for use active power meter	Yes
Standard for shock	IEC 61373
Standard for environmental testing sinusoidal vibrations	IEC 60068-2-6
IP degree of protection, at rear	IP20
IP degree of protection, on the front	IP40
Reference condition for measuring accuracy	in accordance with IEC 61557-12, IEC 62053-22 and IEC 62053-23
Formula for relative total measurement inaccuracy for measured variable current	+/- 0.5%
Formula for relative total measurement inaccuracy for measured variable active energy, reactive energy	+/- 1%
Formula for relative total measurement inaccuracy for measured variable voltage	+/- 0.5%

Designation	Value
Formula for relative total measurement inaccuracy for measured variable active power, reactive power, apparent power	+/- 1%
Measuring method for current measurement	TRMS
Measuring method for voltage measurement	TRMS
Maximum measurable mains voltage between N and L	400 V
Minimum measurable mains voltage between N and L	50 V
Measurable line frequency final value	65 Hz
Measurable line frequency initial value	45 Hz
Power consumption of the power supply	0.4 VA
Measuring category for voltage measurement	CAT III acc. to IEC 61010-2-030
Minimum measurable current at AC	0.04 A
Maximum measurable current at AC	2 x I_r
Radio regulations EU	2014/53/EU
Radio transmission frequency	2400 - 2483.5 MHz
Radio transmission power	10 dBm
RCM product standard (applies only to RCM versions)	RCM function in accordance with IEC 62020-1
RCM type (applies only to RCM versions)	Type F

10.4 5SV6 COM arc fault detection device

Designation	Value
Order number	5SV6016-xMCyy (x= 6 or 7, yy= 06, 10, 13, 16, 20, 25, 32)
Product name	LS-combo 5SV6 COM arc fault detection device
Product version	Compact device
Tripping characteristic class	B and C
Operational current for AC rated value	6 A, 10 A, 13 A, 16 A, 20 A, 25 A, 32 A
Supply voltage for AC rated value	230 V
Supply voltage frequency rated value	50 Hz
Available measurement data	voltage, current, active power, apparent power, reactive power, active energy, line frequency, power factor, temperature, switching cycles, operating hours, tripping operation, warnings
IP degree of protection	IP20, with connected protective device and conductors
Design of electrical connection	Screw connection
Number of poles	1P+N
Mounting type	DIN rail (REG)
Product expansion can be installed	Yes
Breaking capacity current acc. to EN 60898-1 rated value	6 kA
Number of modular width units	1
Power loss [W] (max.)	Between 1 ... 5.7 W depending on rated current and characteristics
Power loss [W] at rated value current with AC at warm operating condition per pole	Between 0.9 ... 4.6 W depending on rated current and characteristics
Minimum supply voltage for testing equipment	195 V

Designation	Value
Touch protection against electric shock according to EN 50274	Finger and back-of-hand safe
Typical mechanical endurance (operating cycles)	10000
Overvoltage category	III
Minimum connectable conductor cross-section, solid	0.75 mm^2
Maximum connectable conductor cross-section, solid	16 mm^2
Minimum connectable conductor cross-section, stranded	0.75 mm^2
Maximum connectable conductor cross-section, stranded	16 mm^2
Maximum connectable conductor cross-section, finely stranded with end sleeve	10 mm^2
Minimum connectable conductor cross-section, finely stranded with end sleeve	0.75 mm^2
Minimum tightening torque	1.2 N.m
Maximum tightening torque	2 N.m
Pollution degree	2
Minimum ambient temperature during storage	-40°C
Maximum ambient temperature during storage	75°C
Maximum ambient temperature during operation	45°C
Minimum ambient temperature during operation	-25°C
Humidity	85% rel. humidity, > 45°C max 0.056 kg/m^3 abs. humidity
Number of test cycles for environmental testing in accordance with IEC 60068-2-30	28
Product feature silicone-free	Yes
Product feature halogen-free	Yes
Minimum tripping current at arcs serial to load	2 A
Maximum tripping current at arcs serial to load	Operating current
Minimum tripping current at arcs parallel to load	50 A
Maximum tripping current at arcs parallel to load	500 A
Mounting position	Any
Mounting depth	70 mm
Height	90 mm
Width	18 mm
Depth	68.5 mm
Net weight	120 g
Suitability for use amperemeter	Yes
Suitability for use reactive power meter	Yes
Suitability for use frequency meter	Yes
Suitability for use voltmeter	Yes
Suitability for use active power meter	Yes
Standard for shock	IEC 61373
Standard for environmental testing sinusoidal vibrations	IEC 60068-2-6
IP degree of protection, at rear	IP20
IP degree of protection, on the front	IP40
Reference condition for measuring accuracy	in accordance with IEC 61557-12, IEC 62053-22 and IEC 62053-23
Formula for relative total measurement inaccuracy for measured variable current	+/- 0.5%

Designation	Value
Formula for relative total measurement inaccuracy for measured variable active energy, reactive energy	+/- 1%
Formula for relative total measurement inaccuracy for measured variable voltage	+/- 0.5%
Formula for relative total measurement inaccuracy for measured variable active power, reactive power, apparent power	+/- 1%
Measuring method for current measurement	TRMS
Measuring method for voltage measurement	TRMS
Maximum measurable mains voltage between N and L	400 V
Minimum measurable mains voltage between N and L	50 V
Measurable line frequency final value	60 Hz
Measurable line frequency initial value	45 Hz
Power consumption of the power supply	0.4 VA
Measuring category for voltage measurement	CAT III acc. to IEC 61010-2-030
Minimum measurable current at AC	0.02x I_n 0.04 A
Maximum measurable current at AC	2x I_n
Radio regulations EU	2014/53/EU
Radio transmission frequency	2400 - 2483.5 MHz
Radio transmission power	10 dBm

10.5 3NA COM fuse

Designation	Value
Order number	3NA32xx-4KK0y (xx = 24, 30, 32, 36, 40, 42, 44, 52; y = 1, 2, 3, 4)
Product name	3NA COM LV HRC fuse link
Tripping characteristic class	gG and gFF
Versions	With and without electronic module
Size	NH2
Operational current for AC rated value	80 A, 100 A, 125 A, 160 A, 200 A, 224 A, 250 A, 315 A
Available measurement data	Current, temperature, operating hours, warnings
Switching capacity current in accordance with IEC 60947-2	100 kA
Supply voltage for AC rated value	400 V
Factor tripping rule for test current	1.45
Measuring accuracy of current measurement at reference temperature 25°C	1% (8 A ... 440 A), 2% (2.5 A ... 8 A)
Measuring accuracy of current measurement in the range -10°C ... +70°C	2.2% (8 A ... 440 A), 3.2% (2.5 A ... 8 A)
Temperature measuring range	+20°C ... +120°C
Measuring accuracy of temperature measurement	+/- 2.5°C
active power input per phase during current measurement	50 mW
Power loss maximum (dependent on rated current)	9 W, 6.6 W, 10 W, 7.5 W, 13.6 W, 10 W, 16.5 W, 12.4 W, 22.3 W, 12.8 W, 13.5 W, 24 W, 17.5 W
Minimum ambient temperature during storage	-10°C
Maximum ambient temperature during storage	70°C

Designation	Value
Maximum ambient temperature during operation	55°C
Minimum ambient temperature during operation	-10°C
Maximum relative humidity / 25°C without condensation during operating	95%
IP degree of protection	IP20
Pollution degree	2
Reference condition for measuring accuracy	IEC 61557-12
Measuring method	TRMS
Power supply	CT Harvesting
Reference ID in accordance with DIN EN 61346-2	F
Reference ID in accordance with IEC 81346-2:2009	F
Design of the indicator	Front indicator
Mounting type	Non-insulated grip lugs
Design of the switching contact	Non-corroding, silver-plated
Mounting position	Any, preferably vertical
Height	149 mm
Width	60 mm
Depth	72 mm
Net weight	710 g (with electronic module) or 540 g (without electronic module)
European standards	
RED Safety	EN 60669-2-5
RED Health	EN 62479
RED EMC or electrical safety	EN 63044-3 /-5-3, EN 301489-17, EN 300480-17
RED Radio Spec	EN 300 328
International standards	
for EMC for industrial fields	EN 63 044-5-3/IEC 61000-6-2
for EMC against power frequency magnetic field	IEC 61000-4-8
for EMC against voltage dips and voltage interruptions	IEC 61000-4-11
for EMC against conducted interferences through HF fields	IEC 61000-4-6
for EMC against voltage surges	IEC 61000-4-5
for EMC against fast transient electrical disturbance variables	IEC 61000-4-4
for EMC against high frequency fields	IEC 61000-4-3
for EMC against discharge	IEC 61000-4-2
for free falling	IEC 60068-2-32
for continuous shocks	IEC 60068-2-29
for shocks	IEC 60068-2-27
for environmental testing sinusoidal vibrations	IEC 60068-2-6
for environmental testing damp heat, cyclic	IEC 60068-2-30
for environmental testing dry heat	IEC 60068-2-2
for environmental testing cold	IEC 60068-2-1
Approvals	VDE, KEMA KEUR

10.6 3RV2 COM wireless auxiliary and signaling switch for 3RV2 motor starter protectors

Designation	Value
Order number	3RV2921-5M
Product name	3RV2 COM wireless auxiliary and signaling switch
Operating voltage rated value	24 V DC
Design of the product factory fitted	No
Available measurement data	Manual ON/OFF, overload, short-circuit tripping operation, tripping operation, temperature, switching cycles, operating hours, warnings
Product expansion can be mounted	3RV2 motor starter protector
Product component auxiliary switch	Yes
Product component fault signal contact	Yes
IP degree of protection	IP20
Maximum operating voltage for DC	28.8 V
Minimum operating voltage for DC	19.2 V
Power loss [V_A] during standby	0.5 W
Maximum current consumption	20 mA
Terminal type	screwless
Maximum connectable conductor cross-section	1.5 mm^2
Minimum connectable conductor cross-section	0.2 mm^2
Type of connectable conductor cross-sections with AWG cables	24 ... 16
Implementation of the LED display	Yes
Typical electrical endurance (operating cycles)	10000
Maximum permissible ambient temperature during storage	85°C
Minimum permissible ambient temperature during storage	-40°C
Minimum ambient temperature during operation	-20°C
Maximum ambient temperature during operation	60°C
Height	90 mm
Width	18 mm
Depth	71 mm
Net weight	44.7 g
Radio regulations EU	2014/53/EU
Radio transmission frequency	2400 - 2483.5 MHz
Radio transmission power	2.5 dBm

10.7 5ST3 COM remote control auxiliary

Designation	Value
Order number	5ST3072-0MC, 5ST3073-0MC
Product name	5ST3 COM remote control auxiliary with ARD also with RCD test (5ST3073-0MC)
Operating voltage rated value	230 V AC (100 ... 240 V AC)
Design of the product factory fitted	No
Available measurement data	Manual ON/OFF, tripping operation, temperature, switching cycles, operating hours, warnings, ARD status, slide status, RCD test
Adjustable	ARD times, warnings, switching command wired or wireless, RCD/IR test
Number of automatic reclose attempts	3 + 3 (in the second cycle); adjustable
Product expansion can be mounted	universal (LS, FI, FI/LS, AFDD ON/OFF 5TL1 remote control auxiliary) with adapter
Number of modular width units	2 (5ST3072-0MC) 2.5 (5ST3073-0MC)
Product component auxiliary switch	Yes 5ST3073-0MC only via COM function
Product component fault signal contact	Yes 5ST3073-0MC only via COM function
Switching command	Wired or via communication
Minimum spacing between two commands	10 s
Minimum response time	1 s
IP degree of protection	IP20
Product standard	IEC 63024
Power loss	1.6 W 1.8 W
Terminal type	screw-on
Tightening torque	0.25 ... 2 Nm
Maximum connectable conductor cross-section	1.5 mm^2
Minimum connectable conductor cross-section	0.5 mm^2
Implementation of the LED display	Yes
Typical electrical endurance (operating cycles)	10000
Ambient temperature during storage	-40 ... 70°C
Ambient temperature during operation	-40 ... 70°C
Operating frequency	50 / 60 Hz
Cable length in control circuit	1500 m
Height	90 mm
Width	36 mm 45 mm
Depth	72 mm
Net weight	195 g 235 g
Withstand voltage	4 kV
Insulation voltage	250 V

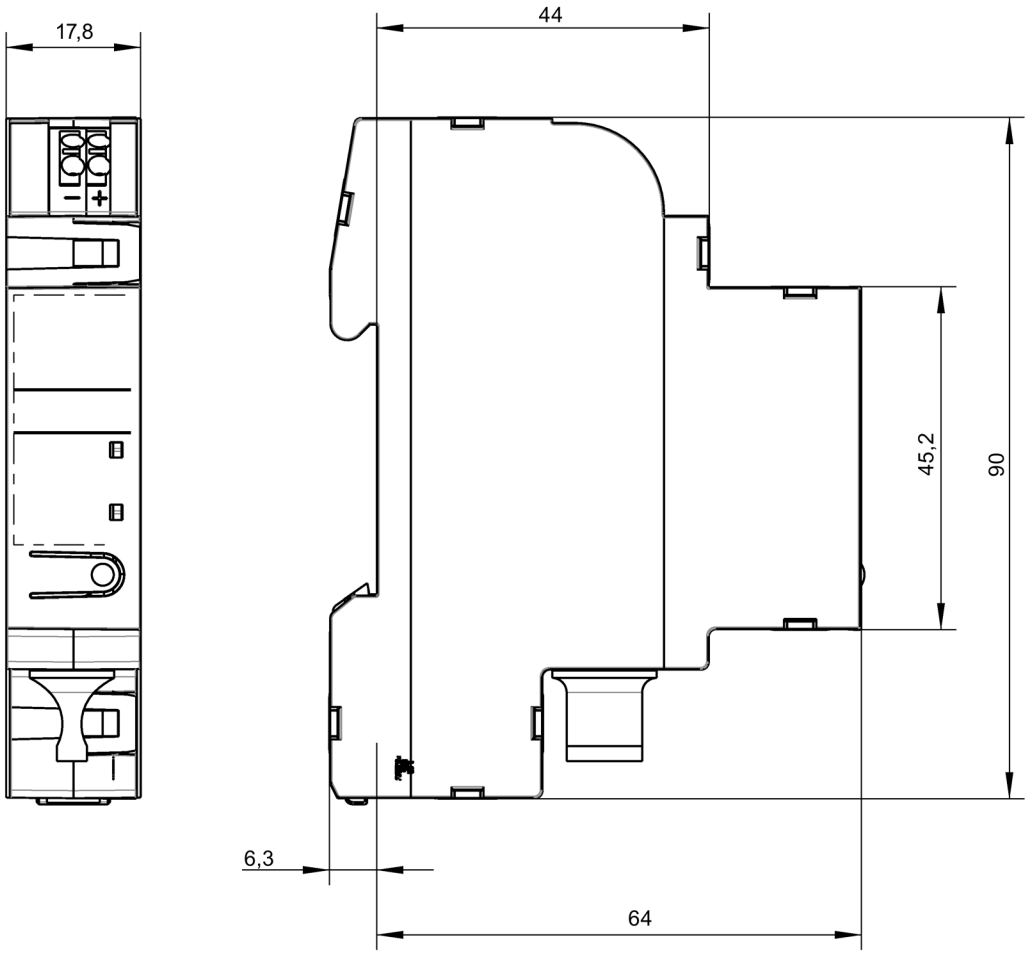
Designation	Value
Overvoltage category	III or AS+FC connections with 5ST3072-0MC: II
Radio regulations EU	2014/53/EU
Radio transmission frequency	2400 - 2483.5 MHz
Radio transmission power	2.5 dBm

Dimension drawings

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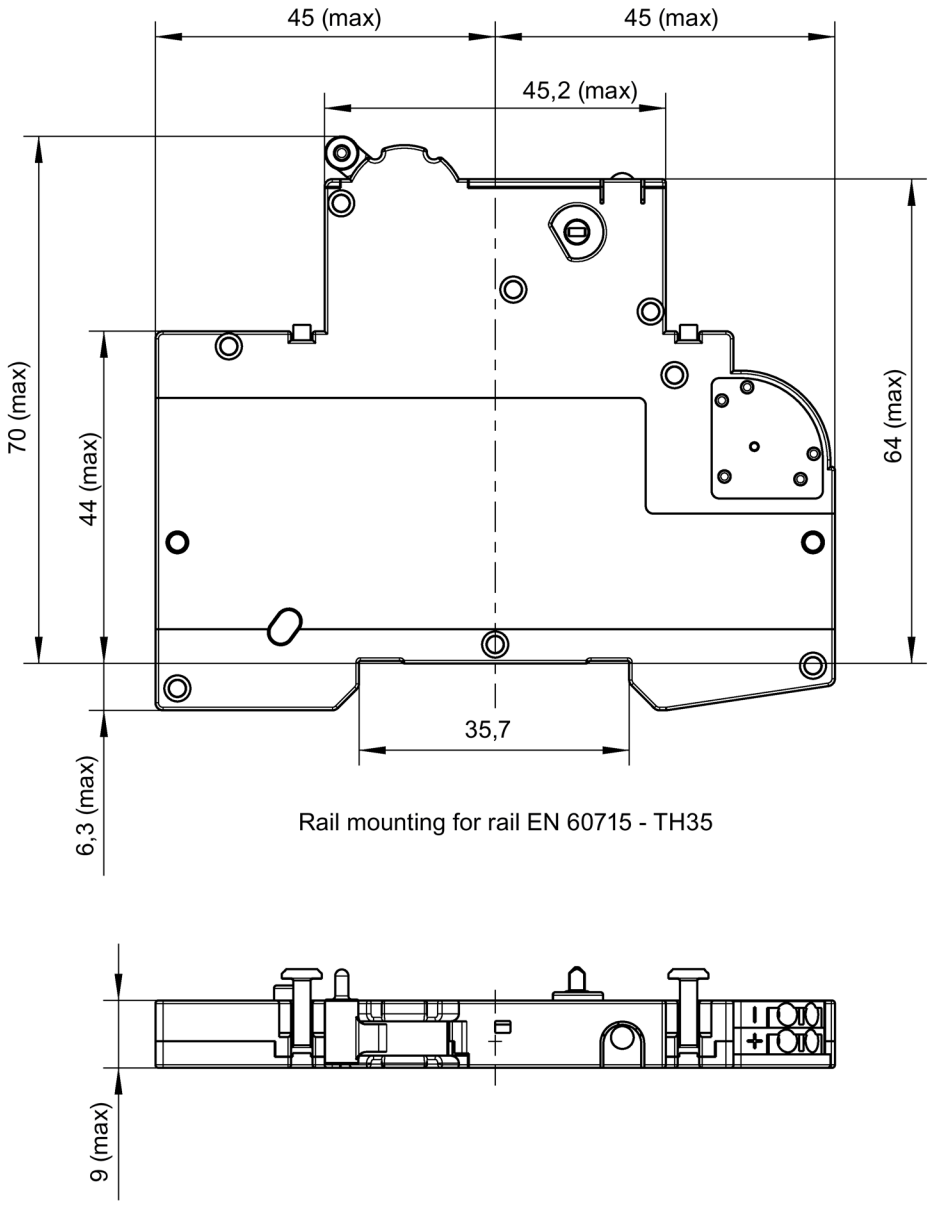
11.1 Powercenter 1000

Dimensions in mm



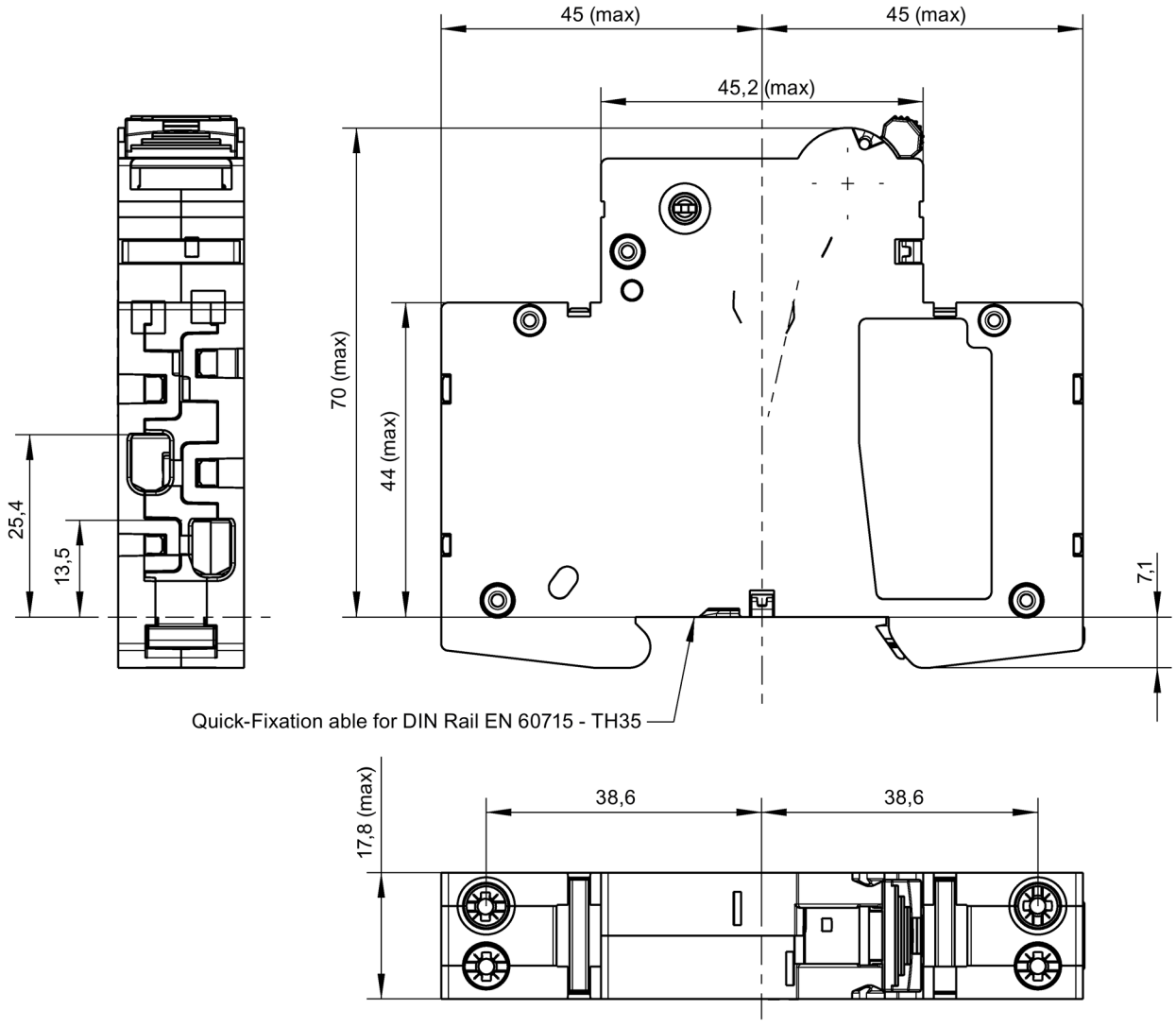
11.2 5ST3 COM auxiliary switch and fault signal contact

Dimensions in mm



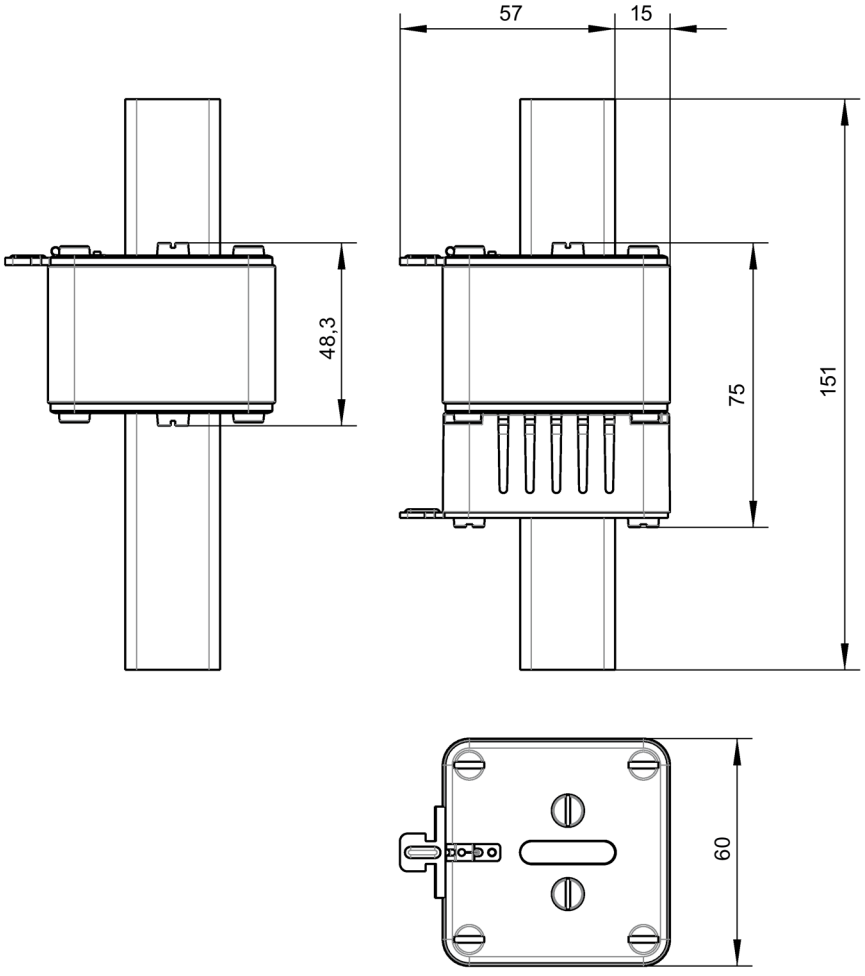
11.3 5SL6 COM / 5SV6 COM miniature circuit breaker and arc fault detection device

Dimensions in mm



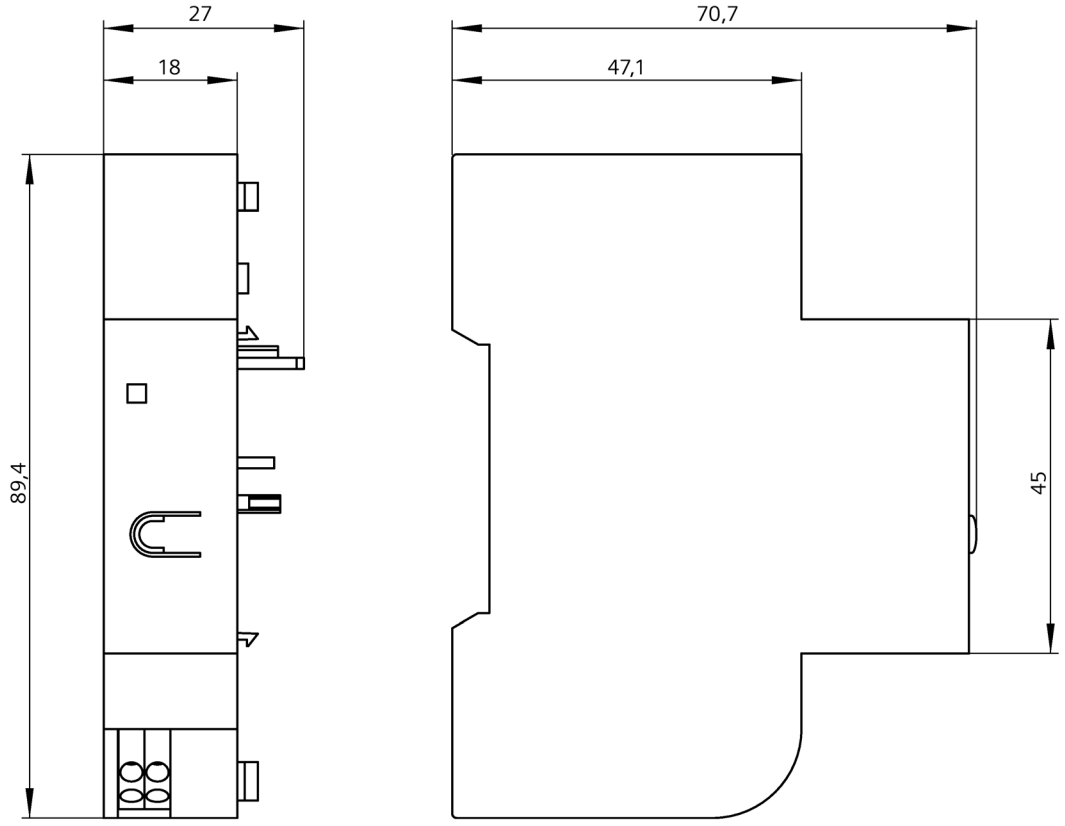
11.4 3NA COM fuse

Dimensions in mm



11.5 3RV2 COM wireless auxiliary and signaling switch for 3RV2 motor starter protectors

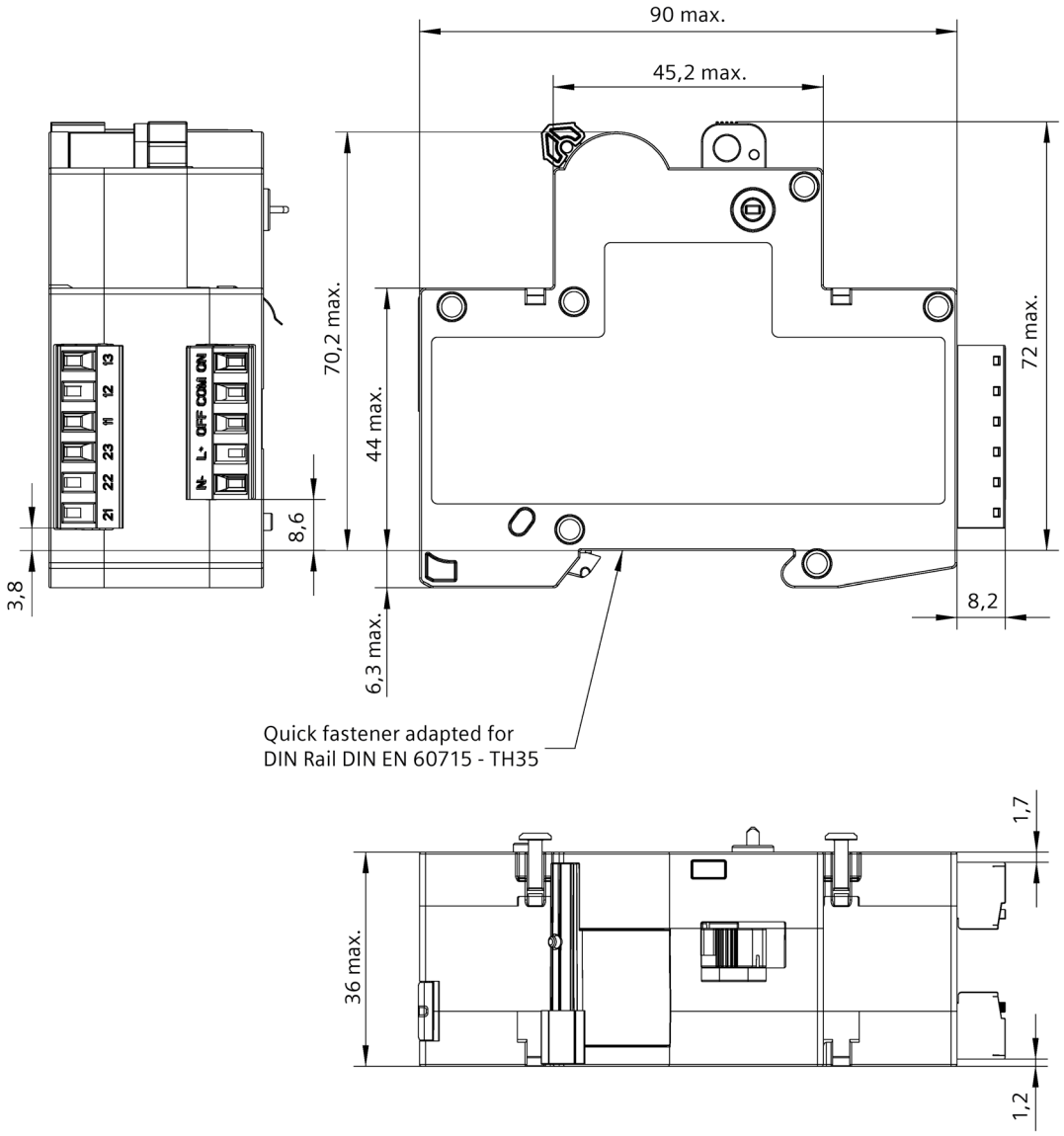
Dimensions in mm



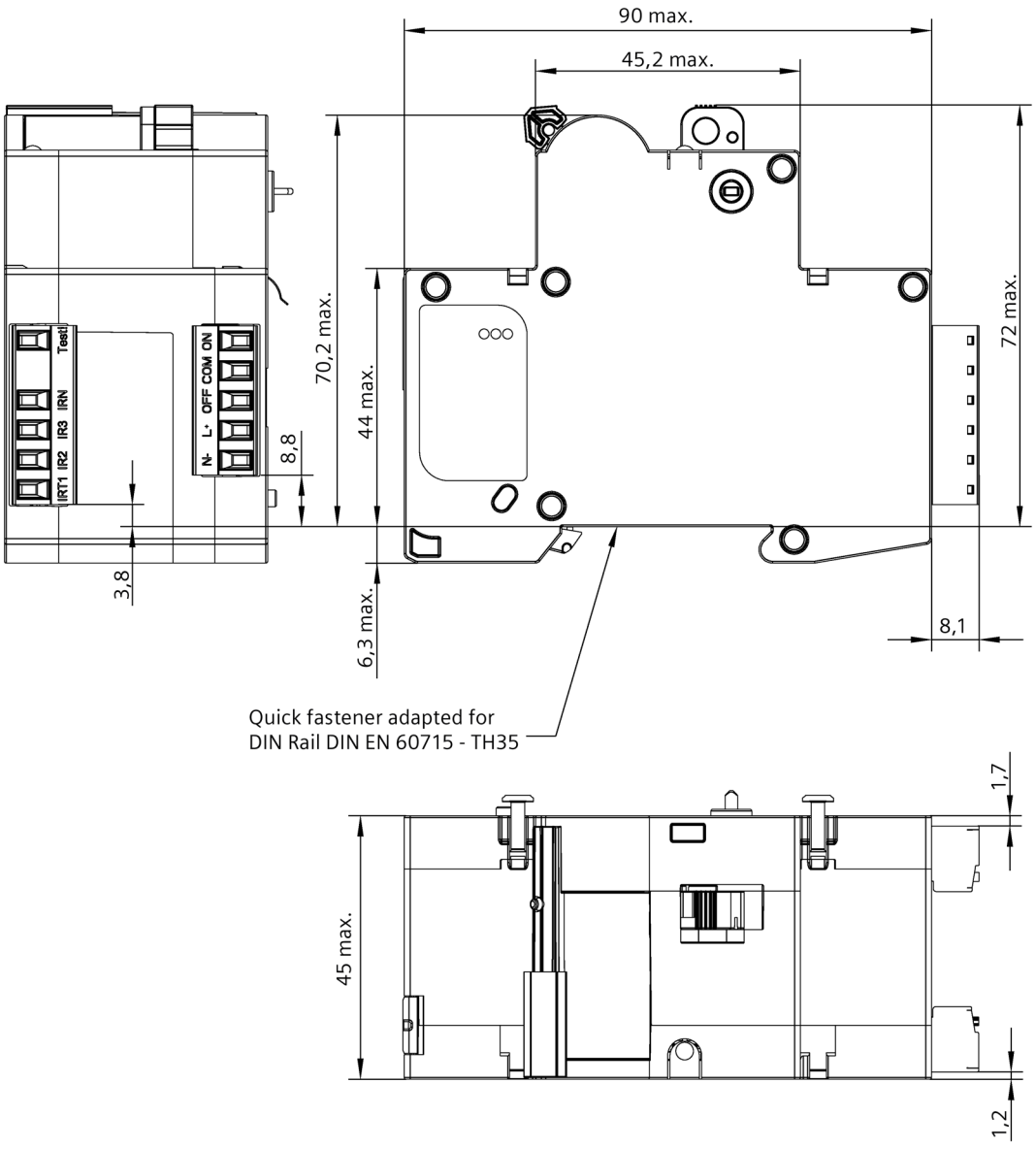
11.6 5ST3 COM remote control auxiliary

Dimensions in mm

5ST3072-0MC (standard version)



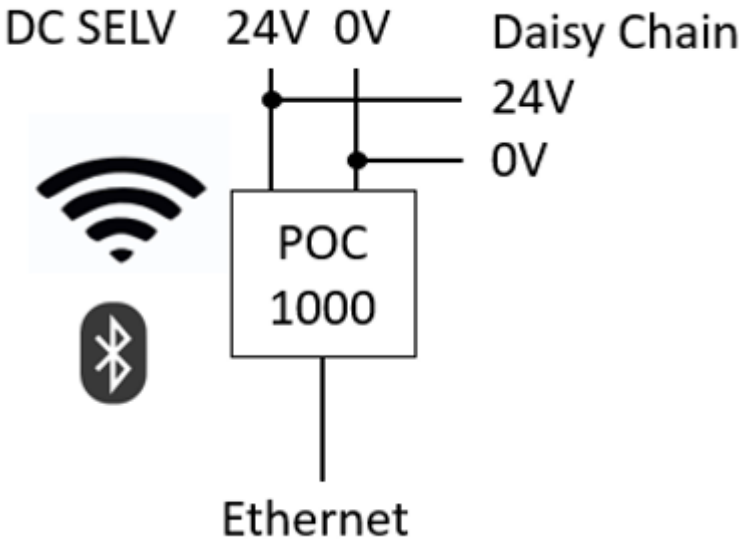
5ST3073-0MC (RCD test version)



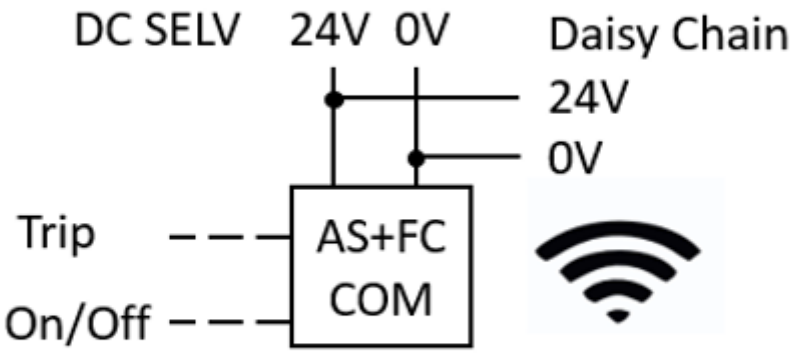
Circuit diagrams

12

12.1 Powercenter 1000

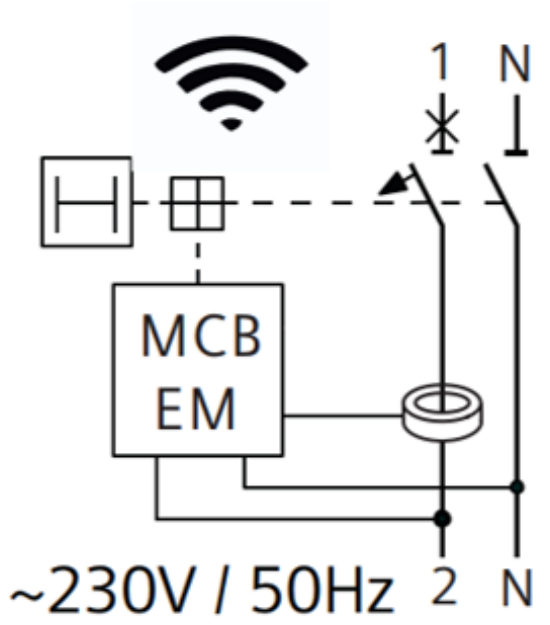


12.2 5ST3 COM auxiliary switch and residual current switch

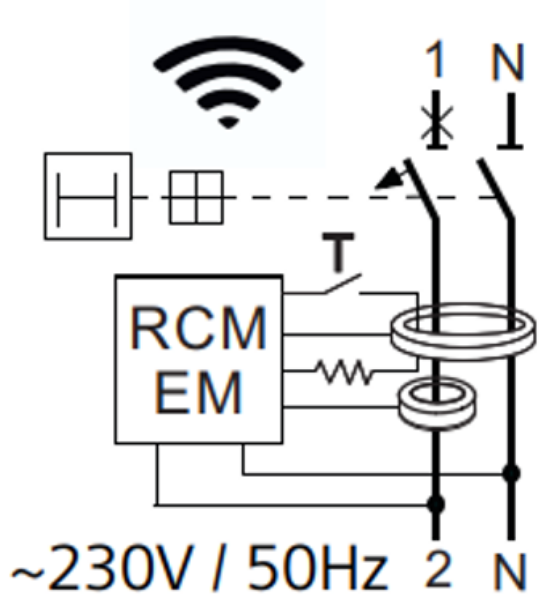


12.3 5SL6 COM miniature circuit breaker

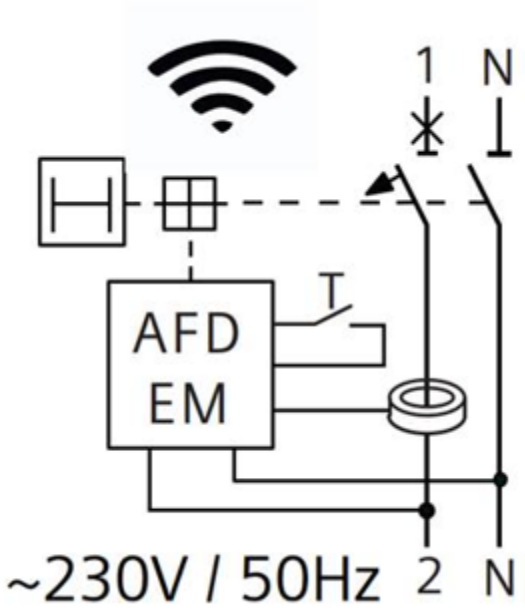
5SL6 COM miniature circuit breaker with power measurement



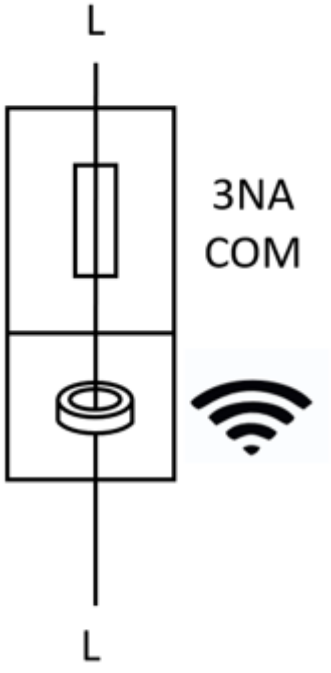
5SL6 COM miniature circuit breaker with power measurement and RCM function



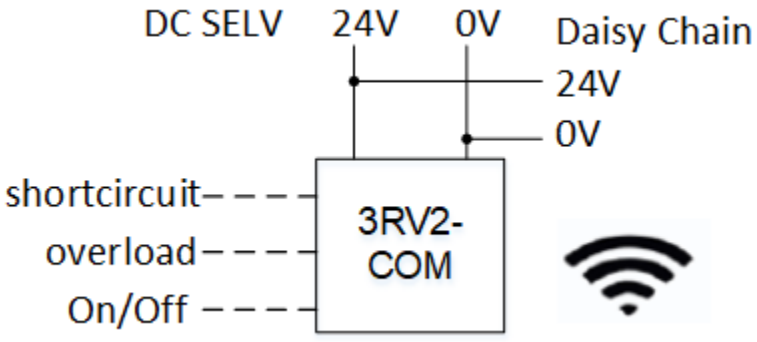
12.4 5SV6 COM arc fault detection device



12.5 3NA COM fuse

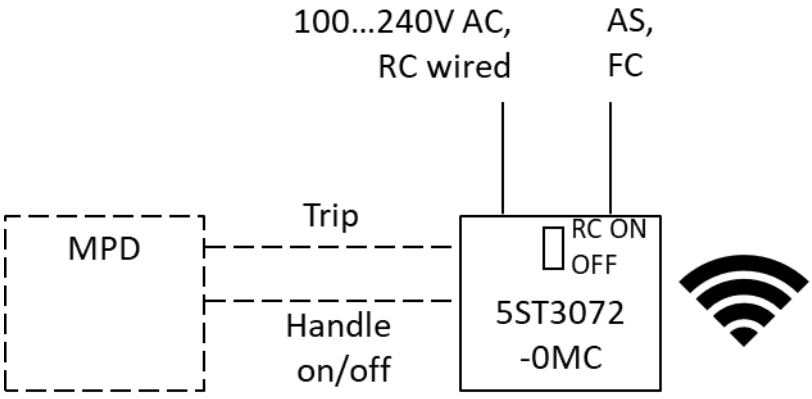


12.6 3RV2 COM wireless auxiliary and signaling switch for 3RV2 motor starter protectors

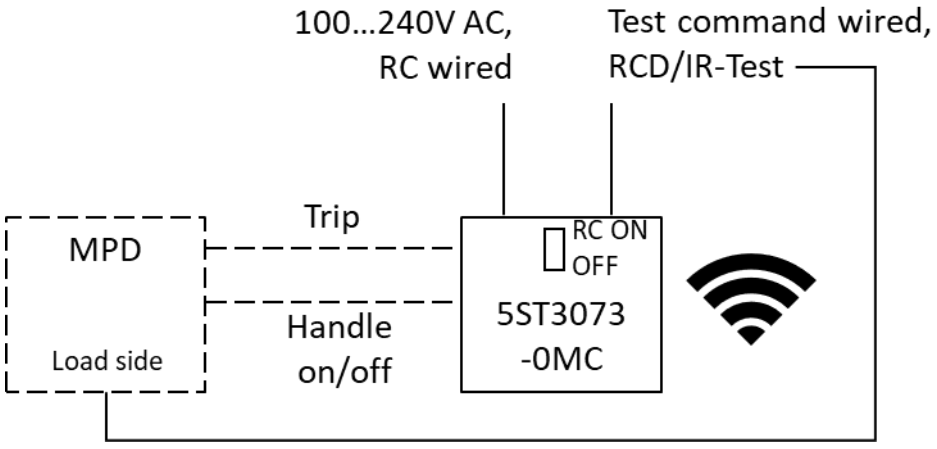


12.7 5ST3 COM remote control auxiliary

5ST3072-0MC



5ST3073-0MC


100...240V AC,
RC wired
Test command wired,
RCD/IR-Test
Trip
MPD
RC ON
OFF
5ST3073
-0MC
Handle
on/off
Load side

ESD guidelines

A

A.1 Electrostatic sensitive devices (ESD)

ESD components are destroyed by voltage and energy far below the limits of human perception. Voltages of this kind occur as soon as a device or an assembly is touched by a person who is not electrostatically discharged. ESD components which have been subject to such voltage are usually not recognized immediately as being defective, because the malfunction does not occur until after a longer period of operation.

ESD Guidelines

NOTICE

Electrostatic sensitive devices

Electronic modules contain components that can be damaged by electrostatic discharge as a result of improper handling.

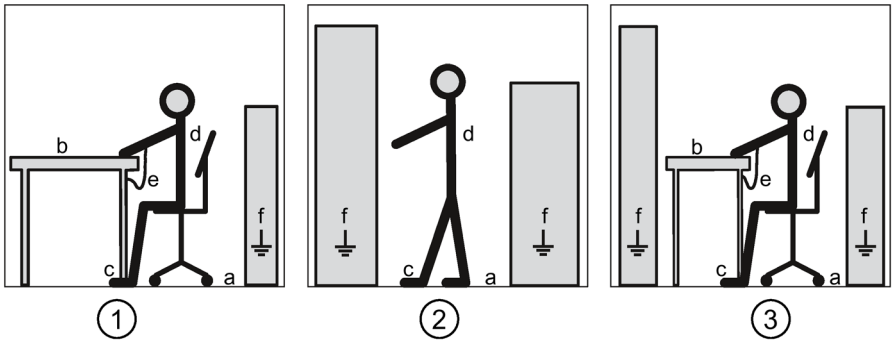
- You must discharge your body electrostatically immediately before touching an electronic module. To do this, touch a conductive, grounded object, e.g., a bare metal part of a switch cabinet or the water pipe.
- Always hold the component by the plastic enclosure.
- Electronic modules should not be brought into contact with electrically insulating materials such as plastic film, plastic parts, insulating table supports or clothing made of synthetic fibers.
- Always place electrostatic sensitive devices on conductive bases.
- Always store and transport electronic modules or components in ESD-safe conductive packaging, e.g. metalized plastic or metal containers. Leave the component in its packaging until installation.

NOTICE

Storage and transport

If you have to store or transport the component in non-conductive packaging, you must first pack the component in ESD-safe, conductive material, e.g., conductive foam rubber, ESD bag.

The diagrams below illustrate the required ESD protective measures for electrostatic sensitive devices.



(1) ESD seat

(2) ESD standing position

(3) ESD seat and ESD standing position

Protective measures

a Conductive floor

b ESD table

c ESD footwear

d ESD smock

e ESD bracelet

f Cubicle ground connection

List of abbreviations

B

Abbreviations

AFDD	Arc Fault Detection Device
ARD	Automatic reclosing function (Auto Reclosing Device)
DA	Delayed Acknowledge
DHCP	Dynamic Host Configuration Protocol
DMC	Data Matrix Code
FI or RCD	Residual-current device
IR	Insulation Resistance
MCB	Miniature Circuit Breaker
RCA	Remote Control Auxiliary
RCM	Residual Current Monitoring
RF	Radio Frequency
RMS	Used here as an abbreviation for the combination of AC & DC measured values for RCM devices (does not indicate mean value)
RSSI	Received Signal Strength Indicator
I_n	Rated current
U_n	Rated voltage

Further Information

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